

New Question Bank

Official College Board Digital SAT Question Bank – SAT Math

Algebra → Advanced Math → Problem-Solving & Data Analysis → Geometry & Trigonometry → June Resit

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Γραμμένα από τον Τσαλαπάτη Σταύρο

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Algebra

Math | 23 questions with full Rule / Tip / Trap / POE / Desmos explanations

The Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Linear equations in one variable

5 questions · Easy 2 / Medium 2 / Hard 1

Linear equations in one variable — Q1 [Easy] (ID: 5eb4918d)

Question ID: 5eb4918d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $9x + 4 = 67$, what is the value of $90x + 40$?

Answer

- A. 7
- B. 70
- C. 130
- D. 670

Correct Answer: D

The Rule: This is a substitution shortcut problem. When a question gives you one equation and asks for the value of a MULTIPLE of that same expression (here, $90x+40$ is exactly 10 times $9x+4$), you almost never need to solve for x itself — you can scale the whole original equation up or down instead.

The Tip: Before solving for x the long way, look closely at the expression you're asked to find ($90x+40$) and compare it to the expression you're given ($9x+4=67$). Notice that $90x$ is exactly 10 times $9x$, and 40 is exactly 10 times 4 . That means you can just multiply the entire original equation by 10, instead of solving for x and then plugging it back in — this saves an entire step and avoids fraction errors.

Why D is right: Multiply both sides of $9x + 4 = 67$ by 10: this gives $90x + 40 = 670$. So without ever finding x by itself, we directly get the value of $90x + 40$, which is 670.

Step-by-Step Solution:

Step 1: Write down the given equation: $9x + 4 = 67$.

Step 2: Notice $90x + 40 = 10 \times (9x + 4)$.

Step 3: Multiply both sides of the original equation by 10: $10(9x + 4) = 10(67)$, which gives $90x + 40 = 670$.

Step 4: The answer is 670, matching choice D.

(For reference, the long way also works: $9x + 4 = 67 \rightarrow 9x = 63 \rightarrow x = 7$. Then $90(7) + 40 = 630 + 40 = 670$ — same answer, just slower.)

The Trap: Choice A (7) is the most commonly picked wrong answer — students solve correctly for $x = 7$, then stop there and select 7 as their final answer, forgetting that the question never asked for x — it asked for the value of $90x + 40$, a completely different quantity.

POE — Eliminate the other three:

X A — 7 is the value of x itself, not the value of $90x + 40$ that the question actually asks for — a classic 'answered the wrong question' trap.

X B — 70 comes from a common arithmetic slip — for example, mistakenly computing $10x$ instead of $90x + 40$, or dividing instead of multiplying somewhere in the process.

X C — 130 doesn't correspond to any correct or nearly-correct path through this problem — it's a distractor value with no clear derivation from the given equation.

Solving with Desmos:

Step 1: Open the Desmos graphing calculator (or the built-in graphing calculator in Bluebook during the actual exam).

Step 2: Click on the first empty input row and type the original equation exactly as written: $9x+4=67$

Step 3: Press Enter. Because this is a one-variable equation, Desmos will automatically plot the solution as a labeled point on the x -axis — it will show the point $(7, 0)$, telling you $x = 7$.

Step 4: Click on the next empty row and type the target expression using that x -value directly: $90*7+40$

Step 5: Press Enter — Desmos will instantly evaluate this and display 670 as the result, which is your final answer, matching choice D.

Step 6 (shortcut for future problems like this): You can skip Steps 3–4 entirely by just typing $10(9x+4)$ in one row and $10*67$ in another to confirm they're equal, or by typing $90x+40$ and 670 as two separate horizontal lines on the same graph — if they're the same value, the lines will coincide perfectly at $x = 7$.

Question ID: 935f3403

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

For a restaurant owner to purchase a pizza oven at a total price of 13,000 dollars, a onetime down payment is required, and then fixed monthly payments are made for the remaining amount owed for the pizza oven. The equation $13,000 = 2,800 + 200t$ represents this situation, where t is the number of fixed monthly payments that are made. Which of the following is the best interpretation of 200 in this context?

Answer

- A. The amount, in dollars, of the down payment
- B. The amount, in dollars, of each fixed monthly payment
- C. The total amount, in dollars, paid for the pizza oven after t fixed monthly payments
- D. The total number of fixed monthly payments

Correct Answer: B

The Rule: In a real-world linear equation of the form (total) = (fixed amount) + (rate)(number of units), the RATE — the number multiplied by the variable — always represents 'how much per one unit,' while the constant by itself represents the one-time fixed amount.

The Tip: Whenever a word problem gives you an equation like $13,000 = 2,800 + 200t$ and asks what a specific number in it 'means,' locate that number in the equation first, then ask: 'Is this number sitting alone (a fixed one-time amount), or is it being multiplied by the variable (a rate per unit)?' The answer choices will usually offer one of each type as a trap.

Why B is right: In the equation $13,000 = 2,800 + 200t$, the number 200 is multiplied by t (the number of monthly payments) — so 200 represents the dollar amount added for EACH additional monthly payment, i.e., the amount of each fixed monthly payment.

Step-by-Step Solution:

Step 1: Identify what t represents: the number of monthly payments made (given directly in the problem).

Step 2: Look at the structure of the equation: $13,000 = 2,800 + 200t$. This matches the general pattern (total price) = (down payment) + (payment amount) $\times$ (number of payments).

Step 3: Since 200 is being multiplied by t , and t counts monthly payments, 200 must be the dollar amount of each individual monthly payment.

Step 4: Match this interpretation to the choices: 'the amount, in dollars, of each fixed monthly payment' is choice B.

The Trap: Choice A is the most commonly picked wrong answer — students sometimes attach '200' to the wrong part of the story, confusing it with the down payment, when the down payment is actually the standalone constant 2,800, not the number being multiplied by t .

POE — Eliminate the other three:

X A — The down payment is the STANDALONE number in the equation (2,800), not the number multiplied by t — 200 is not the down payment.

X C — The total amount paid after t payments would be the ENTIRE right-hand side of the equation ($2,800 + 200t$), not just the coefficient 200 by itself.

X D — The total number of fixed monthly payments is what t itself represents (the variable), not the number 200 that multiplies it.

Solving with Desmos:

This particular question is about interpreting what a number MEANS in a real-world context, not about calculating an unknown value — so Desmos isn't needed to find the answer directly. However, you can use Desmos to double check your understanding of the relationship: Step 1: Type $y=2800+200x$ into Desmos. Step 2: Use the table feature (click the '+' button, then choose the table icon, or just add a table below your equation) and enter $x = 0, 1, 2, 3$ in the x -column. Step 3: Look at how y changes each time x increases by 1 — you'll see y jumps by exactly 200 each time (2800, 3000,

3200, 3400...). This confirms 200 is the amount added per additional payment (per unit of t), verifying that 200 represents each monthly payment's dollar amount, matching choice B.

Linear equations in one variable — Q3 [Medium] (ID: 088bc84e)

Question ID: 088bc84e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $2(5x) = 6$, what is the value of $5x$?

Answer

- A. $\frac{4}{5}$
- B. $\frac{5}{3}$
- C. 3
- D. 4

Correct Answer: C

The Rule: When a number multiplies a full parenthetical expression, like $2(5x) = 6$, you can treat the parenthetical expression ($5x$) as a single unit and divide both sides by the OUTSIDE number to isolate it directly — without ever solving for x by itself first.

The Tip: Whenever the question asks for the value of an expression that's already sitting inside parentheses in the given equation (here, $5x$ is already grouped together), don't break it apart to solve for the single variable first — divide to isolate that exact grouped expression instead. This is faster and avoids unnecessary extra steps.

Why C is right: Starting from $2(5x) = 6$, divide both sides by 2 to isolate the grouped expression ($5x$) directly: $5x = 6 \div 2 = 3$. The question asks for the value of $5x$, which is 3.

Step-by-Step Solution:

Step 1: Start with the given equation: $2(5x) = 6$.

Step 2: Divide both sides by 2 to isolate the grouped expression ($5x$): $5x = 6 \div 2 = 3$.

Step 3: The value of $5x$ is 3, matching choice C.

The Trap: Choice B ($5/3$) is the most commonly picked wrong answer — it comes from solving all the way down to x by itself ($5x = 3$, so $x = 3/5$) and then accidentally flipping that fraction upside down, or from confusing 'the value of $5x$ ' with 'the value of x ' partway through.

POE — Eliminate the other three:

X A — $4/5$ doesn't correspond to a natural error path from this equation — it's a distractor value close in form to the other fraction choice.

X B — $5/3$ typically comes from solving for x alone ($x = 3/5$) and then flipping the fraction by mistake, or from dividing 5 by 3 instead of correctly isolating $5x$ as a single unit.

X D — 4 doesn't correspond to a natural error path from this equation either — likely a distractor value with no clear derivation.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: $2(5x)=6$

Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution as a labeled point on the x -axis, showing the value of x directly ($x = 0.6$, or $3/5$).

Step 3: Since the question asks for $5x$ specifically (not just x), type a second line: $5*0.6$ — Desmos will instantly evaluate this and display 3, confirming the value of $5x$ is 3, matching choice C.

Step 4: Alternatively, skip solving for x entirely — just type $6/2$ directly into Desmos, since dividing both sides of $2(5x)=6$ by 2 gives $5x = 6/2 = 3$ immediately.

Linear equations in one variable — Q4 [Medium] [Grid-In] (ID: f8fa0cf0)

Question ID: f8fa0cf0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$9r = 7(r + 7)$$

What value of r is the solution to the given equation?

Correct Answer: 49/2 or 24.5 (Student-Produced Response — no answer choices given)

The Rule: This is a straightforward one-variable linear equation with a distributed term. The rule is: always distribute (multiply through the parentheses) FIRST, then collect all the variable terms on one side and all the constant terms on the other, before dividing to isolate the variable.

The Tip: When you see a variable on both sides of the equation after distributing (like $9r$ on the left and $7r$ on the right here), move all r -terms to one side by subtracting the smaller coefficient's term from both sides — this keeps your remaining coefficient positive and avoids sign errors.

Why this is the answer: Starting from $9r = 7(r + 7)$, distribute the 7 on the right side: $9r = 7r + 49$. Subtract $7r$ from both sides: $2r = 49$. Divide both sides by 2: $r = 49/2$, or equivalently 24.5.

Step-by-Step Solution:

Step 1: Write the given equation: $9r = 7(r + 7)$.

Step 2: Distribute the 7 across the parentheses on the right side: $9r = 7r + 49$.

Step 3: Subtract $7r$ from both sides to collect the r -terms on the left: $9r - 7r = 49$, which simplifies to $2r = 49$.

Step 4: Divide both sides by 2 to isolate r : $r = 49/2 = 24.5$.

Step 5: This is a grid-in (student-produced response) question, so you would enter either $49/2$ or 24.5 into the answer grid — both are accepted as correct.

The Trap: A common error on this type of problem is distributing incorrectly — for example, writing $7(r+7)$ as $7r + 7$ (forgetting to multiply the 7 by BOTH terms inside the parentheses), which would lead to the wrong equation $9r = 7r + 7$ and an incorrect final answer of $r = 3.5$.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given, using r as the variable: $9r=7(r+7)$

Step 2: Press Enter. Since this is a one-variable equation, Desmos will solve it automatically and plot the solution as a labeled point on the r -axis (which will display as the x -axis in Desmos, since Desmos treats whatever letter you use as its horizontal variable).

Step 3: Click directly on the labeled point to see its exact coordinates — it will show $r = 24.5$ (which Desmos may also display as the fraction $49/2$ if you click to toggle the coordinate display format).

Step 4: Enter 24.5 or $49/2$ into the grid-in answer box on the actual test — both forms are accepted.

Linear equations in one variable — Q5 [Hard] (ID: 55ea0659)

Question ID: 55ea0659

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$x(r - 9) + 4 = 19x + 27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r ?

Answer

- A. 4
- B. 9
- C. 23
- D. 28

Correct Answer: D

The Rule: For an equation of the form $x(r - a) = bx + c$ to have exactly ONE solution, the coefficient of x after simplifying must NOT be zero. If the coefficient becomes zero, the equation either has NO solution (if the constants don't match) or INFINITELY MANY solutions (if they do) — never exactly one. So the question is really asking: for what value of r does the x -coefficient become zero?

The Tip: Whenever a question asks what value 'CANNOT' make an equation have exactly one solution, your first move should be to simplify the equation into the form (coefficient) x = (constant), then figure out what makes that coefficient equal to zero — that specific value is almost always the 'cannot be' answer.

Why D is right: Starting from $x(r - 9) + 4 = 19x + 27$, distribute and rearrange: $xr - 9x + 4 = 19x + 27 \rightarrow xr - 9x - 19x = 27 - 4 \rightarrow xr - 28x = 23 \rightarrow x(r - 28) = 23$. This equation has exactly one solution UNLESS the coefficient $(r - 28)$ equals zero — which happens when $r = 28$. If $r = 28$, the equation becomes $0 = 23$, which is never true (no solution, not one solution). So r cannot equal 28.

Step-by-Step Solution:

Step 1: Start with $x(r - 9) + 4 = 19x + 27$.

Step 2: Distribute x across $(r - 9)$: $xr - 9x + 4 = 19x + 27$.

Step 3: Move all x -terms to the left side and constants to the right: $xr - 9x - 19x = 27 - 4$.

Step 4: Combine like terms: $xr - 28x = 23$, which factors to $x(r - 28) = 23$.

Step 5: For this equation to have exactly one solution, the coefficient $(r - 28)$ must NOT be zero — because if it were zero, the left side would be 0, and $0 = 23$ is impossible (no solution at all, not one).

Step 6: Set $r - 28 = 0$ to find the forbidden value: $r = 28$. This is the value r cannot be, matching choice D.

The Trap: Choice A (4) is the most commonly picked wrong answer for students who correctly simplify to $x(r - 28) = 23$ but then make an arithmetic slip while solving for the forbidden r -value, perhaps confusing which number to set the coefficient equal to zero against.

POE — Eliminate the other three:

X A — 4 does not make the coefficient $(r - 28)$ equal zero — plugging $r = 4$ gives $x(4 - 28) = 23$, or $-24x = 23$, which has exactly one clean solution, so $r = 4$ is perfectly fine.

X B — 9 does not make $(r - 28)$ equal zero either — this value doesn't create any problem with the equation having one solution.

X C — 23 also does not make $(r - 28)$ equal zero — like the other wrong choices, this value still allows exactly one solution.

Solving with Desmos:

Step 1: Open Desmos. Since this problem involves TWO variables (x and r) and asks about a special condition, direct equation-solving won't show the answer as neatly as it does for simpler problems — but you can still verify your algebra.

Step 2: First, simplify the equation by hand (or using Desmos as a calculator) to the form $x(r - 28) = 23$, as shown in the steps above. Step 3: To CONFIRM $r = 28$ breaks the equation, type this test directly into Desmos as a check: $x*(28-28)=23$ — Desmos will show this simplifies to $0 = 23$, which Desmos will indicate has no solution (no point will plot), confirming $r = 28$ is impossible. Step 4: To confirm any of the wrong answers (like $r = 4$) DO work, type $x*(4-28)=23$ into Desmos — it will solve this immediately and plot a valid point, confirming $r = 4$ is a value that r CAN be, ruling it out as the answer.

Linear equations in two variables

2 questions · Easy 0 / Medium 1 / Hard 1

Linear equations in two variables — Q1 [Medium] (ID: 4722abd2)

Question ID: 4722abd2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question
Line p is defined by $2y + 8x = 11$. Line r is perpendicular to line p in the xy -plane. What is the slope of line r ?

- Answer**
- A. -4
 - B. $-\frac{1}{4}$
 - C. $\frac{1}{4}$
 - D. 4

Correct Answer: C

The Rule: For two lines to be perpendicular in the xy -plane, their slopes must be negative reciprocals of each other — meaning if one line has slope m , the perpendicular line has slope $-1/m$. Always rewrite a given equation into slope-intercept form ($y = mx + b$) first to clearly identify its slope before finding a perpendicular slope.

The Tip: When an equation is given in a mixed-up form like $2y + 8x = 11$, always isolate y completely first (getting it into $y = mx + b$ form) before trying to identify the slope — trying to read the slope directly off the unsimplified equation is a common source of sign errors.

Why C is right: Rewrite $2y + 8x = 11$ by isolating y : $2y = -8x + 11$, then divide everything by 2: $y = -4x + 11/2$. This shows line p has a slope of -4 . A line perpendicular to p must have a slope equal to the negative reciprocal of -4 , which is $1/4$.

Step-by-Step Solution:

- Step 1: Start with the given equation for line p : $2y + 8x = 11$.
- Step 2: Subtract $8x$ from both sides: $2y = -8x + 11$.
- Step 3: Divide every term by 2 to fully isolate y : $y = -4x + 11/2$.
- Step 4: Identify the slope of line p from this slope-intercept form: the slope is -4 (the coefficient of x).
- Step 5: To find the slope of a line perpendicular to p , take the negative reciprocal of -4 . Flip the fraction ($-4 = -4/1$ becomes $-1/4$) and then flip the sign: the negative reciprocal is $1/4$.
- Step 6: The slope of line r is $1/4$, matching choice C.

The Trap: Choice A (-4) is the most commonly picked wrong answer — students correctly find the slope of line p but then forget the question asks for the PERPENDICULAR line's slope, mistakenly selecting line p 's own slope instead of its negative reciprocal.

POE — Eliminate the other three:

- A** — -4 is the slope of line p itself, not the slope of the perpendicular line r that the question actually asks for.
- B** — -1/4 is the negative of the reciprocal's sign but keeps the same sign as the original slope — this comes from only flipping the fraction without also flipping the sign, an incomplete negative-reciprocal calculation.
- D** — 4 comes from taking the reciprocal (flipping the fraction) but forgetting to change the sign — since the original slope is negative, the perpendicular slope must be positive, but the magnitude here (4 instead of 1/4) is also flipped incorrectly.

Solving with Desmos:

- Step 1: Open Desmos and type the original equation for line p exactly as given: $2y+8x=11$
- Step 2: Press Enter — Desmos will graph the line automatically, and you can visually confirm it's a steep, downward-sloping line.
- Step 3: Click on the line itself, or hover over it, and Desmos may show you its slope directly, or you can click the wrench/settings icon to view its equation in slope-intercept form.
- Step 4: To find and verify the perpendicular slope, type a new line using the negative reciprocal you calculated: $y=(1/4)x+0$ (the y-intercept doesn't matter for checking perpendicularity, just the slope).
- Step 5: Visually confirm on the graph that these two lines cross at a perfect right angle (90 degrees) — Desmos graphs are drawn to scale, so a true perpendicular intersection will visibly look like a plus sign (+) where the lines cross, confirming slope 1/4 is correct.

Linear equations in two variables — Q2 [Hard] [Grid-In] (ID: 107d0c62)

Question ID: 107d0c62

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

$$0.10x + 0.20y = 0.17(x + y)$$

The equation gives a volume x , in gallons, of a 10% solution that could be mixed with a volume y , in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Correct Answer: 210 (Student-Produced Response — no answer choices given)

The Rule: Mixture problems give you a general equation relating two unknown quantities (here, x and y). When the problem later tells you a SPECIFIC value for one variable, substitute that specific number in for that variable FIRST, then solve the resulting one-variable equation for the other unknown.

The Tip: Don't try to solve a mixture equation with two unknowns in general form — as soon as the problem gives you a specific number for one of the variables (here, 'mixed with 90.0 gallons of the 10% solution' tells you $x = 90$), plug that number in immediately to reduce the problem to a simple one-variable equation.

Why this is the answer: Substitute $x = 90$ into the equation $0.10x + 0.20y = 0.17(x + y)$: $0.10(90) + 0.20y = 0.17(90 + y)$. This simplifies to $9 + 0.20y = 15.3 + 0.17y$. Subtracting $0.17y$ from both sides and 9 from both sides gives $0.03y = 6.3$, so $y = 6.3 / 0.03 = 210$.

Step-by-Step Solution:

- Step 1: Start with the given equation: $0.10x + 0.20y = 0.17(x + y)$.
- Step 2: Substitute the known value $x = 90$ into the equation: $0.10(90) + 0.20y = 0.17(90 + y)$.
- Step 3: Simplify the left side: $0.10 \times 90 = 9$, so the left side becomes $9 + 0.20y$.
- Step 4: Distribute the 0.17 on the right side: $0.17 \times 90 = 15.3$, and $0.17 \times y = 0.17y$, so the right side becomes $15.3 + 0.17y$.

Step 5: Now the equation reads: $9 + 0.20y = 15.3 + 0.17y$.
 Step 6: Subtract $0.17y$ from both sides: $9 + 0.03y = 15.3$.
 Step 7: Subtract 9 from both sides: $0.03y = 6.3$.
 Step 8: Divide both sides by 0.03: $y = 6.3 / 0.03 = 210$.
 Step 9: This is a grid-in question — enter 210 as your final answer, representing 210 gallons of the 20% solution.

The Trap: A common error on this problem is forgetting to distribute the 0.17 across BOTH terms inside the parentheses $(90 + y)$ — students sometimes only multiply 0.17 by y and forget to also multiply it by 90, leading to an incorrect final answer.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the original equation, substituting $x = 90$ directly: $0.10(90)+0.20y=0.17(90+y)$
 Step 3: Press Enter — since y is now the only variable, Desmos will solve this one-variable equation automatically and plot the solution as a labeled point on the y -axis.
 Step 4: Click on the labeled point to see its exact coordinates — it will display $y = 210$, confirming your answer.
 Step 5: Enter 210 into the grid-in answer box on the actual test.

Systems of two linear equations in two variables

7 questions · Easy 1 / Medium 3 / Hard 3

Systems of two linear equations in two variables — Q1 [Easy] (ID: 0e926898)

Question ID: 0e926898

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

Students and chaperones from a middle school are going on a field trip to a museum. Admission to the museum will cost \$8.50 for each student and \$12 for each chaperone. It will cost a total of \$730 for admission to the museum for all 83 people on the field trip. How many students are going on the field trip?

Answer

A. 31

B. 70

C. 76

D. 86

Correct Answer: C

The Rule: Word problems describing two different groups (students and chaperones) with a combined total count and a combined total cost always translate into a SYSTEM of two equations: one equation for the total COUNT of people, and one equation for the total COST. Solve the system using substitution or elimination.

The Tip: When setting up a system from a word problem, write out both equations in words first before converting to symbols: 'number of students + number of chaperones = total people' and 'cost per student × students + cost per chaperone × chaperones = total cost.' This helps prevent mixing up which number goes where.

Why C is right: Let s = number of students and c = number of chaperones. The two equations are: $s + c = 83$ (total people) and $8.50s + 12c = 730$ (total cost). Solving this system by substitution: from the first equation, $c = 83 - s$. Substitute into the second equation: $8.50s + 12(83 - s) = 730$, which simplifies to $8.50s + 996 - 12s = 730$, then $-3.5s = -266$, so $s = 76$.

Step-by-Step Solution:

Step 1: Define variables: let s = number of students, c = number of chaperones.

Step 2: Write the total-people equation: $s + c = 83$.

Step 3: Write the total-cost equation: $8.50s + 12c = 730$.

Step 4: Solve the first equation for c : $c = 83 - s$.

Step 5: Substitute this expression for c into the second equation: $8.50s + 12(83 - s) = 730$.

Step 6: Distribute the 12: $8.50s + 996 - 12s = 730$.

Step 7: Combine the s -terms: $8.50s - 12s = -3.5s$, so the equation becomes $-3.5s + 996 = 730$.

Step 8: Subtract 996 from both sides: $-3.5s = -266$.

Step 9: Divide both sides by -3.5 : $s = 76$.

Step 10: This matches choice C — 76 students are going on the field trip.

The Trap: Choice D (86) is the most commonly picked wrong answer — it often results from a sign error while distributing the 12 across $(83 - s)$, such as writing $12(83 - s)$ as $996 + 12s$ instead of the correct $996 - 12s$, which flips the direction of the final subtraction.

POE — Eliminate the other three:

X A — 31 would be the value of c (the number of chaperones), not s (the number of students) — this is likely selected by students who solve the system correctly but grab the wrong variable's value as their final answer.

X B — 70 doesn't correspond to a clean derivation from this system — likely an arithmetic slip somewhere in the distribution or combination step.

X D — 86 typically results from a sign error while distributing 12 across $(83 - s)$, flipping a subtraction into an addition partway through the solution.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the first equation exactly as derived: $s+c=83$ — but since Desmos uses x and y by default for graphing two-variable equations, it's easiest to substitute x for s and y for c : $x+y=83$

Step 3: Press Enter — Desmos will graph this as a line.

Step 4: Click on the next empty row and type the second equation, again using x for students and y for chaperones: $8.50x+12y=730$

Step 5: Press Enter — Desmos will graph this second line as well.

Step 6: Look for the point where the two lines cross (intersect) — this is the solution to the system. Click directly on that intersection point, and Desmos will display its exact coordinates.

Step 7: The x -coordinate of that intersection point is the number of students (76), and the y -coordinate is the number of chaperones (7) — confirming $s = 76$ matches choice C.

Question ID: ea8bd66f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 4x$$

$$5x - 3y = -14$$

The solution to the given system of equations is (x, y) . What is the value of $x + y$?

Answer

- A. 4
- B. 6
- C. 8
- D. 10

Correct Answer: D

The Rule: When one equation in a system is already solved for one variable (like $y = 4x$), the fastest solving method is **SUBSTITUTION** — plug that expression directly into the other equation in place of that variable, turning the system into a single one-variable equation.

The Tip: Whenever you see a system where one equation looks like ' $y = (\text{something with } x)$ ' or ' $x = (\text{something with } y)$,' that's your signal to use substitution rather than elimination — it's almost always faster in that specific setup.

Why D is right: Since $y = 4x$, substitute $4x$ in place of y in the second equation: $5x - 3(4x) = -14$. This simplifies to $5x - 12x = -14$, so $-7x = -14$, giving $x = 2$. Then $y = 4x = 4(2) = 8$. Finally, $x + y = 2 + 8 = 10$.

Step-by-Step Solution:

- Step 1: Note that the first equation is already solved for y : $y = 4x$.
- Step 2: Substitute $4x$ in place of y into the second equation: $5x - 3(4x) = -14$.
- Step 3: Distribute the -3 : $5x - 12x = -14$.
- Step 4: Combine like terms: $-7x = -14$.
- Step 5: Divide both sides by -7 : $x = 2$.
- Step 6: Substitute $x = 2$ back into $y = 4x$ to find y : $y = 4(2) = 8$.
- Step 7: The question asks for $x + y$, so calculate: $2 + 8 = 10$.
- Step 8: This matches choice D.

The Trap: Choice A (4) is the most commonly picked wrong answer — students sometimes solve correctly for $x = 2$ but then mistakenly report a value related only to the coefficient in the first equation, or make an early substitution error that happens to produce 4 as a partial (incorrect) result.

POE — Eliminate the other three:

- X A** — 4 doesn't match either the correct x -value (2), y -value (8), or their sum (10) — likely a partial calculation error along the way.
- X B** — 6 doesn't correspond to a clean derivation from this system either.
- X C** — 8 is the correct value of y ALONE, not $x + y$ — a student who correctly finds $y = 8$ but forgets to add $x = 2$ would mistakenly select this.

Solving with Desmos:

- Step 1: Open Desmos. Step 2: Type the first equation exactly as given: $y=4x$
- Step 3: Press Enter — Desmos graphs this line immediately.
- Step 4: Type the second equation on the next line: $5x-3y=-14$
- Step 5: Press Enter — Desmos graphs this second line.

Step 6: Click directly on the point where the two lines intersect — Desmos will display the exact coordinates of that point, which will show (2, 8), meaning $x = 2$ and $y = 8$.

Step 7: Since the question asks for $x + y$, type a third line: $2+8$ (or click and read both coordinates and add them yourself) — this gives 10, confirming choice D.

Systems of two linear equations in two variables — Q3 [Medium] (ID: 340afb12)

Question ID: 340afb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

Ellen bought two types of snack bags for a school event. Each box of chips Ellen bought contained 24 bags and cost \$11. Each box of crackers Ellen bought contained 16 bags and cost \$9. It cost Ellen \$497 before tax to buy 968 bags. If x is the number of boxes of chips and y is the number of boxes of crackers that Ellen bought, which of the following systems of equations represents this situation?

Answer

- A. $11x + 16y = 968$
 $24x + 9y = 497$
 B. $11x + 9y = 968$
 $24x + 16y = 497$
 C. $11x + 9y = 497$
 $24x + 16y = 968$
 D. $9x + 11y = 497$
 $16x + 24y = 968$

Correct Answer: C

The Rule: When translating a word problem into a system of equations, carefully match EACH number to what it's actually describing — cost-per-box goes with a cost total, and bags-per-box goes with a bags total. Mixing these up (using cost numbers in the bags equation, or vice versa) is the most common error in this question type.

The Tip: Make a small table before writing your equations: list each item (chips, crackers), its cost per box, its bags per box, and how many boxes were bought. This visual organization prevents the classic mistake of swapping which number belongs to which equation.

Why C is right: Chips cost \$11 per box and contain 24 bags per box; crackers cost \$9 per box and contain 16 bags per box. The COST equation is $11x + 9y = 497$ (matching the total cost of \$497). The BAGS equation is $24x + 16y = 968$ (matching the total of 968 bags). This matches choice C exactly.

Step-by-Step Solution:

Step 1: Identify what x and y represent: x = number of boxes of chips, y = number of boxes of crackers.

Step 2: Build the COST equation using the price per box: chips cost \$11 per box, crackers cost \$9 per box, and the total cost was \$497. This gives: $11x + 9y = 497$.

Step 3: Build the BAGS equation using the bags per box: chips have 24 bags per box, crackers have 16 bags per box, and the total was 968 bags. This gives: $24x + 16y = 968$.

Step 4: Compare these two equations to the answer choices — choice C lists exactly $11x + 9y = 497$ and $24x + 16y = 968$, matching perfectly.

The Trap: Choice B is the most commonly picked wrong answer — it swaps the numbers between the two equations, incorrectly pairing the bags-per-box numbers (24 and 16) with the cost total (968) and the cost-per-box numbers (11 and 9) with... actually it uses $11x+9y=968$ and $24x+16y=497$, which mismatches which total belongs to which set of per-box numbers.

POE — Eliminate the other three:

- ✗ A** — This equation set pairs the wrong price (\$9) with chips and the wrong price (\$24, which is actually the bag count) with crackers in the cost equation — the numbers are scrambled between the two equations.
- ✗ B** — This equation set swaps which TOTAL (968 bags vs. \$497 cost) is paired with which set of per-box numbers — the cost equation incorrectly uses 968 and the bags equation incorrectly uses 497.
- ✗ D** — This equation set swaps the roles of x and y entirely (putting crackers' numbers where chips' numbers should be, and vice versa) while also mismatching the totals — multiple errors combined.

Solving with Desmos:

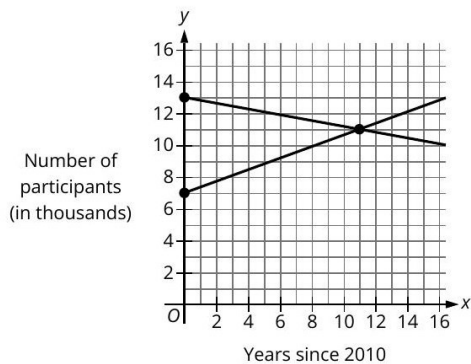
- Step 1: Open Desmos. Step 2: Type the cost equation you derived: $11x+9y=497$
- Step 3: Press Enter — Desmos graphs this line.
- Step 4: Type the bags equation on the next line: $24x+16y=968$
- Step 5: Press Enter — Desmos graphs this second line.
- Step 6: Click on the intersection point of the two lines — Desmos will display the exact (x, y) coordinates, which should both be positive whole numbers (since you can't buy a fractional box) — this confirms your two equations correctly model the real-world situation, verifying choice C represents the system correctly, since it produces a sensible, whole-number solution.

Systems of two linear equations in two variables — Q4 [Medium] (ID: 080e75c2)

Question ID: 080e75c2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question



The functions f and g model the number of participants, in thousands, in two different programs x years since 2010. The graphs of $y = f(x)$ and $y = g(x)$ are shown. Which of the following could represent functions f and g ?

Answer

- A. $f(x) = -\frac{4}{11}x + 13$
 $g(x) = \frac{2}{11}x + 7$
- B. $f(x) = -\frac{4}{11}x + 7$
 $g(x) = \frac{2}{11}x + 13$
- C. $f(x) = -\frac{2}{11}x + 13$
 $g(x) = \frac{4}{11}x + 7$
- D. $f(x) = -\frac{2}{11}x + 7$
 $g(x) = \frac{4}{11}x + 13$

Correct Answer: C

The Rule: To find the equation of a line from its graph, identify two clear points on the line, calculate the slope using $(\text{change in } y) \div (\text{change in } x)$, then use the y-intercept (where the line crosses the y-axis) to write the full equation in slope-intercept form: $y = mx + b$.

The Tip: When a graph shows two lines and gives you their y-intercepts as clearly marked dots on the y-axis, start there — the y-intercept immediately tells you the 'b' value, narrowing down your answer choices before you even calculate the slope.

Why C is right: The decreasing line (f) starts at y-intercept 13 and passes through the intersection point near (11, 11); its slope is $(11 - 13)/(11 - 0) = -2/11$, giving $f(x) = -2/11x + 13$. The increasing line (g) starts at y-intercept 7 and passes through the same intersection point; its slope is $(11 - 7)/(11 - 0) = 4/11$, giving $g(x) = 4/11x + 7$. This matches choice C exactly.

Step-by-Step Solution:

Step 1: Identify the y-intercepts directly from the graph: the decreasing line crosses the y-axis at 13, and the increasing line crosses the y-axis at 7.

Step 2: Identify a second clear point on each line — both lines appear to cross paths at approximately (11, 11).

Step 3: Calculate the slope of the decreasing line using the two points (0, 13) and (11, 11): slope = $(11 - 13) / (11 - 0) = -2/11$.

Step 4: Calculate the slope of the increasing line using the two points (0, 7) and (11, 11): slope = $(11 - 7) / (11 - 0) = 4/11$.

Step 5: Write each equation in slope-intercept form: $f(x) = -2/11x + 13$ (the decreasing line) and $g(x) = 4/11x + 7$ (the increasing line).

Step 6: Compare to the answer choices — choice C lists $f(x) = -2/11x + 13$ and $g(x) = 4/11x + 7$, matching exactly.

The Trap: Choice D is the most commonly picked wrong answer — it swaps the y-intercepts between f and g (using 7 for f and 13 for g), which happens when a student correctly calculates both slopes but then mismatches which slope-intercept pair belongs to the increasing vs. decreasing line.

POE — Eliminate the other three:

X A — This uses the correct y-intercepts (13 for f, 7 for g) but the wrong slope fractions ($-4/11$ and $2/11$ instead of $-2/11$ and $4/11$) — the numerator and denominator relationship is reversed for both slopes.

X B — This uses the wrong y-intercepts (7 for f instead of 13, and 13 for g instead of 7) while also using the wrong slope magnitudes — multiple errors combined.

X D — This uses the correct slope magnitudes ($-2/11$ and $4/11$) but swaps which y-intercept (7 or 13) goes with which line, incorrectly pairing the decreasing line with intercept 7 and the increasing line with intercept 13.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Since you're estimating a line from a picture, first type each answer choice's pair of equations as a test — start with choice C: $y = -2/11x + 13$ and on the next line $y = 4/11x + 7$

Step 3: Press Enter after each — Desmos will graph both lines.

Step 4: Compare Desmos's graph to the original picture in the question: check that one line starts at (0, 13) and slopes downward, the other starts at (0, 7) and slopes upward, and that they cross near $x = 11$.

Step 5: If everything matches the original graph's shape and intercepts, you've confirmed choice C is correct. If you're unsure between two choices, graph both candidate pairs in different colors (Desmos automatically assigns different colors to each line) and see which pair visually matches the original picture's intercepts and steepness most closely.

Question ID: 59813abf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard ■■■■■

Question

$$38(x - n) = 38y + 38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

Answer

- A. $4x - 4y = 8n$
- B. $4x + 4y = 4n$
- C. $4x + 4y = 8n$
- D. $4x - 4y = 4n$

Correct Answer: D

The Rule: A system of two linear equations has NO solution only when the two lines are PARALLEL but not identical — meaning they have the exact same slope (same ratio of x-coefficient to y-coefficient) but different y-intercepts (so they never touch).

The Tip: To quickly check whether two equations represent parallel, non-identical lines, first simplify both equations into the same basic form, then compare the ratio of their x and y coefficients. If those ratios match but the constant term doesn't scale the same way, you have a 'no solution' pair.

Why D is right: Simplify the given equation $38(x - n) = 38y + 38n$ by dividing every term by 38: $x - n = y + n$, which rearranges to $x - y = 2n$. For the system to have NO solution, the second equation must have the same left side pattern (same x and y coefficients) but a DIFFERENT constant relationship. Choice D, $4x - 4y = 4n$, simplifies (dividing by 4) to $x - y = n$. Since n is a positive constant, $n \neq 2n$ (they can only be equal if $n = 0$, but n is specified to be positive), so these two equations represent parallel lines with different intercepts — confirming no solution.

Step-by-Step Solution:

Step 1: Simplify the given equation: $38(x - n) = 38y + 38n$. Divide every term by 38: $x - n = y + n$.

Step 2: Rearrange into standard form: $x - y = n + n = 2n$.

Step 3: For a system to have NO solution, the second equation needs the exact same coefficient ratio for x and y (so the lines are parallel) but a DIFFERENT constant value relative to that same scaling (so the lines never actually meet).

Step 4: Test each answer choice by simplifying it to the same ' $x - y = \dots$ ' form. Choice D: $4x - 4y = 4n$. Divide by 4: $x - y = n$.

Step 5: Compare: the original equation says $x - y = 2n$, and choice D says $x - y = n$. Since n is a positive constant, $2n$ and n are always different values — confirming these are parallel, non-identical lines with NO solution.

Step 6: Double-check the other choices don't also produce this pattern (see POE below) — only choice D fits.

The Trap: Choice A ($4x - 4y = 8n$) is the most commonly picked wrong answer — dividing this by 4 gives $x - y = 2n$, which is IDENTICAL to the original simplified equation, meaning these two equations actually represent the exact SAME line, producing infinitely many solutions, not no solution.

POE — Eliminate the other three:

X A — Simplifying $4x - 4y = 8n$ by dividing by 4 gives $x - y = 2n$ — this is the exact same equation as the original ($x - y = 2n$), meaning these represent the SAME line, giving infinitely many solutions, not no solution.

X B — $4x + 4y = 4n$ has a different coefficient pattern (+4y instead of -4y), meaning this line has a different (positive) slope than the original — different slopes mean the lines will cross at exactly one point, not zero.

X C — $4x + 4y = 8n$ also has a +4y term, giving it a different slope than the original equation — again, different slopes mean exactly one solution, not no solution.

Solving with Desmos:

Step 1: Since this problem involves the unknown constant n , first pick a simple test value for n to make the equations concrete — let's use $n = 1$ (any positive number works the same way). Step 2: Open Desmos and type the original equation with $n = 1$ substituted in: $38(x-1)=38y+38$

Step 3: Press Enter — Desmos graphs this line.

Step 4: Now type each answer choice with $n = 1$ substituted in, one at a time, starting with choice D: $4x-4y=4$

Step 5: Press Enter — look at the graph. If the new line is perfectly parallel to (never touches) the first line, that choice gives NO solution, confirming it's correct.

Step 6: For comparison, also try choice A with $n=1$: $4x-4y=8$ — you'll see this line lands EXACTLY on top of the original line (they're the same line), showing infinitely many solutions, which confirms choice A is wrong and helps you see the difference visually.

Systems of two linear equations in two variables — Q6 [Hard] (ID: 5e9b6079)

Question ID: 5e9b6079

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 7x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

Answer

- A. $-21x + y = 54$
- B. $-21x + y = 18$
- C. $-7x + y = 21$
- D. $-7x + y = 18$

Correct Answer: C

The Rule: For a system to have NO solution, the second equation's line must be PARALLEL to the first (same slope) but have a DIFFERENT y -intercept (so the lines run side by side forever without crossing).

The Tip: Rewrite the given equation into the standard form '(number) $x + y =$ (number)' to easily compare its coefficients to each answer choice — matching coefficients with a different constant means no solution; matching coefficients AND constant means infinite solutions; different coefficients mean exactly one solution.

Why C is right: The given equation $y = 7x + 18$ can be rewritten as $-7x + y = 18$. For no solution, we need an equation with the same $-7x + y$ pattern but a DIFFERENT constant than 18. Choice C, $-7x + y = 21$, has exactly this same coefficient pattern (-7 and 1) but a different constant (21 instead of 18), confirming these lines are parallel and never intersect.

Step-by-Step Solution:

Step 1: Rewrite the given equation $y = 7x + 18$ into standard form by subtracting $7x$ from both sides: $-7x + y = 18$.

Step 2: For no solution, look for an answer choice with the identical coefficients on x and y (-7 and 1) but a constant term different from 18 .

Step 3: Check choice C: $-7x + y = 21$. The coefficients (-7 and 1) match exactly, and the constant (21) is different from 18 — this is the parallel-line pattern needed for no solution.

Step 4: Confirm this is correct and check the other choices don't also fit this pattern (see POE below).

The Trap: Choice D ($-7x + y = 18$) is the most commonly picked wrong answer — this has the exact same coefficients AND the exact same constant as the original equation, meaning it's not a second, different equation at all — it's the SAME line, giving infinitely many solutions instead of no solution.

POE — Eliminate the other three:

- A** — $-21x + y = 54$ has a different x-coefficient (-21 instead of -7) than the original equation, meaning this line has a different, steeper slope — different slopes always produce exactly one solution, not zero.
- B** — $-21x + y = 18$ also has a different x-coefficient (-21 instead of -7), giving it a different slope than the original line — this means exactly one solution, not no solution.
- D** — $-7x + y = 18$ has identical coefficients AND an identical constant to the original equation — this isn't a different line at all, it's the exact same line, producing infinitely many solutions rather than none.

Solving with Desmos:

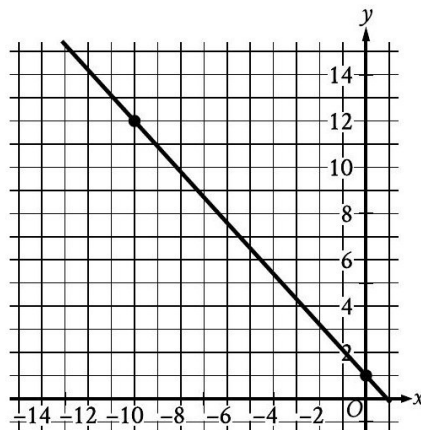
- Step 1: Open Desmos and type the original equation: $y=7x+18$
- Step 2: Press Enter — Desmos graphs this line.
- Step 3: Type choice C on the next line, rearranged to match: $y=7x+21$ (this is the same as $-7x+y=21$, just solved for y to make graphing easier)
- Step 4: Press Enter — look at the graph. You should see two lines that are perfectly parallel (same steepness, same direction) but shifted apart vertically, never touching no matter how far you zoom out — this visually confirms no solution.
- Step 5: To double check, try graphing choice D ($y=7x+18$) on a third line — you'll see it lands EXACTLY on top of the original line, since it's actually the identical equation, confirming why D gives infinitely many solutions instead of zero.

Systems of two linear equations in two variables — Q7 [Hard] [Grid-In] (ID: 18ac8354)

Question ID: 18ac8354

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



The graph of line g is shown in the xy -plane. Line k is defined by $165x + py = w$, where p and w are constants. If line k is graphed in this xy -plane, resulting in the graph of a system of two linear equations, the system of two linear equations will have infinitely many solutions. What is the value of $p + w$?

Correct Answer: 495 (Student-Produced Response — no answer choices given)

The Rule: For a system of two equations to have INFINITELY MANY solutions, the two equations must represent the EXACT SAME line — meaning every coefficient and constant in one equation is the same multiple of the corresponding value in the other equation.

The Tip: When you're given a graph and told a second equation (with unknown constants) must produce infinitely many solutions, first find the equation of the graphed line yourself in simple whole-number form, then figure out what single 'scale factor' turns your simple equation into the given equation's exact form.

Why this is the answer: From the graph, line g passes through $(0, 2)$ and $(-10, 12)$, giving a slope of $(12 - 2)/(-10 - 0) = 10/(-10) = -1$, so the line's equation is $y = -x + 2$, or equivalently $x + y = 2$. To make this match $165x + py = w$ exactly (for infinitely many solutions), multiply the entire equation $x + y = 2$ by 165: $165x + 165y = 330$. So $p = 165$ and $w = 330$, giving $p + w = 165 + 330 = 495$.

Step-by-Step Solution:

Step 1: Read two clear points off the graph of line g : it appears to pass through $(0, 2)$ — the y -intercept — and $(-10, 12)$.

Step 2: Calculate the slope using these two points: $\text{slope} = (12 - 2) / (-10 - 0) = 10 / (-10) = -1$.

Step 3: Write the equation of line g in slope-intercept form using the y -intercept (2) and slope (-1) : $y = -x + 2$.

Step 4: Rearrange this into standard form to match the format of line k : $x + y = 2$.

Step 5: For line k ($165x + py = w$) to represent the exact same line as g ($x + y = 2$) — which is required for infinitely many solutions — every term must be scaled by the same factor. Since the x -coefficient goes from 1 to 165, the scale factor is 165.

Step 6: Apply this scale factor to the other terms: the y -coefficient becomes $1 \times 165 = 165$, so $p = 165$. The constant becomes $2 \times 165 = 330$, so $w = 330$.

Step 7: Calculate $p + w = 165 + 330 = 495$.

Step 8: This is a grid-in question — enter 495 as your final answer.

The Trap: A common error on this problem is misreading the graph's slope or y -intercept (for example, reading the y -intercept as 3 instead of 2, or misjudging the second point), which throws off the entire scale-factor calculation and produces a wrong final sum.

Solving with Desmos:

Step 1: Open Desmos. Step 2: To verify your reading of the graph, type $y = -x + 2$ into Desmos and compare the resulting line to the picture given in the question — check that it passes through $(0, 2)$ and $(-10, 12)$ exactly as shown.

Step 3: Once confirmed, type the scaled version: $165x + 165y = 330$ on a second line.

Step 4: Press Enter — Desmos should show this second line landing EXACTLY on top of the first line (they're the same line, just written differently), confirming that $p = 165$ and $w = 330$ make line k identical to line g .

Step 5: Calculate $p + w = 165 + 330 = 495$ and enter this into the grid-in box.

Linear functions

6 questions · Easy 1 / Medium 2 / Hard 3

Linear functions — Q1 [Easy] (ID: 0c490cd5)

Question ID: 0c490cd5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Sam has a flag with an area of 110 square inches (in^2). Sam plans to enlarge the flag by sewing on pieces of colored nylon, where each piece would increase the flag's area by 64 in^2 . Which equation gives the flag's total area y , in square inches, after Sam sews on x pieces of nylon?

Answer

- A. $y = 64x$
- B. $y = 64 + 110x$
- C. $y = 110 + 64x$
- D. $y = 174x$

Correct Answer: C

The Rule: In a real-world 'starting amount plus additional amount per unit' scenario, the equation always takes the form: (total) = (starting amount) + (rate per unit)(number of units) — the starting amount stays fixed, and the rate is multiplied by however many units are added.

The Tip: Identify the 'starting point' (the value when $x = 0$) and the 'rate of change' (how much y increases for each additional unit of x) separately before writing the equation — the starting point is always the constant added alone, and the rate is always the number multiplied by x .

Why C is right: The flag starts with an area of 110 square inches (before any nylon pieces are added), so 110 is the fixed starting amount. Each piece of nylon adds 64 square inches, so 64 is the rate multiplied by x , the number of pieces added. This gives the equation $y = 110 + 64x$, matching choice C.

Step-by-Step Solution:

- Step 1: Identify the fixed starting value: the flag begins with an area of 110 square inches before Sam adds any nylon pieces.
- Step 2: Identify the rate of change: each nylon piece adds 64 square inches to the area.
- Step 3: Identify the variable: x represents the number of nylon pieces added.
- Step 4: Combine these into the equation: total area = starting area + (rate)(number of pieces), which gives $y = 110 + 64x$.
- Step 5: This matches choice C.

The Trap: Choice B ($y = 64 + 110x$) is the most commonly picked wrong answer — it swaps which number is the starting amount and which is the rate, incorrectly treating 64 as the fixed starting area and 110 as the per-piece rate, when it's actually the reverse.

POE — Eliminate the other three:

- X A** — $y = 64x$ completely omits the starting area of 110 square inches — this equation would incorrectly suggest the flag starts with zero area before any nylon is added.
- X B** — This swaps the roles of 110 and 64 — it uses 64 as the fixed starting amount and 110 as the rate per piece, which is backwards from what the problem describes.
- X D** — $y = 174x$ incorrectly combines 110 and 64 into a single rate ($174 = 110 + 64$) and multiplies the entire sum by x , which doesn't correctly represent a fixed starting amount plus a separate per-unit rate.

Solving with Desmos:

- Step 1: Open Desmos and type the equation you derived: $y=110+64x$
- Step 2: Press Enter — Desmos will graph this line.
- Step 3: Check the y-intercept by looking at where the line crosses the y-axis (or use the table feature to see the value when $x = 0$) — it should show $y = 110$, confirming the flag's starting area before any nylon is added.

Step 4: Use the table feature (click the '+' icon and select table, or add a table below the equation) and enter $x = 1$ — you should see $y = 174$, confirming that adding one piece of nylon increases the area by exactly 64 square inches ($174 - 110 = 64$), which matches the problem's description and confirms choice C.

Linear functions — Q2 [Medium] (ID: fc4a0d2a)

Question ID: fc4a0d2a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$g(x) = 3(14x - 15)$$

What is the y -coordinate of the y -intercept of the graph of $y = g(x) - 2$ in the xy -plane?

Answer

- A. -47
- B. -45
- C. -17
- D. -15

Correct Answer: A

The Rule: The y -intercept of any function is the value of the function when $x = 0$. If a new function is defined as the original function shifted by a constant (like $y = g(x) - 2$), you find its y -intercept by first finding $g(0)$, and then applying that same shift.

The Tip: Whenever you're asked about a graph like ' $y = g(x) - 2$ ' or ' $y = g(x) + 5$,' don't try to re-derive a whole new equation — just find the value of the ORIGINAL function at the point you care about (here, $x = 0$), and then simply add or subtract the shift amount at the very end.

Why A is right: First expand $g(x) = 3(14x - 15) = 42x - 45$. The y -intercept of g itself is $g(0) = 42(0) - 45 = -45$. Since we want the y -intercept of $y = g(x) - 2$, subtract 2 from $g(0)$: $-45 - 2 = -47$.

Step-by-Step Solution:

- Step 1: Expand the given function: $g(x) = 3(14x - 15) = 3(14x) - 3(15) = 42x - 45$.
- Step 2: Find the y -intercept of $g(x)$ itself by evaluating $g(0)$: $g(0) = 42(0) - 45 = -45$.
- Step 3: The question asks about the NEW function $y = g(x) - 2$, which shifts every output of g down by 2.
- Step 4: Apply this shift to the y -intercept you just found: $-45 - 2 = -47$.
- Step 5: This matches choice A.

The Trap: Choice C (-17) is the most commonly picked wrong answer — it typically results from a distribution error while expanding $3(14x - 15)$, such as computing 3×15 incorrectly or forgetting to distribute the 3 across BOTH terms inside the parentheses.

POE — Eliminate the other three:

- X B** — -45 is the y -intercept of the ORIGINAL function $g(x)$ alone — this is the value BEFORE subtracting 2 for the shifted function $y = g(x) - 2$. A student who forgets to apply the final shift would stop here.
- X C** — -17 typically comes from a distribution error, such as computing $3(14x - 15)$ incorrectly and only partially multiplying through the parentheses.
- X D** — -15 is simply the constant that appears inside the original parentheses ($14x - 15$) — selecting this means mistaking a piece of the unexpanded expression for the final y -intercept, without doing any of the actual distribution or shifting work.

Solving with Desmos:

Step 1: Open Desmos and type the original function exactly as given: $g(x)=3(14x-15)$

Step 2: Press Enter — Desmos will graph this line and also let you use 'g(x)' in later expressions.

Step 3: On the next line, type the new shifted function: $y=g(x)-2$

Step 4: Press Enter — Desmos graphs this new line.

Step 5: To find its y-intercept, either look at where this new line crosses the y-axis on the graph, or click the wrench/settings icon and use the table feature: create a table, enter $x = 0$, and read the resulting y-value — it will show -47 , confirming choice A.

Linear functions — Q3 [Medium] (ID: 2493c767)

Question ID: 2493c767

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$s(x) = 133 - 4x$$

The function s shown gives the number of spoons, $s(x)$, remaining in a dispenser x minutes after the dispenser had been loaded with spoons. Which statement is the best interpretation of $s(3) = 121$?

Answer

- A. The number of spoons remaining in the dispenser decreased by a total of 3 spoons after 121 minutes.
- B. There were 3 spoons remaining in the dispenser 121 minutes after the dispenser had been loaded with spoons.
- C. There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons.
- D. The number of spoons remaining in the dispenser decreased by a total of 121 spoons after 3 minutes.

Correct Answer: C

The Rule: When a function $s(x)$ models a real-world quantity over time, evaluating $s(\text{some number}) = \text{some result}$ means: 'INPUT this number of [units] gives OUTPUT this amount of [quantity].' Always identify what the input variable represents and what the output represents before matching to an answer choice.

The Tip: Underline what x represents (here, minutes) and what $s(x)$ represents (here, spoons remaining) directly in the problem before reading the answer choices — many wrong choices deliberately swap these two roles to test whether you're reading carefully.

Why C is right: In the function $s(x) = 133 - 4x$, x represents minutes after loading, and $s(x)$ represents the number of spoons remaining. So $s(3) = 121$ means: when $x = 3$ (3 minutes have passed), $s(x) = 121$ (121 spoons remain in the dispenser).

Step-by-Step Solution:

- Step 1: Identify what the input variable x represents in this context: minutes after the dispenser was loaded with spoons.
- Step 2: Identify what the output $s(x)$ represents: the number of spoons remaining in the dispenser at that time.
- Step 3: Interpret $s(3) = 121$ by substituting these roles: the INPUT is 3 (meaning 3 minutes have passed), and the OUTPUT is 121 (meaning 121 spoons remain at that time).
- Step 4: Match this interpretation to the answer choices: 'There were 121 spoons remaining in the dispenser 3 minutes after the dispenser had been loaded with spoons' is choice C.

The Trap: Choice B is the most commonly picked wrong answer — it swaps the input and output values, incorrectly stating there were '3 spoons remaining... 121 minutes' after loading, when it's actually the reverse: 121 spoons remaining after 3 minutes.

POE — Eliminate the other three:

✗ A — This choice invents a 'decrease' framing that doesn't match how function notation works — $s(3) = 121$ isn't describing a decrease of 3 spoons over 121 minutes; it's a direct input-output pair.

✗ B — This swaps which number is the input (minutes) and which is the output (spoons remaining) — it should be 121 spoons remaining after 3 minutes, not 3 spoons remaining after 121 minutes.

✗ D — This also swaps the input and output values (using 121 as if it were the time elapsed and 3 as if it were the spoon count), plus incorrectly frames it as a 'decrease' rather than a direct count remaining.

Solving with Desmos:

Step 1: Open Desmos and type the function exactly as given: $s(x)=133-4x$

Step 2: Press Enter — Desmos will graph this line and remember the function $s(x)$ for later use.

Step 3: On the next line, type: $s(3)$ directly — Desmos will instantly calculate and display the output value, showing 121.

Step 4: This confirms that when $x = 3$ (3 minutes), $s(x) = 121$ (121 spoons remain), matching the direct input-output interpretation in choice C. You can also click on the graphed line at $x = 3$ to visually see the point (3, 121) marked on the graph.

Linear functions — Q4 [Hard] (ID: d4572f55)

Question ID: d4572f55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-34	t
-17	$t + 23$
0	$t + 46$

For a linear relationship between x and y , the table gives three values of x and their corresponding values of y , where t is a constant. Which equation represents this relationship?

Answer

A. $y = -2x + t + 23$

B. $y = 2x + t + 23$

C. $y = -\frac{23}{17}x + t + 46$

D. $y = \frac{23}{17}x + t + 46$

Correct Answer: D

The Rule: When a table gives you values that include an unknown constant (like t), find the slope using the differences between rows that DON'T involve solving for t directly — subtracting two rows that both contain ' t ' will cancel it out, leaving you with a clean numeric slope calculation.

The Tip: Look for two rows in the table where subtracting their y -values makes the unknown constant ' t ' cancel out completely — this lets you calculate the slope using only clean numbers, without ever needing to know what t actually equals.

Why D is right: Using the rows $x = -17$ ($y = t + 23$) and $x = 0$ ($y = t + 46$): the change in y is $(t + 46) - (t + 23) = 23$ (the t 's cancel out), and the change in x is $0 - (-17) = 17$. So the slope is $23/17$. Using the y -intercept from the $x = 0$ row ($y = t + 46$), the equation is $y = (23/17)x + t + 46$, matching choice D.

Step-by-Step Solution:

Step 1: Choose two rows from the table where subtracting the y -values will cancel out the unknown t . Use $x = -17$ ($y = t + 23$) and $x = 0$ ($y = t + 46$).

Step 2: Calculate the change in y : $(t + 46) - (t + 23) = t + 46 - t - 23 = 23$ (the t cancels out completely).

Step 3: Calculate the change in x : $0 - (-17) = 17$.

Step 4: Calculate the slope: $(\text{change in } y) / (\text{change in } x) = 23/17$.

Step 5: Since $x = 0$ gives $y = t + 46$ directly from the table, this row already tells you the y -intercept of the line: $t + 46$.

Step 6: Write the full equation in slope-intercept form: $y = (23/17)x + t + 46$.

Step 7: This matches choice D. As a check, verify with the third row ($x = -34$, $y = t$): plug $x = -34$ into the equation: $y = (23/17)(-34) + t + 46 = 23(-2) + t + 46 = -46 + t + 46 = t$. This matches the table exactly, confirming the equation is correct.

The Trap: Choice C is the most commonly picked wrong answer — it has the correct constant term ($t + 46$) but an incorrect slope with the fraction flipped and the wrong sign ($-23/17$ instead of the correct $23/17$), likely from a sign error while calculating the change in y or change in x .

POE — Eliminate the other three:

X A — $y = -2x + t + 23$ uses a whole-number slope (-2) instead of the correct fraction ($23/17$), and pairs it with the wrong constant term ($t + 23$ instead of $t + 46$) — likely from using the wrong pair of rows or miscalculating entirely.

X B — $y = 2x + t + 23$ has the same issues as choice A but with a positive whole-number slope instead of negative — still doesn't match the correct fractional slope of $23/17$.

X C — $y = -23/17 x + t + 46$ has the correct constant term but the wrong sign on the slope — the line should be increasing (positive slope), not decreasing, based on the table values.

Solving with Desmos:

Step 1: Since this table involves the unknown constant t , first pick a simple test value for t to make things concrete — let's use $t = 0$ for simplicity. Step 2: Open Desmos and create a table: click the '+' icon, select the table option, and enter the x -values from the problem ($-34, -17, 0$) in the x -column, and the corresponding y -values WITH $t = 0$ substituted ($0, 23, 46$) in the y -column.

Step 3: Desmos will plot these three points. Click the line-fitting or regression feature if available, or simply type $y=mx+b$ below the table and adjust m and b using sliders until the line passes through all three points.

Step 4: You should find the line that fits is $y = (23/17)x + 46$ (when $t = 0$, this matches $y = 23/17x + t + 46$ exactly, since $t + 46 = 0 + 46 = 46$).

Step 5: This confirms the slope is $23/17$ and the constant term follows the pattern $t + 46$, matching choice D.

Linear functions — Q5 [Hard] (ID: 1a2a0f59)

Question ID: 1a2a0f59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is \$25.40, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

Answer

A. $f(x) = 25.40(x - 1) + 12.70$

B. $f(x) = 25.40(x - 2) + 12.70$

C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$

D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

Correct Answer: B

The Rule: When a real-world pricing deal has DIFFERENT rules for the first few units and a STANDARD rule for all remaining units, build the equation by keeping the special-case amounts as separate fixed constants, and only multiply the standard rate by the units that follow the standard rule (not the total number of units).

The Tip: Carefully identify exactly which sessions get the SPECIAL treatment (free or discounted) and keep those as separate fixed dollar amounts — then multiply the regular price only by the number of REMAINING sessions (x minus however many special sessions there were), not by x itself.

Why B is right: The first session is free (\$0) and the second session is half off ($\$25.40 / 2 = \12.70) — these two special sessions contribute a fixed \$12.70 total ($0 + 12.70$), regardless of x . Every session from the 3rd session onward costs the full regular price of \$25.40, and there are $(x - 2)$ such sessions when $x \geq 2$. This gives the total cost function $f(x) = 25.40(x - 2) + 12.70$, matching choice B.

Step-by-Step Solution:

Step 1: Identify the special pricing for the first two sessions: session 1 is free (\$0), and session 2 is half price ($\$25.40 \div 2 = \12.70).

Step 2: Add these two special amounts together as a fixed constant: $0 + 12.70 = 12.70$. This fixed amount doesn't change no matter how many total sessions (x) are purchased.

Step 3: Determine how many sessions are charged at the FULL regular price: since sessions 1 and 2 are special, all sessions from the 3rd onward are regular price — that's $(x - 2)$ sessions when x is the total number of sessions and $x \geq 2$.

Step 4: Build the full cost function: (fixed special-session cost) + (regular price) $\times$ (number of regular-price sessions) = $12.70 + 25.40(x - 2)$.

Step 5: This matches choice B: $f(x) = 25.40(x - 2) + 12.70$.

The Trap: Choice A is the most commonly picked wrong answer — it uses $(x - 1)$ instead of $(x - 2)$, which would be correct if only the FIRST session were special, but forgets that the SECOND session is ALSO discounted (not full price), so only sessions from the 3rd onward should be counted at full price.

POE — Eliminate the other three:

X A — Using $(x - 1)$ instead of $(x - 2)$ only accounts for the first session being free, but forgets that the second session is ALSO not at full price (it's half off) — this would overcharge for the second session.

X C — Multiplying both $(x - 1)$ and $(x - 2)$ together creates a nonsensical quadratic relationship — this doesn't match the simple linear 'fixed cost plus per-session rate' structure the problem describes.

X D — Similarly, multiplying $(x - 2)$ and $(x - 1)$ together creates an unnecessary quadratic term that doesn't match the linear pricing structure described in the problem.

Solving with Desmos:

Step 1: Open Desmos and type the function you derived: $f(x)=25.40(x-2)+12.70$

Step 2: Press Enter — Desmos graphs this line.

Step 3: Use the table feature to check specific values: click the '+' icon, add a table, and enter $x = 2$ (meaning exactly 2 sessions purchased). The output should show $f(2) = 25.40(0) + 12.70 = 12.70$, which makes sense — 2 sessions costing free + half-off = \$12.70 total, with no full-price sessions yet.

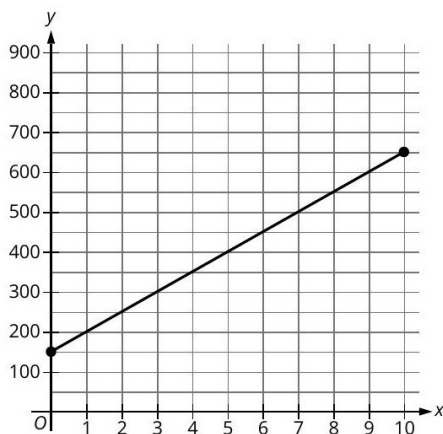
Step 4: Now enter $x = 3$ in the table — the output should jump to $f(3) = 25.40(1) + 12.70 = 38.10$, confirming that the 3rd session correctly adds one full \$25.40 charge on top of the \$12.70 base, verifying choice B is structured correctly.

Question ID: 590d662d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y , in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

Correct Answer: 500 (Student-Produced Response — no answer choices given)

The Rule: When a graph shows TOTAL accumulated distance over time, the question 'how much distance was covered DURING a specific time interval' is asking for the CHANGE in the y-value over that interval — not the final y-value itself. Always subtract the starting y-value from the ending y-value.

The Tip: Be very careful with wording like 'distance walked in the 10 minutes after' — this is NOT simply asking for the y-value at $x = 10$; it's asking for the DIFFERENCE between the y-value at the end of the interval and the y-value at the start of the interval.

Why this is the answer: From the graph, at $x = 0$ (right when Mary left the trail sign), she had already walked 150 meters total (the y-intercept). At $x = 10$ minutes later, she had walked 650 meters total. The distance walked DURING those 10 minutes is the difference: $650 - 150 = 500$ meters.

Step-by-Step Solution:

Step 1: Read the starting point directly from the graph: at $x = 0$ minutes (right when Mary left the sign), the graph shows $y = 150$ meters already walked.

Step 2: Read the ending point from the graph: at $x = 10$ minutes, the graph shows $y = 650$ meters total walked.

Step 3: Since the question asks for the distance walked DURING the 10-minute interval (not the total distance at the 10-minute mark), calculate the CHANGE in y : $650 - 150 = 500$.

Step 4: This is a grid-in question — enter 500 as your final answer, representing 500 meters walked during those 10 minutes.

The Trap: A common error on this problem is reading the final y-value (650) directly off the graph at $x = 10$ and submitting that as the answer, without realizing the question is asking for the CHANGE in distance during the interval, not the total accumulated distance at that point.

Solving with Desmos:

Step 1: Since this is a graph-reading problem, Desmos is most useful here for finding the exact equation of the line first. Estimate two clear points from the graph: $(0, 150)$ and $(10, 650)$. Step 2: Open Desmos and type these two points as a check: $(0, 150)$ and $(10, 650)$ on separate lines — Desmos will plot them as dots.

Step 3: Calculate the slope between them: type $(650-150)/(10-0)$ on a new line — Desmos will compute this as 50, confirming Mary walks 50 meters per minute.

Step 4: To find the distance walked specifically during the 10-minute interval, type 650-150 directly — Desmos will show 500, confirming your answer. Enter 500 into the grid-in box.

Linear inequalities in one or two variables

3 questions · Easy 0 / Medium 3 / Hard 0

Linear inequalities in one or two variables — Q1 [Medium] (ID: 30b1e186)

Question ID: 30b1e186

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

In the xy -plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

Answer

- A. The x -axis
- B. The region where $x > 0$ and $y > 0$
- C. The region where $x < 0$ and $y < 0$
- D. The region where $x < 0$ and $y > 0$

Correct Answer: D

The Rule: To determine which region of the xy -plane satisfies a linear inequality, first rewrite the inequality in slope-intercept form ($y < mx + b$ or $y > mx + b$) — this tells you immediately whether the solution region is ABOVE or BELOW the boundary line, which you can then compare against each answer choice's described region.

The Tip: After rewriting the inequality with y isolated, pick a few simple test points from each answer choice's described region (like the origin, or a point clearly in that quadrant) and plug them into the ORIGINAL inequality to quickly check whether that region could contain solution points.

Why D is right: Rewrite $5x - 7y > 35$ by isolating y : $-7y > 35 - 5x$, then divide by -7 (flipping the inequality sign since we're dividing by a negative number): $y < (5/7)x - 5$. This means the solution region is entirely BELOW the line $y = (5/7)x - 5$. In the region where $x < 0$ and $y > 0$ (choice D), the line itself has already dropped below -5 for any negative x , meaning y would need to be even MORE negative than that to satisfy the inequality — but this region requires $y > 0$, which can never be low enough. So this region contains no solution points.

Step-by-Step Solution:

Step 1: Start with the given inequality: $5x - 7y > 35$.

Step 2: Isolate y by first subtracting $5x$ from both sides: $-7y > 35 - 5x$, or equivalently $-7y > -5x + 35$.

Step 3: Divide both sides by -7 . Remember: dividing an inequality by a NEGATIVE number flips the inequality sign. This gives: $y < (5/7)x - 5$.

Step 4: This tells you the solution set is every point BELOW the line $y = (5/7)x - 5$.

Step 5: Test choice D's region ($x < 0$ and $y > 0$): for any negative x -value, the boundary line $y = (5/7)x - 5$ is already less than -5 (since $(5/7)$ times a negative number is negative, minus 5 makes it even more negative). For a point to satisfy the inequality, y would need to be even smaller (more negative) than this already-negative boundary value.

Step 6: But choice D requires $y > 0$ (positive), which can never be less than a negative boundary value — so this region contains zero solution points, confirming choice D is correct.

The Trap: Choice C is the most commonly picked wrong answer — students sometimes assume that because BOTH x and y are negative, the inequality (which involves subtraction) can't be satisfied, but testing an actual point (like $x = -1$, y

$= -100)$ shows $5(-1) - 7(-100) = -5 + 700 = 695$, which IS greater than 35 — so this region actually DOES contain solution points.

POE — Eliminate the other three:

- X A** — The x-axis (where $y = 0$) does contain solution points — for example, at $x = 10, y = 0$: $5(10) - 7(0) = 50$, which is greater than 35, satisfying the inequality.
- X B** — The region where $x > 0$ and $y > 0$ does contain solution points — for example, at $x = 100, y = 1$: $5(100) - 7(1) = 500 - 7 = 493$, which is greater than 35.
- X C** — The region where $x < 0$ and $y < 0$ does contain solution points — for example, at $x = -1, y = -100$: $5(-1) - 7(-100) = -5 + 700 = 695$, which is greater than 35.

Solving with Desmos:

- Step 1: Open Desmos and type the inequality exactly as given: $5x-7y>35$
- Step 2: Press Enter — Desmos will shade the entire region of the plane that satisfies this inequality (this is one of Desmos's most useful features for inequality problems — it does the region-testing for you visually).
- Step 3: Look specifically at the upper-left area of the graph (where $x < 0$ and $y > 0$, matching choice D's description) — you'll see this entire area is UNSHADED, meaning no points there satisfy the inequality.
- Step 4: For comparison, check the other three regions described in choices A, B, and C — you'll see the shaded (solution) region does extend into those areas, confirming they DO contain solution points, while only the upper-left region (choice D) is completely free of shading.

Linear inequalities in one or two variables — Q2 [Medium] (ID: d914abec)

Question ID: d914abec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

To complete a landscaping project, Adam charges a fee of \$20.00 for his equipment and \$9.50 per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of \$17.00 for her equipment and \$10.00 per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

Answer

- A. $5 \leq x \leq 6$
- B. $6 \leq x \leq 7$
- C. $x < 5$
- D. $x > 6$

Correct Answer: D

The Rule: To compare two people's or companies' pricing structures and find when one is greater than the other, set up an expression for each total charge, then write an inequality with one total GREATER THAN the other exactly as described in the question, and solve for the variable.

The Tip: Write out each person's total charge as its own separate expression first (fixed fee + rate $\times$ hours), clearly labeled, before combining them into an inequality — this prevents mixing up whose fee and whose rate belong in which part of the inequality.

Why D is right: Caroline's total charge is $17 + 10x$ (her \$17 fee plus \$10 per hour). Adam's total charge is $20 + 9.5x$ (his \$20 fee plus \$9.50 per hour). The question asks when Caroline's charge is GREATER than Adam's: $17 + 10x > 20 + 9.5x$. Subtracting $9.5x$ from both sides and 17 from both sides gives $0.5x > 3$, so $x > 6$.

Step-by-Step Solution:

Step 1: Write Caroline's total charge as an expression: her fee is \$17, plus \$10 for every hour (x), giving $17 + 10x$.

Step 2: Write Adam's total charge as an expression: his fee is \$20, plus \$9.50 for every hour (x), giving $20 + 9.5x$.

Step 3: Set up the inequality for when Caroline's charge is GREATER than Adam's: $17 + 10x > 20 + 9.5x$.

Step 4: Subtract $9.5x$ from both sides to gather the x-terms on one side: $17 + 0.5x > 20$.

Step 5: Subtract 17 from both sides: $0.5x > 3$.

Step 6: Divide both sides by 0.5: $x > 6$.

Step 7: This matches choice D.

The Trap: Choice C ($x < 5$) is the most commonly picked wrong answer — it results from setting up the inequality backwards (solving for when ADAM's charge is greater than Caroline's instead of the reverse) combined with an arithmetic slip, flipping both the inequality direction and the resulting boundary number.

POE — Eliminate the other three:

- X A** — $5 \leq x \leq 6$ introduces an upper bound that doesn't exist in this problem — there's no reason x would be capped at 6; the inequality is simply $x > 6$ with no upper limit.
- X B** — $6 \leq x \leq 7$ similarly introduces a nonexistent upper bound, and the boundary values don't align with the correct solution $x > 6$.
- X C** — $x < 5$ reverses the direction of the inequality (suggesting Caroline's charge is greater when x is SMALL, not large) and uses an incorrect boundary number — this would actually represent when Adam's charge exceeds Caroline's, not the reverse.

Solving with Desmos:

Step 1: Open Desmos and type Caroline's total charge as a function: $c(x)=17+10x$

Step 2: Press Enter, then type Adam's total charge on the next line: $a(x)=20+9.5x$

Step 3: Press Enter — Desmos graphs both lines. Look for the point where they cross (intersect) — click on it to see its exact coordinates, which should show $x = 6$.

Step 4: To confirm which side of $x = 6$ has Caroline's charge GREATER than Adam's, look at the graph to the RIGHT of $x = 6$ (larger x-values) — you'll see Caroline's line ($c(x)$) is now ABOVE Adam's line ($a(x)$), confirming that for $x > 6$, Caroline charges more, matching choice D.

Step 5: As an extra check, type $c(x)>a(x)$ directly into Desmos on a new line — Desmos will shade the x-values where this is true, visually confirming the region is $x > 6$.

Linear inequalities in one or two variables — Q3 [Medium] (ID: e8fd6d99)

Question ID: e8fd6d99

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

Terry has a goal to collect at least 16 items per week for a donation drive. This week, Terry has collected 4 items. If x represents the additional number of items Terry needs to collect this week to meet the goal, which inequality represents this situation?

- Answer**
- A. $4 - x \leq 16$
 - B. $4 + x \leq 16$
 - C. $4 - x \geq 16$
 - D. $4 + x \geq 16$

Correct Answer: D

The Rule: Phrases like 'at least' in a word problem always translate to a 'greater than or equal to' ($\geq$) inequality symbol, never a strict 'greater than' ($>$) — because 'at least 16' includes the possibility of exactly reaching 16, not just exceeding it.

The Tip: Build a mental translation table for common inequality phrases before setting up any word problem: 'at least' means $\geq$, 'at most' means $\leq$, 'more than' means strictly $>$, and 'fewer than' means strictly $<$. Misreading these phrases is one of the most common sources of errors on inequality word problems.

Why D is right: Terry has already collected 4 items and needs x more items, so the total collected will be $4 + x$. Since the goal is to collect 'at least 16 items,' this total must be greater than or equal to 16, giving the inequality $4 + x \geq 16$.

Step-by-Step Solution:

Step 1: Identify the two quantities being added together: the 4 items Terry has already collected, and x , the additional items still needed.

Step 2: Write an expression for the total number of items Terry will have collected by the end of the week: $4 + x$.

Step 3: Translate the phrase 'at least 16 items' into an inequality symbol: 'at least' means the total must be greater than OR EQUAL TO 16, so use the $\geq$ symbol.

Step 4: Combine these pieces into the final inequality: $4 + x \geq 16$.

Step 5: This matches choice D.

The Trap: Choice B ($4 + x \leq 16$) is the most commonly picked wrong answer — it uses the correct expression ($4 + x$) but the WRONG inequality direction, mistakenly treating the goal as an upper limit ('at most 16') instead of a minimum threshold ('at least 16').

POE — Eliminate the other three:

✗ A — $4 - x \leq 16$ incorrectly subtracts x instead of adding it, which would represent items being taken away rather than items being collected — this doesn't match the situation of Terry ADDING more items toward the goal.

✗ B — $4 + x \leq 16$ uses the correct addition but the wrong inequality direction — 'at least' means $\geq$ (a minimum), not $\leq$ (a maximum), so this reverses the intended meaning of the goal.

✗ C — $4 - x \geq 16$ incorrectly subtracts x AND uses it in a way that would require x to be a large negative number to ever reach 16, which doesn't logically fit collecting additional items toward a goal.

Solving with Desmos:

Step 1: Open Desmos and type the inequality you derived: $4+x \geq 16$ (in Desmos, type the greater-than-or-equal-to symbol by typing the $>$ key followed by $=$, and Desmos will automatically convert it to the $\geq$ symbol on screen).

Step 2: Press Enter — Desmos will display a shaded region or highlighted portion of the number line (since this is a one-variable inequality) showing all values of x that make the inequality true.

Step 3: You should see the shading starts at $x = 12$ and continues to the right (toward positive infinity) with a solid/filled boundary point at $x = 12$ (the filled-in point confirms $x = 12$ itself IS included, matching the 'or equal to' part of $\geq$).

Step 4: This confirms that Terry needs to collect AT LEAST 12 more items ($x \geq 12$) to reach the goal of 16 total, and the inequality $4 + x \geq 16$ correctly models this, matching choice D.

Advanced Math

Math | 27 questions with full Rule / Tip / Trap / POE / Desmos explanations

The Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Nonlinear functions

11 questions · Easy 2 / Medium 2 / Hard 7

Nonlinear functions — Q1 [Easy] (ID: 3cf2698e)

Question ID: 3cf2698e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The number of cells of a certain bacteria doubles every 25 minutes during its growth period when placed in a medium of milk in a controlled laboratory environment. If a sample of 830 cells of this bacteria is placed in milk in a controlled laboratory environment, which function f best represents the number of cells of the bacteria x minutes after the growth period begins, where x is less than the number of minutes in the growth period?

Answer

- A. $f(x) = 830\left(\frac{1}{2}\right)^{\frac{x}{25}}$
- B. $f(x) = 830(2)^{\frac{x}{25}}$
- C. $f(x) = 830\left(\frac{1}{3}\right)^{\frac{x}{25}}$
- D. $f(x) = 830(3)^{\frac{x}{25}}$

Correct Answer: B

The Rule: For exponential growth or decay problems, the general model is: (final amount) = (initial amount) $\times$ (growth factor)^(number of periods elapsed). When something **DOUBLES** every fixed time interval, the growth factor is exactly 2, and the exponent should represent 'how many complete doubling periods have occurred' — usually written as x divided by the length of one period.

The Tip: Translate the words into the model piece by piece before looking at answer choices: 'doubles' tells you the base of the exponent is 2 (not $1/2$, which would be halving/decay). 'Every 25 minutes' tells you the exponent must be $x/25$, so that when $x = 25$, the exponent equals exactly 1 (one full doubling).

Why B is right: The initial amount is 830 cells. Since the population doubles (multiplies by 2) every 25 minutes, the growth factor per minute is $2^{(1/25)}$, and after x minutes the population has gone through $x/25$ doubling periods. This gives $f(x) = 830(2)^{(x/25)}$, matching choice B.

Step-by-Step Solution:

Step 1: Identify the initial amount: 830 cells (given directly in the problem).

Step 2: Identify the growth factor: the population 'doubles,' so the base of the exponential is 2 (not $1/2$ — that would represent something halving, or decaying, not growing).

Step 3: Identify the time period: doubling happens 'every 25 minutes,' so the exponent must be $x/25$ — this way, when $x = 25$ minutes have passed, the exponent is $25/25 = 1$, giving exactly one full doubling.

Step 4: Assemble the full function: $f(x) = (\text{initial amount}) \times (\text{growth factor})^{(x / \text{period})} = 830(2)^{(x/25)}$.

Step 5: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it uses $1/2$ as the base instead of 2, which would model the population being cut **IN HALF** every 25 minutes (decay) rather than doubling (growth) — a direct misreading of 'doubles' as 'halves.'

POE — Eliminate the other three:

X A — Using $1/2$ as the base models decay (the population shrinking by half each period), which is the opposite of 'doubles' — this misreads growth as decay.

X C — Using $1/3$ as the base doesn't correspond to anything in the problem — there's no tripling or thirding described, and it's also a decay-style base (less than 1).

X D — Using 3 as the base would model the population **TRIPLING** every 25 minutes, not doubling — this doesn't match the given growth rate.

Solving with Desmos:

Step 1: Open Desmos and type the function you derived: $f(x)=830(2)^{(x/25)}$

Step 2: Press Enter — Desmos will graph this exponential curve, which should start at (0, 830) and rise steeply as x increases.

Step 3: Use the table feature (click '+' then table) and enter $x = 25$ — the output should show $f(25) = 830(2)^1 = 1660$, exactly double the starting amount, confirming the doubling behavior described in the problem.

Step 4: Enter $x = 50$ in the table as well — you should see $f(50) = 830(2)^2 = 3320$, which is double 1660, confirming the population doubles again after another 25 minutes, verifying your equation and choice B.

Question ID: fe62f031

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Easy

Question

The function f is defined by $f(x) = 3x^5$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. Which equation defines the function g ?

Answer

- A. $g(x) = 3x^5 + 4$
- B. $g(x) = 3x^5 - 4$
- C. $g(x) = 3x^{20}$
- D. $g(x) = 12x^5$

Correct Answer: A

The Rule: Shifting a graph UP by a certain number of units means adding that number directly to the entire function's output — for any function $y = f(x)$, shifting up k units gives the new function $y = f(x) + k$. Shifting DOWN would instead subtract k .

The Tip: Remember the direction rule: shifting UP always means ADDING a positive constant to the whole function (outside any exponents or parentheses affecting x), and shifting DOWN always means SUBTRACTING. Don't confuse this with horizontal shifts, which affect x INSIDE the function and work in the opposite, 'backwards' direction.

Why A is right: Since g is the result of shifting f up by 4 units, and $f(x) = 3x^5$, the new function is $g(x) = f(x) + 4 = 3x^5 + 4$, matching choice A.

Step-by-Step Solution:

Step 1: Write down the original function: $f(x) = 3x^5$.

Step 2: Recall the rule for vertical shifts: shifting a graph UP by k units means adding k to the entire function, giving $g(x) = f(x) + k$.

Step 3: Substitute $k = 4$ (since the graph shifts up 4 units) and the original function: $g(x) = 3x^5 + 4$.

Step 4: This matches choice A.

The Trap: Choice B is the most commonly picked wrong answer — it subtracts 4 instead of adding it, which would represent shifting the graph DOWN 4 units rather than UP, a direct sign confusion between the two vertical shift directions.

POE — Eliminate the other three:

X B — Subtracting 4 ($3x^5 - 4$) represents a shift DOWN 4 units, not UP — this reverses the direction described in the problem.

X C — $3x^{20}$ changes the EXPONENT of x rather than adding a constant — this would represent a completely different type of transformation (not a simple vertical shift at all), and doesn't match 'shifting up.'

X D — $12x^5$ multiplies the entire function by 4 (a vertical stretch) rather than adding 4 (a vertical shift) — stretching and shifting are different transformations.

Solving with Desmos:

Step 1: Open Desmos and type the original function: $f(x)=3x^5$

Step 2: Press Enter — Desmos graphs this curve.

Step 3: On the next line, type the proposed answer: $g(x)=f(x)+4$

Step 4: Press Enter — Desmos graphs $g(x)$ as well, and you'll visually see it's the exact same S-shaped curve as $f(x)$, just moved straight up by 4 units (compare where each curve crosses the y -axis: f crosses at 0, g should cross at 4).

Step 5: To double check, use the table feature for both functions at the same x -value (like $x = 1$): $f(1) = 3$, and $g(1)$ should be exactly $3 + 4 = 7$ — confirming the vertical shift is working correctly and matches choice A.

Nonlinear functions — Q3 [Medium] (ID: 71b74a44)

Question ID: 71b74a44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$y = 7,400(0.87)^x$$

The given equation estimates the value, in dollars, of a certain piece of equipment x years after it was purchased. According to the given equation, what was the estimated value, in dollars, of the piece of equipment at the time it was purchased?

Answer

- A. 740
- B. 6,438
- C. 7,400
- D. 8,700

Correct Answer: C

The Rule: In an exponential model of the form $y = a(b)^x$, the constant 'a' (the number OUTSIDE the parentheses, not raised to any power) always represents the STARTING value — the value of y when $x = 0$, which corresponds to 'time zero,' i.e., right at the very beginning before any time has passed.

The Tip: Whenever a question asks for the value 'at the time it was purchased' or 'initially' or 'at the start,' that's asking for the value at $x = 0$ — and in any exponential equation, plugging in $x = 0$ makes the entire exponent term equal 1 (since anything to the power of 0 is 1), leaving just the leading constant 'a' by itself.

Why C is right: At the time of purchase, $x = 0$ years have passed. Substituting $x = 0$ into $y = 7,400(0.87)^x$ gives $y = 7,400(0.87)^0 = 7,400 \times 1 = 7,400$, since any nonzero number raised to the power of 0 equals 1.

Step-by-Step Solution:

- Step 1: Recognize that 'at the time it was purchased' means $x = 0$ (no years have passed yet).
- Step 2: Substitute $x = 0$ into the given equation: $y = 7,400(0.87)^0$.
- Step 3: Recall that any nonzero base raised to the power of 0 equals 1: $(0.87)^0 = 1$.
- Step 4: Simplify: $y = 7,400 \times 1 = 7,400$.
- Step 5: This matches choice C.

The Trap: Choice A (740) is the most commonly picked wrong answer — it results from a decimal-place error, such as accidentally dividing 7,400 by 10, rather than correctly evaluating $(0.87)^0$ as exactly 1.

POE — Eliminate the other three:

- X A** — 740 is off by a factor of 10 from the correct starting value — likely from a decimal-point slip rather than correctly evaluating the zero exponent.
- X B** — 6,438 comes from actually multiplying 7,400 by 0.87 (as if $x = 1$, one year later), rather than using $x = 0$ for the very start.
- X D** — 8,700 doesn't correspond to a natural calculation from this equation — likely confused with a different number from the problem or a distractor.

Solving with Desmos:

- Step 1: Open Desmos and type the given equation: $y=7400(0.87)^x$

Step 2: Press Enter — Desmos graphs this decreasing exponential curve.

Step 3: Use the table feature (click '+' then table) and enter $x = 0$ in the x-column.

Step 4: Look at the corresponding y-value in the table — it will show exactly 7400, confirming that the value at the time of purchase ($x = 0$) is \$7,400, matching choice C.

Step 5: As an extra visual check, look at where the curve crosses the y-axis on the graph — this crossing point IS the y-intercept, which always represents the $x = 0$ value, and you should see it lines up with $y = 7400$.

Nonlinear functions — Q4 [Medium] (ID: 769b612d)

Question ID: 769b612d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function g is defined by $g(x) = 10^{5x-8}$. What is the value of x when $g(x)$ is equal to 10,000?

Answer

- A. $\frac{11}{5}$
- B. $\frac{12}{5}$
- C. 3
- D. 4

Correct Answer: B

The Rule: When solving an equation where the variable is in the exponent and both sides can be written as powers of the SAME base, rewrite both sides using that common base, then set the exponents equal to each other — this converts an exponential equation into a simple linear equation you can solve directly.

The Tip: Before trying logarithms or a calculator, check whether the number on the other side of the equation (here, 10,000) can be written as a power of the SAME base already in the equation (here, base 10). If it can, this shortcut is much faster than any other method.

Why B is right: Rewrite 10,000 as a power of 10: $10,000 = 10^4$. Now the equation $10^{(5x-8)} = 10^4$ has matching bases, so the exponents must be equal: $5x - 8 = 4$. Solving: $5x = 12$, so $x = 12/5$.

Step-by-Step Solution:

Step 1: Start with the given equation: $g(x) = 10^{(5x-8)}$, and we want $g(x) = 10,000$.

Step 2: Set up the equation: $10^{(5x-8)} = 10,000$.

Step 3: Rewrite 10,000 as a power of 10: $10,000 = 10 \times 10 \times 10 \times 10 = 10^4$.

Step 4: Now both sides have the same base (10): $10^{(5x-8)} = 10^4$.

Step 5: Since the bases match, set the exponents equal to each other: $5x - 8 = 4$.

Step 6: Add 8 to both sides: $5x = 12$.

Step 7: Divide both sides by 5: $x = 12/5$.

Step 8: This matches choice B.

The Trap: Choice A ($11/5$) is the most commonly picked wrong answer — it typically results from an arithmetic slip while solving $5x - 8 = 4$, such as adding 8 incorrectly or misreading the exponent 4 as a different number during the final steps.

POE — Eliminate the other three:

X A — $11/5$ doesn't satisfy $5x - 8 = 4$ — plugging $x = 11/5$ back in gives $5(11/5) - 8 = 11 - 8 = 3$, not 4, so this doesn't actually solve the equation correctly.

X C — 3 would come from a completely different (incorrect) exponent equation, not from correctly setting $5x - 8 = 4$.

X D — 4 is the value of the EXPONENT (4) itself, not the value of x — a student who correctly identifies $10,000 = 10^4$ but then stops there, forgetting to actually solve $5x - 8 = 4$ for x , might mistakenly select 4.

Solving with Desmos:

Step 1: Open Desmos and type the original function: $g(x)=10^{(5x-8)}$

Step 2: Press Enter — Desmos graphs this steep exponential curve.

Step 3: To find where $g(x) = 10,000$, type a second equation on the next line: $y=10000$ — this creates a flat horizontal line at height 10,000.

Step 4: Look for where the two graphs (the curve and the horizontal line) intersect — click directly on that intersection point, and Desmos will display its exact coordinates.

Step 5: The x -coordinate of that intersection point will show 2.4 (which is the decimal form of $12/5$), confirming $x = 12/5$ and matching choice B.

Nonlinear functions — Q5 [Hard] (ID: f01108a8)

Question ID: f01108a8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$y = 9\left(\frac{a}{7}\right)^{x+c} - b$$

How many times does the graph of the given equation in the xy -plane cross the x -axis, where a , b , and c are positive constants such that $a > 7$ and $b > c$?

Answer

- A. Zero
- B. One
- C. Two
- D. Three

Correct Answer: B

The Rule: For an exponential function of the form $y = A(\text{base})^{(x+c)} - b$ (where the base is greater than 1, making it an increasing exponential), the graph has a horizontal asymptote at $y = -b$ and is always strictly increasing. A strictly increasing curve with a single horizontal asymptote crosses the x -axis EXACTLY ONCE if the asymptote is below the x -axis (negative), and NEVER crosses if the asymptote is above or on the x -axis.

The Tip: For these 'how many times does the graph cross the x -axis' questions with exponential functions, you don't need to solve any equation — just figure out the sign of the horizontal asymptote (the value the function approaches but never reaches) and whether the function is increasing or decreasing. This alone tells you the crossing count.

Why B is right: Since $a > 7$, the base $a/7$ is greater than 1, meaning the exponential term $(a/7)^{(x+c)}$ is always positive and grows as x increases — this makes the whole function strictly increasing. As x approaches negative infinity, the exponential term approaches 0, so y approaches $-b$. Since b is a positive constant, $-b$ is negative, meaning the graph's horizontal asymptote sits below the x -axis. A strictly increasing curve that starts near a negative asymptote and increases without bound must cross the x -axis exactly once.

Step-by-Step Solution:

Step 1: Identify the base of the exponential term: $a/7$. Since the problem states $a > 7$, this means $a/7 > 1$, so the exponential term is a GROWING (increasing) exponential, not a decaying one.

Step 2: Determine the horizontal asymptote: as $x \rightarrow -\infty$, $(a/7)^{(x+c)} \rightarrow 0$, so $y \rightarrow 0 - b = -b$.

Step 3: Since b is stated to be a positive constant, $-b$ is a negative number — meaning the graph's horizontal asymptote is below the x -axis (in negative y territory).

Step 4: Since the function is strictly increasing (from Step 1) and approaches a NEGATIVE value as $x \rightarrow -\infty$ but grows without bound as $x \rightarrow +\infty$, the graph must cross from below the x -axis to above it — and because it's strictly increasing (never doubles back), this crossing happens exactly once.

Step 5: The condition $b > c$ given in the problem is extra information not needed to answer this particular question — it doesn't change the crossing count, only potentially the crossing's exact location.

Step 6: This matches choice B: One.

The Trap: Choice A (Zero) is the most commonly picked wrong answer — students sometimes assume that because the function has a vertical shift ($-b$), it might never reach zero, without realizing that since the function grows without bound in the positive direction, it's guaranteed to eventually cross the x -axis exactly once.

POE — Eliminate the other three:

- X A** — The function grows without bound as x increases (since the base $a/7 > 1$), so it will always eventually rise above the x -axis, guaranteeing at least one crossing — 'zero crossings' isn't possible here.
- X C** — An exponential function of this form is always STRICTLY increasing (monotonic) when its base is greater than 1 — it can never curve back down to cross the x -axis a second time.
- X D** — Like choice C, three crossings would require the function to change direction multiple times, but a simple exponential function like this one only ever increases or only ever decreases — it can never reverse direction to cross more than once.

Solving with Desmos:

Step 1: Since this problem involves unknown constants a , b , and c , pick simple test values that satisfy the given conditions ($a > 7$, $b > c > 0$) to make the function concrete — let's use $a = 14$ (so $a/7 = 2$), $c = 1$, and $b = 5$. Step 2: Open Desmos and type: $y=9(14/7)^{(x+1)}-5$, which simplifies to $y=9(2)^{(x+1)}-5$

Step 3: Press Enter — Desmos graphs this curve.

Step 4: Look at how many times the curve crosses the horizontal x -axis — you should see it cross exactly ONCE, confirming choice B.

Step 5: Try a few different valid values for a , b , and c (always keeping $a > 7$ and $b > c > 0$) and re-graph — you'll find the graph always crosses the x -axis exactly once, no matter which valid values you choose, confirming this is a general property of the function's structure, not just a coincidence of your specific numbers.

Nonlinear functions — Q6 [Hard] (ID: 58b4c6f3)

Question ID: 58b4c6f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard ██████████

Question

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 10)$ and $(2, 13)$. What is the value of ab ?

Answer

- A. 13
- B. 18
- C. 20
- D. 26

Correct Answer: B

The Rule: When a function has unknown constants and you're given two specific points the graph passes through, substitute each point's x and y values into the function to create a system of two equations — then solve that system for the unknown constants.

The Tip: Always substitute the point with $x = 0$ first if one is given — since anything raised to the power of 0 equals 1, this usually eliminates the exponential term entirely and gives you a much simpler equation to start solving with.

Why B is right: Substituting $(0, 10)$: $h(0) = a^0 + b = 1 + b = 10$, so $b = 9$. Substituting $(2, 13)$: $h(2) = a^2 + b = a^2 + 9 = 13$, so $a^2 = 4$, giving $a = 2$ (taking the positive root, since a is stated to be a positive constant). Therefore $ab = 2 \times 9 = 18$.

Step-by-Step Solution:

Step 1: Write down the general function: $h(x) = a^x + b$.

Step 2: Substitute the first point $(0, 10)$: $h(0) = a^0 + b$. Since any nonzero number to the power of 0 equals 1, this simplifies to $1 + b = 10$.

Step 3: Solve for b : $b = 10 - 1 = 9$.

Step 4: Substitute the second point $(2, 13)$: $h(2) = a^2 + b = a^2 + 9 = 13$.

Step 5: Solve for a^2 : $a^2 = 13 - 9 = 4$.

Step 6: Take the square root of both sides: $a = 2$ (using the positive root, since the problem states a is a positive constant).

Step 7: Calculate the requested value: $ab = 2 \times 9 = 18$.

Step 8: This matches choice B.

The Trap: Choice A (13) is the most commonly picked wrong answer — it comes from confusing the requested product ab with one of the given y -coordinates (13) from the original points, rather than actually solving for and multiplying the two constants.

POE — Eliminate the other three:

X A — 13 is one of the given y -coordinates from the original points $(0, 10)$ and $(2, 13)$ — it's not the product ab , which requires solving for both constants first.

X C — 20 doesn't correspond to a natural calculation error from this problem — likely a distractor value.

X D — 26 might result from adding a and b incorrectly derived values instead of multiplying them, or from a sign/arithmetic slip while solving for a^2 .

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the two given points directly to visualize them: $(0,10)$ and $(2,13)$ — Desmos will plot these as dots.

Step 3: Type the general function with sliders: $h(x)=a^x+b$ — Desmos will offer to create sliders for a and b automatically (click 'all' when prompted, or add sliders manually).

Step 4: Adjust the sliders for a and b until the curve $h(x)$ passes exactly through both plotted points — you should find this happens when $a = 2$ and $b = 9$.

Step 5: With the sliders confirmed at $a = 2$ and $b = 9$, type a new line: $2*9$ (or reference the slider values directly) — Desmos will show 18, confirming $ab = 18$ and matching choice B.

Question ID: 7bbfbe70

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A lab analyst observes a sample of a substance. An exponential model estimates that the mass, in grams, of the sample decreases by 24% every 22.96 minutes. Which of the following equations could represent this model, where M is the estimated mass, in grams, of the sample t minutes after the lab analyst began observing the sample?

Answer

A. $M = 100(0.24)^{t+22.96}$

B. $M = 100(0.24)^{\frac{t}{22.96}}$

C. $M = 100(0.76)^{t+22.96}$

D. $M = 100(0.76)^{\frac{t}{22.96}}$

Correct Answer: D

The Rule: When a quantity 'decreases by p%' repeatedly, the correct decay factor to use in an exponential model is $(1 - p/100)$, NOT $p/100$ itself — because after each period, the quantity retains $(100 - p)\%$ of its previous value, not just p%.

The Tip: Never use the percentage given in a 'decreases by p%' problem directly as the base of your exponential — always convert it first by subtracting from 100% (or from 1, in decimal form). A common trap answer will use the raw percentage as if it were the retained amount instead of the LOST amount.

Why D is right: Since the mass decreases by 24% every 22.96 minutes, it RETAINS $100\% - 24\% = 76\%$ of its mass each period, which as a decimal is 0.76. The exponent must be $t/22.96$ so that after $t = 22.96$ minutes, exactly one full decay period has occurred. This gives $M = 100(0.76)^{(t/22.96)}$, matching choice D.

Step-by-Step Solution:

Step 1: Identify the percentage decrease: the mass decreases by 24% every 22.96 minutes.

Step 2: Convert this to a retention rate (the fraction of mass that REMAINS after each period): $100\% - 24\% = 76\%$, or as a decimal, 0.76.

Step 3: Identify the time period for the exponent: since the decay happens 'every 22.96 minutes,' the exponent should be $t/22.96$, so that after exactly 22.96 minutes, the exponent equals 1 (one full decay period).

Step 4: Assemble the model: $M = (\text{initial mass}) \times (\text{retention rate})^{(t / \text{period})} = 100(0.76)^{(t/22.96)}$.

Step 5: This matches choice D.

The Trap: Choice B is the most commonly picked wrong answer — it correctly uses $t/22.96$ as the exponent, but incorrectly uses 0.24 (the percentage LOST) as the base instead of 0.76 (the percentage RETAINED), which would model the mass shrinking to 24% of its PREVIOUS value every period — an extremely fast decay, not a 24% decrease.

POE — Eliminate the other three:

X A — This uses 0.24 as the base (should be 0.76) AND adds 22.96 directly to the exponent instead of dividing by it — both the base and the exponent structure are incorrect.

X B — This uses 0.24 as the base — but 0.24 represents the percentage LOST, not the percentage RETAINED; using it directly would model the mass dropping to just 24% of its value every period, a much faster decay than described.

X C — This uses the correct base (0.76) but the wrong exponent structure — adding 22.96 directly to t instead of dividing t by 22.96 doesn't correctly represent 'one full decay period every 22.96 minutes.'

Solving with Desmos:

Step 1: Open Desmos and type the equation you derived: $M=100(0.76)^{(t/22.96)}$

Step 2: Press Enter — Desmos graphs this decreasing exponential curve.

Step 3: Use the table feature and enter $t = 22.96$ — the output should show $M = 100(0.76)^1 = 76$, confirming the mass has dropped to 76% of its original value (a 24% decrease) after exactly one period, matching the problem's description.

Step 4: Enter $t = 45.92$ (double the period) in the table as well — you should see $M = 100(0.76)^2 = 57.76$, confirming the mass decreases by another 24% of its NEW value during the second period, verifying the exponential decay model and choice D.

Nonlinear functions — Q8 [Hard] (ID: a1e65979)

Question ID: a1e65979

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is a quadratic function. In the xy -plane, the graph of $y = f(x)$ has a vertex at $(1, 9)$ and passes through the points $(2, 33)$ and $(-1, 105)$. What is the value of $f(-2) - f(0)$?

Answer

- A. 81
- B. 129
- C. 192
- D. 225

Correct Answer: C

The Rule: For a quadratic function given in vertex form $y = a(x - h)^2 + k$, where (h, k) is the vertex, use ONE additional known point on the graph to solve for the leading coefficient 'a' — then you have the complete equation and can evaluate the function at any x -value needed.

The Tip: When a quadratic's vertex is given directly, always start by writing the function in VERTEX FORM (using the vertex coordinates), not standard form — vertex form requires solving for only ONE unknown (the coefficient a) using a single additional point, which is much more efficient than solving a system with three unknowns.

Why C is right: With vertex $(1, 9)$, write $f(x) = a(x - 1)^2 + 9$. Use the point $(2, 33)$: $33 = a(2 - 1)^2 + 9 = a(1) + 9$, so $a = 24$. This gives $f(x) = 24(x - 1)^2 + 9$. Now calculate $f(-2) = 24(-2 - 1)^2 + 9 = 24(9) + 9 = 216 + 9 = 225$, and $f(0) = 24(0 - 1)^2 + 9 = 24(1) + 9 = 33$. Finally, $f(-2) - f(0) = 225 - 33 = 192$.

Step-by-Step Solution:

Step 1: Write the quadratic in vertex form using the given vertex $(1, 9)$: $f(x) = a(x - 1)^2 + 9$.

Step 2: Use the given point $(2, 33)$ to solve for a : substitute $x = 2$ and $f(x) = 33$: $33 = a(2 - 1)^2 + 9$, which simplifies to $33 = a(1) + 9$, so $33 = a + 9$.

Step 3: Solve for a : $a = 33 - 9 = 24$.

Step 4: Write the complete function: $f(x) = 24(x - 1)^2 + 9$.

Step 5: (Optional check using the third point $(-1, 105)$): $f(-1) = 24(-1 - 1)^2 + 9 = 24(4) + 9 = 96 + 9 = 105$. This matches the given point, confirming the equation is correct.

Step 6: Calculate $f(-2)$: $f(-2) = 24(-2 - 1)^2 + 9 = 24(-3)^2 + 9 = 24(9) + 9 = 216 + 9 = 225$.

Step 7: Calculate $f(0)$: $f(0) = 24(0 - 1)^2 + 9 = 24(-1)^2 + 9 = 24(1) + 9 = 24 + 9 = 33$.

Step 8: Calculate the requested difference: $f(-2) - f(0) = 225 - 33 = 192$.

Step 9: This matches choice C.

The Trap: Choice A (81) is the most commonly picked wrong answer — it results from subtracting the x -values instead of the function outputs, or from a similar mix-up between the INPUT difference and the actual OUTPUT (y -value) difference the question is asking for.

POE — Eliminate the other three:

X A — 81 doesn't come from correctly evaluating $f(-2)$ and $f(0)$ and subtracting — it likely results from a calculation error, such as using the wrong vertex offset value.

X B — 129 similarly doesn't align with the correct values of $f(-2) = 225$ and $f(0) = 33$ — likely from an arithmetic slip in one of the intermediate steps.

X D — 225 is the value of $f(-2)$ ALONE, not the full difference $f(-2) - f(0)$ — a student who correctly calculates $f(-2)$ but forgets to subtract $f(0)$ would mistakenly select this.

Solving with Desmos:

Step 1: Open Desmos and type the function in vertex form using the given vertex: $f(x) = a(x-1)^2 + 9$ — Desmos will offer to create a slider for 'a' (click to add it).

Step 2: Also type the two given points to plot them: (2,33) and (-1,105)

Step 3: Adjust the slider for 'a' until the curve $f(x)$ passes through both plotted points — you should find this happens at $a = 24$.

Step 4: Once confirmed, type $f(-2)$ directly into Desmos — it will evaluate to 225.

Step 5: Type $f(0)$ on the next line — it will evaluate to 33.

Step 6: Type $f(-2) - f(0)$ on a final line — Desmos will compute this as 192, confirming choice C.

Nonlinear functions — Q9 [Hard] (ID: 5acbd30)

Question ID: 5acbd30

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

The function f is defined by $f(x) = 56(0.19)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?

Answer

- A. 19
- B. 44
- C. 56
- D. 81

Correct Answer: D

The Rule: In an exponential function $y = a(b)^x$ where the base b is a decimal between 0 and 1, the percent DECREASE from one integer x -value to the next is always $(1 - b) \times 100\%$, regardless of which specific integer values of x you compare — this percent change is constant for all exponential functions of this form.

The Tip: For 'percent change from $f(n-1)$ to $f(n)$ ' questions with an exponential function, you never need to plug in actual numbers for n — the percent change only depends on the base of the exponential (the number being raised to the x power), which you can read directly from the function.

Why D is right: The function $f(x) = 56(0.19)^x$ has a base of 0.19, meaning each time x increases by 1, the output is multiplied by 0.19 — so the new value retains only 19% of the previous value. This means the value DECREASES by $100\% - 19\% = 81\%$ each time, so $p = 81$.

Step-by-Step Solution:

Step 1: Identify the base of the exponential function: 0.19 (the number being raised to the power of x).

Step 2: Understand what multiplying by 0.19 means: each time x increases by 1, the new value is only 19% (0.19) of the previous value — meaning 19% of the value is RETAINED.

Step 3: Calculate the percent that is LOST (the percent decrease): $100\% - 19\% = 81\%$.

Step 4: Since $f(n)$ is $p\%$ less than $f(n-1)$, and we just calculated this percent decrease is 81%, we have $p = 81$.

Step 5: This matches choice D.

(Algebraic verification: $f(n)/f(n-1) = 56(0.19)^n / 56(0.19)^{(n-1)} = 0.19^1 = 0.19$. This ratio means $f(n) = 0.19 \times f(n-1)$, so $f(n)$ is 19% OF $f(n-1)$, meaning it decreased by $100\% - 19\% = 81\%$ from $f(n-1)$.)

The Trap: Choice A (19) is the most commonly picked wrong answer — it directly uses the base of the exponential (0.19, or 19%) as the answer, without realizing that 19% represents the amount **RETAINED** after each step, not the amount **DECREASED** — these are two different, complementary percentages.

POE — Eliminate the other three:

X A — 19 is the percent of the value that **REMAINS** (is retained) after each step — but the question asks for the percent **DECREASE**, which is the complement: $100 - 19 = 81$, not 19 itself.

X B — 44 doesn't correspond to any natural calculation from the base 0.19 — likely a distractor value.

X C — 56 is simply the leading coefficient (the initial value) of the function — it has nothing to do with the percent change between consecutive terms, which depends only on the base.

Solving with Desmos:

Step 1: Open Desmos and type the function: $f(x)=56(0.19)^x$

Step 2: Press Enter — Desmos graphs this rapidly decreasing exponential curve.

Step 3: Use the table feature and enter $x = 0$ and $x = 1$ to see two consecutive values — you'll see $f(0) = 56$ and $f(1) = 10.64$.

Step 4: Calculate the percent decrease directly: type $(56-10.64)/56*100$ into Desmos — it will evaluate to 81, confirming $p = 81$ and matching choice D.

Step 5: To confirm this percent decrease is **CONSTANT** (the same between any two consecutive integers, not just 0 and 1), try the same calculation with $x = 5$ and $x = 6$ in the table — you'll find the percent decrease between $f(5)$ and $f(6)$ is also exactly 81%, confirming this is a general property of the exponential base, not specific to any particular starting point.

Nonlinear functions — Q10 [Hard] [Grid-In] (ID: c95686e9)

Question ID: c95686e9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$$h(t) = -16t^2 + b$$

The function h estimates a ball's height, in feet, above a floor t seconds after the ball is released, where b is a constant. The function estimates that the ball is 19.36 feet above the floor when it is released at $t = 0$. How many seconds after being released does the function estimate the ball will reach the floor?

Correct Answer: 1.1 (Student-Produced Response — no answer choices given)

The Rule: When a real-world quadratic model gives you height above the ground over time, 'reaching the floor' means the height equals 0 — so set the function equal to 0 and solve for the positive value of t (since negative time doesn't make sense in this context).

The Tip: First use the given starting condition (the height at $t = 0$) to solve for any unknown constant in the function, THEN set the complete function equal to 0 and solve for t — don't try to do both steps at once.

Why this is the answer: Since the ball is 19.36 feet above the floor at $t = 0$, substitute into $h(t) = -16t^2 + b$: $h(0) = -16(0)^2 + b = b = 19.36$, so $b = 19.36$. Now set the height to 0 to find when the ball reaches the floor: $0 = -16t^2 + 19.36$. Solving: $16t^2 = 19.36$, so $t^2 = 19.36/16 = 1.21$, giving $t = \sqrt{1.21} = 1.1$ (taking the positive square root, since time must be positive).

Step-by-Step Solution:

Step 1: Use the given initial condition to find the constant b : at $t = 0$, the height is 19.36 feet, so substitute into $h(t) = -16t^2 + b$: $h(0) = -16(0)^2 + b = b$. Since $h(0) = 19.36$, this means $b = 19.36$.

Step 2: Write the complete function: $h(t) = -16t^2 + 19.36$.

Step 3: The ball reaches the floor when its height equals 0, so set $h(t) = 0$: $-16t^2 + 19.36 = 0$.

Step 4: Add $16t^2$ to both sides: $19.36 = 16t^2$.

Step 5: Divide both sides by 16: $t^2 = 19.36/16 = 1.21$.

Step 6: Take the square root of both sides: $t = \sqrt{1.21} = 1.1$ (using the positive root, since time elapsed must be a positive value).

Step 7: This is a grid-in question — enter 1.1 as your final answer.

The Trap: A common error on this problem is forgetting to take the SQUARE ROOT at the final step, mistakenly submitting $t^2 = 1.21$ as the final answer instead of continuing to solve for $t = 1.1$, or alternatively including the negative root (-1.1), which doesn't make sense for a time value.

Solving with Desmos:

Step 1: Open Desmos and type the equation with the constant already substituted: $y = -16x^2 + 19.36$ (using x in place of t for graphing purposes).

Step 2: Press Enter — Desmos graphs this downward-opening parabola.

Step 3: Look for where the curve crosses the x -axis (where height = 0) on the POSITIVE side of the x -axis (since negative time doesn't make sense here) — click on that point.

Step 4: Desmos will display the exact coordinates of that crossing point, showing $x = 1.1$ (with $y = 0$), confirming the ball reaches the floor at $t = 1.1$ seconds.

Step 5: Enter 1.1 into the grid-in answer box.

Nonlinear functions — Q11 [Hard] (ID: 7bf77c19)

Question ID: 7bf77c19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question
 The functions p and r are defined by the equations shown, where a and b are integer constants such that $a < b$ and $b < 0$. If $y = p(x)$ and $y = r(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

- I. $p(x) = a(4.1)^x + b$
- II. $r(x) = a(4.1)^{x+b}$

- Answer**
- A. I only
 - B. II only
 - C. I and II
 - D. Neither I nor II

Correct Answer: D

The Rule: The y -intercept of any function is its value when $x = 0$. For it to be directly 'displayed as a constant or coefficient' in an equation (without needing any further calculation), the equation must be written so that setting $x = 0$ leaves ONLY one of the already-visible numbers by itself — not a combination or product of multiple constants.

The Tip: For each candidate equation, mentally plug in $x = 0$ and simplify as far as possible. If what's left over is just ONE of the letters/constants already sitting alone in the original equation (not added to or multiplied by anything else),

that equation displays its y-intercept directly. If you still need to compute a sum or product to get the final value, it does NOT count as being 'displayed.'

Why D is right: For equation I, $p(0) = a(4.1)^0 + b = a(1) + b = a + b$ — this requires ADDING two constants together, so the y-intercept isn't a single constant or coefficient already sitting in the equation by itself. For equation II, $r(0) = a(4.1)^{(0+b)} = a(4.1)^b$ — this requires MULTIPLYING a by a power of 4.1, again not a single displayed constant. Since neither equation shows its y-intercept as a single already-visible constant or coefficient, the answer is 'Neither I nor II.'

Step-by-Step Solution:

Step 1: For equation I, $p(x) = a(4.1)^x + b$, substitute $x = 0$: $p(0) = a(4.1)^0 + b$.

Step 2: Simplify using the rule that anything to the power of 0 equals 1: $p(0) = a(1) + b = a + b$.

Step 3: Notice that $a + b$ is a SUM of two separate constants — it's not simply 'a' or simply 'b' displayed alone; you'd need to actually calculate this sum to know the y-intercept's value. So equation I does NOT directly display its y-intercept as a single constant or coefficient.

Step 4: For equation II, $r(x) = a(4.1)^{(x+b)}$, substitute $x = 0$: $r(0) = a(4.1)^{(0+b)} = a(4.1)^b$.

Step 5: This is a PRODUCT of a and a power of 4.1 raised to b — again, not simply 'a' or simply 'b' by itself; you'd need to calculate this product to find the actual y-intercept value. So equation II also does NOT directly display its y-intercept.

Step 6: Since neither equation shows its y-intercept as an already-visible single constant or coefficient, the correct answer is D: Neither I nor II.

The Trap: Choice C (I and II) is the most commonly picked wrong answer — students sometimes assume that because both equations CONTAIN the letters a and b, one of those letters must represent the y-intercept directly, without carefully checking whether evaluating at $x = 0$ actually isolates just one constant or requires further calculation (addition or multiplication).

POE — Eliminate the other three:

X A — Equation I's y-intercept is $a + b$ (a sum), not simply 'a' or 'b' displayed alone — you still need to add two numbers together to find it, so it's not directly 'displayed.'

X B — Equation II's y-intercept is $a(4.1)^b$ (a product involving an exponent), not simply 'a' or 'b' displayed alone — significant calculation is still needed to find the actual value.

X C — Since NEITHER equation isolates its y-intercept as a single already-visible constant (both require additional calculation — addition for I, multiplication with an exponent for II), 'I and II' incorrectly claims both work when neither actually does.

Solving with Desmos:

Step 1: Since this problem involves abstract reasoning about GENERAL constants a and b rather than specific numbers, pick simple test values to verify your reasoning — let's use $a = 2$ and $b = -3$ (remembering $b < 0$ as given). Step 2: Open Desmos and type equation I with these values: $y = 2(4.1)^x + (-3)$

Step 3: Use the table feature and enter $x = 0$ — the output will show $y = 2(1) - 3 = -1$. Notice this value (-1) doesn't match either a (2) or b (-3) directly — it required computing $a + b$, confirming equation I doesn't display its y-intercept as a simple constant.

Step 4: Now type equation II with the same values: $y = 2(4.1)^{(x+(-3))}$

Step 5: Enter $x = 0$ in the table — the output will show some computed decimal value (from 2×4.1^{-3}), which again doesn't match a or b directly, confirming equation II also requires calculation, not direct display — verifying choice D.

Nonlinear equations in one variable and systems of equations in two variables

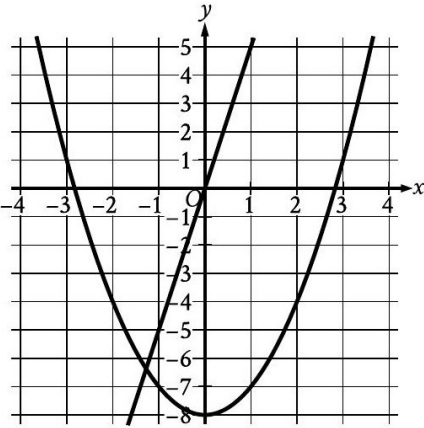
9 questions · Easy 1 / Medium 2 / Hard 6

Nonlinear equations in one variable and systems of equations in two variables — Q1 [Easy] (ID: f123e039)

Question ID: f123e039

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question



The graph of a system of a linear and a quadratic equation is shown. Which system of equations is represented by the graph?

Answer

- A. $y = -5x$
 $y = x^2 + 8$
- B. $y = -5x$
 $y = x^2 - 8$
- C. $y = 5x$
 $y = x^2 + 8$
- D. $y = 5x$
 $y = x^2 - 8$

Correct Answer: B

The Rule: To match a graphed parabola or line to its equation, first read the KEY FEATURES directly off the graph — for a parabola, its vertex (lowest or highest point); for a line, its y-intercept and slope — then compare these features to each answer choice.

The Tip: Start with the parabola's vertex — it's usually the easiest feature to spot exactly on a grid. The vertex's y-coordinate directly tells you the constant term in the parabola's equation when written as $y = x^2 + (\text{constant})$, instantly narrowing your choices.

Why B is right: The parabola's vertex sits at $(0, -8)$, meaning its equation is $y = x^2 - 8$ (matching the pattern $y = x^2 + k$, where $k = -8$ is the vertical shift). The line passes through the origin $(0, 0)$ and through approximately $(-1, 5)$, giving a slope of $(5-0)/(-1-0) = -5$, so its equation is $y = -5x$. Together, these match choice B.

Step-by-Step Solution:

- Step 1: Identify the parabola's vertex directly from the graph: it appears to be at $(0, -8)$, the lowest point of the U-shaped curve.
- Step 2: Since a basic parabola $y = x^2$ shifted down by 8 units has equation $y = x^2 - 8$, and the vertex confirms this shift, the parabola's equation is $y = x^2 - 8$.
- Step 3: Identify a second clear point on the line: it appears to pass through the origin $(0, 0)$ and through approximately $(-1, 5)$.
- Step 4: Calculate the line's slope using these two points: $\text{slope} = (5 - 0) / (-1 - 0) = 5 / (-1) = -5$.
- Step 5: Since the line passes through the origin with slope -5 , its equation is $y = -5x$.
- Step 6: Compare both equations to the answer choices: $y = -5x$ and $y = x^2 - 8$ matches choice B exactly.

The Trap: Choice A is the most commonly picked wrong answer — it uses the correct line equation but the **WRONG** sign on the parabola's vertical shift (+8 instead of -8), which would place the parabola's vertex **ABOVE** the x-axis instead of below it, not matching the graph shown.

POE — Eliminate the other three:

- X A** — This uses $y = x^2 + 8$, which would place the parabola's vertex at $(0, 8)$ — well above the x-axis — but the graph clearly shows the vertex below the x-axis, around $y = -8$.
- X C** — This uses $y = 5x$ for the line, which has a **POSITIVE** slope — but the line in the graph is clearly decreasing (going down from left to right), meaning it needs a **NEGATIVE** slope.
- X D** — This combines the wrong-sign line (positive slope $5x$ instead of $-5x$) with the correct parabola — but the line's direction doesn't match the graph, which shows a decreasing line.

Solving with Desmos:

- Step 1: Open Desmos and type the system you determined: $y = -5x$ and on a second line $y = x^2 - 8$
- Step 2: Press Enter after each — Desmos will graph both.
- Step 3: Compare this graph directly to the picture given in the question — check that the parabola's lowest point matches $(0, -8)$ and that the line passes through the origin, sloping steeply downward to the right.
- Step 4: If everything matches the original picture, you've confirmed choice B is correct. If you're ever unsure between two candidate systems, graph both pairs in Desmos (they'll appear in different colors) and see which pair visually overlaps the original picture most closely.

Nonlinear equations in one variable and systems of equations in two variables — Q2 [Medium] [Grid-In] (ID: 6bd40794)

Question ID: 6bd40794

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$(x - 16)(x - 10)(x + 7)(x + 17) = 0$$

What is a positive solution to the given equation?

Correct Answer: 16 (or 10) (Student-Produced Response — no answer choices given)

The Rule: For an equation already written as a product of factors set equal to zero, the Zero Product Property says the equation is true whenever **ANY ONE** of the individual factors equals zero — so you can find every solution by setting each factor equal to zero separately and solving.

The Tip: Never expand or multiply out a fully-factored equation like this one — that would create unnecessary work. Instead, immediately set each individual factor equal to zero, one at a time, to instantly find each solution.

Why this is the answer: Setting each factor equal to zero: $(x - 16) = 0$ gives $x = 16$; $(x - 10) = 0$ gives $x = 10$; $(x + 7) = 0$ gives $x = -7$; $(x + 17) = 0$ gives $x = -17$. The positive solutions are 16 and 10 — either one is an acceptable grid-in answer.

Step-by-Step Solution:

Step 1: Recognize that the equation is already fully factored: $(x - 16)(x - 10)(x + 7)(x + 17) = 0$.

Step 2: Apply the Zero Product Property: since the product of these four factors equals 0, at least one of the factors itself must equal 0.

Step 3: Set each factor equal to 0 individually and solve: $x - 16 = 0$ gives $x = 16$; $x - 10 = 0$ gives $x = 10$; $x + 7 = 0$ gives $x = -7$; $x + 17 = 0$ gives $x = -17$.

Step 4: The complete solution set is $\{16, 10, -7, -17\}$. The question asks for a POSITIVE solution — both 16 and 10 qualify.

Step 5: This is a grid-in question — enter either 16 or 10 as your final answer; both are accepted as correct.

The Trap: A common error on this type of problem is accidentally flipping the sign when setting a factor to zero — for example, incorrectly solving $(x - 16) = 0$ as $x = -16$ instead of $x = 16$, which would produce a negative value instead of the correct positive solution.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: $(x-16)(x-10)(x+7)(x+17)=0$

Step 2: Press Enter — since this is a one-variable equation, Desmos will find and plot ALL solutions as labeled points along the x-axis.

Step 3: You should see four labeled points: $x = -17$, $x = -7$, $x = 10$, and $x = 16$.

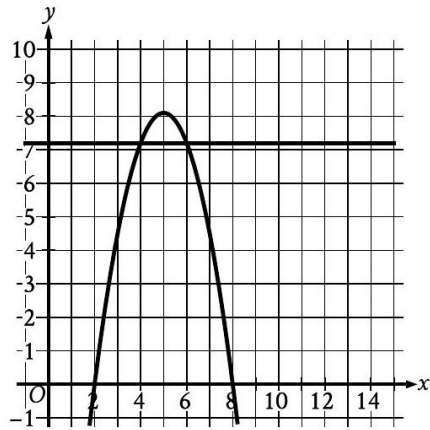
Step 4: Since the question asks for a POSITIVE solution, choose either $x = 10$ or $x = 16$ and enter it into the grid-in box — both are correct answers.

Nonlinear equations in one variable and systems of equations in two variables — Q3 [Medium] (ID: 5f10c095)

Question ID: 5f10c095

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question



The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

Answer

- A. 4
- B. 5
- C. 7.2
- D. 8

Correct Answer: A

The Rule: When a graph shows the intersection of a line and a curve, the x-coordinate of ANY intersection point is automatically a valid solution to the system those two graphs represent — you can read this directly off the graph without doing any algebra, as long as you can identify the intersection points clearly.

The Tip: When answer choices give you SPECIFIC numbers to check against a graph, don't try to find the exact equations of the curves first — just look directly at the graph for where the two curves cross, and check which of the given x-values lines up with a crossing point.

Why A is right: The horizontal line $y = 7$ intersects the parabola at two points, occurring at approximately $x = 4$ and $x = 6$ (visually symmetric around the parabola's vertex, which appears to be at $x = 5$). Since $x = 4$ is one of these intersection points, it's a valid solution to the system, matching choice A.

Step-by-Step Solution:

- Step 1: Identify the horizontal line's equation from the graph: it appears to be a flat line at $y = 7$.
- Step 2: Look for where this horizontal line crosses the parabola — these crossing points are the solutions to the system.
- Step 3: Reading the graph carefully, the line and parabola appear to intersect at approximately $x = 4$ and $x = 6$ (the parabola's vertex is at approximately $(5, 8)$, and the two intersection points are symmetric around this vertex).
- Step 4: Compare these x-values to the answer choices: $x = 4$ is listed as choice A, confirming it as a valid solution.

The Trap: Choice C (7.2) is the most commonly picked wrong answer — it confuses the y-coordinate of the horizontal line ($y = 7$) with a required x-coordinate, creating a plausible-looking decimal that doesn't actually correspond to any point where the graphs intersect.

POE — Eliminate the other three:

X B — $x = 5$ corresponds to the parabola's VERTEX (its highest point, around $y = 8$), not to where it crosses the horizontal line $y = 7$ — at $x = 5$, the parabola's y -value doesn't equal 7.

X C — 7.2 seems to borrow the '7' from the horizontal line's y -value but doesn't correspond to any actual intersection point visible on the graph.

X D — $x = 8$ is close to where the parabola comes back down to the x -axis ($y = 0$), not where it crosses the horizontal line at $y = 7$ — this is reading the wrong feature of the graph.

Solving with Desmos:

Step 1: Since this is a graph-reading problem, you can use Desmos to find the actual equations first, if you want to double check. Estimate the parabola's vertex from the picture — it looks like $(5, 8)$ — and that it passes through the origin area near $(2, 0)$ and $(8, 0)$. Step 2: Open Desmos and type a test parabola: $y = -(x-5)^2 + 8$

Step 3: Compare this to the picture — if it matches the shape shown, you've found the right equation.

Step 4: Type the horizontal line: $y = 7$

Step 5: Press Enter — Desmos will show the exact intersection points when you click on them, confirming the parabola and line cross at $x = 4$ and $x = 6$, verifying choice A is a valid solution.

Nonlinear equations in one variable and systems of equations in two variables — Q4 [Hard] [Grid-In] (ID: 46308566)

Question ID: 46308566

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$r^2 + qr = 2r - 55$$

In the given equation, q is an integer constant. The given equation has no real solutions. What is the largest possible value of q ?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

The Rule: For a quadratic equation to have NO real solutions, its discriminant (the part under the square root in the quadratic formula, $b^2 - 4ac$) must be NEGATIVE. Rearrange any given equation into standard form ($ax^2 + bx + c = 0$) first, then set up and solve the discriminant inequality to find the range of values for an unknown constant.

The Tip: Whenever a problem tells you an equation 'has no real solutions' and asks about a range of possible values for a constant, that's your signal to use the discriminant inequality ($b^2 - 4ac < 0$) rather than trying to solve the equation directly — you can't solve it directly precisely because it has no real solutions.

Why this is the answer: Rearrange $r^2 + qr = 2r - 55$ into standard form: $r^2 + qr - 2r + 55 = 0$, which combines to $r^2 + (q-2)r + 55 = 0$. For no real solutions, the discriminant must be negative: $(q-2)^2 - 4(1)(55) < 0$, so $(q-2)^2 < 220$. Taking the square root: $-\sqrt{220} < q-2 < \sqrt{220}$. Since $\sqrt{220} \approx 14.83$, this gives approximately $-12.83 < q < 16.83$. Since q must be an integer, the largest possible value is 16.

Step-by-Step Solution:

Step 1: Start with the given equation: $r^2 + qr = 2r - 55$.

Step 2: Move all terms to one side to get standard form: $r^2 + qr - 2r + 55 = 0$.

Step 3: Combine the like terms involving r : $r^2 + (q-2)r + 55 = 0$.

Step 4: Identify the coefficients for the discriminant formula: $a = 1$, $b = (q-2)$, $c = 55$.

Step 5: For no real solutions, set up the discriminant inequality: $b^2 - 4ac < 0$, which becomes $(q-2)^2 - 4(1)(55) < 0$.

Step 6: Simplify: $(q-2)^2 - 220 < 0$, so $(q-2)^2 < 220$.

Step 7: Take the square root of both sides (remembering this creates a two-sided inequality): $-\sqrt{220} < q-2 < \sqrt{220}$.

Step 8: Calculate $\sqrt{220} \approx 14.83$.

Step 9: Add 2 to all three parts of the inequality: $2 - 14.83 < q < 2 + 14.83$, which gives approximately $-12.83 < q < 16.83$.

Step 10: Since q is stated to be an INTEGER constant, find the largest integer within this range: the largest integer less than 16.83 is 16.

Step 11: This is a grid-in question — enter 16 as your final answer.

The Trap: A common error on this problem is forgetting to properly rearrange the equation into standard form BEFORE identifying a , b , and c — for example, using $b = q$ instead of the correct $b = (q - 2)$, which comes from combining the qr term with the $-2r$ term after moving everything to one side.

Solving with Desmos:

Step 1: Open Desmos. Step 2: Type the rearranged equation with q as a slider variable: $y = x^2 + (q-2)x + 55$ (using x in place of r for graphing).

Step 3: Desmos will offer to create a slider for q — click to add it.

Step 4: Drag the slider for q and watch the parabola — when the parabola does NOT touch or cross the x -axis at all, the equation has no real solutions for that value of q .

Step 5: Slowly increase the q slider until the parabola is JUST ABOUT to touch the x -axis (this is the boundary case) — check the q value at this point; it should be very close to 16.83.

Step 6: Since q must be an integer and must stay strictly below this boundary (so the parabola never touches the x -axis), test $q = 16$ directly by typing $y = x^2 + 14x + 55$ — confirm this parabola stays entirely above the x -axis, verifying $q = 16$ works and is the largest such integer.

Nonlinear equations in one variable and systems of equations in two variables — Q5 [Hard] (ID: e833de9a)

Question ID: e833de9a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$x^2 + y^2 = 36$$

$$y = mx + \frac{b}{4}$$

In the given system of equations, m and b are negative constants. In the xy -plane, the graphs of the equations in the given system intersect at the point $(-5, y)$, where $y < 0$. Which expression represents the value of b ?

Answer

A. $-\frac{5m}{4} + \frac{\sqrt{11}}{4}$

B. $\frac{5m}{4} - \frac{\sqrt{11}}{4}$

C. $-20m + 4\sqrt{11}$

D. $20m - 4\sqrt{11}$

Correct Answer: D

The Rule: To find where a line intersects a circle (or any curve) at a SPECIFIC given x -value, first use the circle's equation to find the corresponding y -value(s) at that x , then substitute BOTH the x and y values into the line's equation to solve for any remaining unknown constant.

The Tip: When a problem gives you a specific x -coordinate for an intersection point but leaves the y -coordinate as an unknown with a stated condition (like ' $y < 0$ '), always solve for y using the OTHER equation in the system first (the one that doesn't contain the unknown you're solving for) — this narrows things down before you touch the equation with the unknown.

Why D is right: Substituting $x = -5$ into the circle's equation: $(-5)^2 + y^2 = 36$, which gives $25 + y^2 = 36$, so $y^2 = 11$, meaning $y = \pm\sqrt{11}$. Since the problem specifies $y < 0$, we take $y = -\sqrt{11}$. Now substitute $x = -5$ and $y = -\sqrt{11}$ into the line's equation to solve for b : $-\sqrt{11} = m(-5) + b/4$, which gives $-\sqrt{11} = -5m + b/4$. Adding $5m$ to both sides: $5m - \sqrt{11} = b/4$. Multiplying both sides by 4 : $b = 20m - 4\sqrt{11}$.

Step-by-Step Solution:

Step 1: Substitute the known x -coordinate ($x = -5$) into the CIRCLE equation to find y : $(-5)^2 + y^2 = 36$.

Step 2: Simplify: $25 + y^2 = 36$.

Step 3: Solve for y^2 : $y^2 = 36 - 25 = 11$.

Step 4: Take the square root of both sides: $y = \pm\sqrt{11}$. Since the problem states $y < 0$, choose the negative root: $y = -\sqrt{11}$.

Step 5: Now substitute both $x = -5$ and $y = -\sqrt{11}$ into the LINE equation: $y = mx + b/4$, giving $-\sqrt{11} = m(-5) + b/4$.

Step 6: Simplify the right side: $-\sqrt{11} = -5m + b/4$.

Step 7: Add $5m$ to both sides to isolate the $b/4$ term: $-\sqrt{11} + 5m = b/4$, or equivalently $5m - \sqrt{11} = b/4$.

Step 8: Multiply both sides by 4 to solve for b : $b = 4(5m - \sqrt{11}) = 20m - 4\sqrt{11}$.

Step 9: This matches choice D.

The Trap: Choice C ($-20m + 4\sqrt{11}$) is the most commonly picked wrong answer — it has the correct terms but with both signs flipped, typically resulting from a sign error while moving terms across the equals sign during Step 7, such as subtracting $5m$ instead of adding it.

POE — Eliminate the other three:

X A — This choice has both incorrect coefficients on m (should be 20 , not 5) and an incorrect overall structure — it appears to divide by 4 in the wrong place.

X B — Similarly, this uses the wrong coefficient on m ($5m$ instead of $20m$) — likely from forgetting to multiply through by 4 in the final step.

X C — This has the correct magnitude for both terms but with the signs flipped ($-20m$ instead of $20m$, and $+4\sqrt{11}$ instead of $-4\sqrt{11}$) — a sign error somewhere in isolating b likely while moving $-5m$ across the equation.

Solving with Desmos:

Step 1: Open Desmos and type the circle equation: $x^2 + y^2 = 36$

Step 2: Press Enter — Desmos graphs a circle centered at the origin with radius 6 .

Step 3: To find the y -value at $x = -5$, use the table feature on a new expression: type $y = \sqrt{36 - x^2}$ and $y = -\sqrt{36 - x^2}$ as two separate expressions (representing the top and bottom halves of the circle), then create a table and enter $x = -5$ for each — you'll see the negative version gives $y \approx -3.317$, matching $-\sqrt{11}$.

Step 4: Since this problem involves the unknown constants m and b , pick a simple test value for m (like $m = 1$) to make the line concrete, then type: $y = 1x + b/4$ with a slider for b .

Step 5: Adjust the b slider until the line passes through the point $(-5, -\sqrt{11})$ (which you can also type directly as a point: $(-5, -\sqrt{11})$ to see it plotted) — the b value where this happens should match $20(1) - 4\sqrt{11} \approx 6.73$, confirming your formula for b in terms of m matches choice D.

Nonlinear equations in one variable and systems of equations in two variables — Q6 [Hard] (ID: 5da5c665)

Question ID: 5da5c665

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$4x^2 - 5x - 5 = 0$$

What is the greatest solution to the given equation?

Answer

- A. $\frac{5}{8} - \frac{\sqrt{105}}{8}$
- B. $\frac{5}{8} + \frac{\sqrt{105}}{8}$
- C. $\frac{5}{4} - \frac{\sqrt{105}}{4}$
- D. $\frac{5}{4} + \frac{\sqrt{105}}{4}$

Correct Answer: B

The Rule: For a quadratic equation that does NOT factor into nice whole numbers, use the Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The '+' version of this formula always gives the GREATER of the two solutions, and the '-' version always gives the smaller one.

The Tip: Before reaching for the quadratic formula, quickly check whether the equation factors nicely with whole numbers (it usually doesn't when the answer choices involve square roots, like here) — square roots in the answer choices are a strong signal you should go straight to the quadratic formula rather than wasting time trying to factor.

Why B is right: Using the quadratic formula on $4x^2 - 5x - 5 = 0$ with $a = 4$, $b = -5$, $c = -5$: $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}}{2(4)} = \frac{5 \pm \sqrt{25 + 80}}{8} = \frac{5 \pm \sqrt{105}}{8}$. The GREATEST solution uses the '+' sign: $x = \frac{5}{8} + \frac{\sqrt{105}}{8}$.

Step-by-Step Solution:

Step 1: Identify the coefficients from the standard form $ax^2 + bx + c = 0$: $a = 4$, $b = -5$, $c = -5$.

Step 2: Write out the quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Step 3: Substitute the coefficients: $x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(4)(-5)}}{2(4)}$.

Step 4: Simplify the numerator's first term: $-(-5) = 5$.

Step 5: Simplify inside the square root: $(-5)^2 = 25$, and $4(4)(-5) = -80$, so subtracting a negative becomes addition: $25 - (-80) = 25 + 80 = 105$.

Step 6: Simplify the denominator: $2(4) = 8$.

Step 7: The formula now reads: $x = \frac{5 \pm \sqrt{105}}{8}$.

Step 8: Since the question asks for the GREATEST solution, use the '+' version: $x = \frac{5}{8} + \frac{\sqrt{105}}{8}$.

Step 9: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it correctly computes the formula but selects the '-' (minus) version instead of the '+' version, giving the SMALLER solution instead of the greatest one the question actually asks for.

POE — Eliminate the other three:

X A — This uses the '-' version of the quadratic formula ($\frac{5}{8} - \frac{\sqrt{105}}{8}$), which gives the SMALLER of the two solutions, not the greatest one the question asks for.

X C — This has the correct '-' structure but with the wrong denominator (4 instead of 8 under the square root term) — a mismatched-denominator error, and also selects the smaller solution.

X D — This has the correct '+' structure but the wrong denominator under the square root term (4 instead of 8) — the square root term should be divided by 8 (matching $2a$), not 4.

Solving with Desmos:

Step 1: Open Desmos and type the equation exactly as given: $4x^2-5x-5=0$

Step 2: Press Enter — since this is a one-variable equation, Desmos will solve it and plot BOTH solutions as labeled points on the x-axis.

Step 3: You should see two points — one around $x \approx -0.66$ and another around $x \approx 1.9$.

Step 4: Click on the point with the LARGER x-value (the one farther to the right, around $x \approx 1.9$) — this is the greatest solution.

Step 5: To verify this matches $5/8 + \sqrt{105}/8$ exactly (not just as a decimal), type $(5+\sqrt{105})/8$ into Desmos on a new line — it should evaluate to the same decimal value (≈ 1.9), confirming choice B is correct.

Nonlinear equations in one variable and systems of equations in two variables — Q7 [Hard] (ID: c8db0e19)

Question ID: c8db0e19

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{4xy} + xyz = \frac{1}{3yz}$$

In the given equation, x , y , and z are positive numbers. Which expression is equivalent to y ?

Answer

- A. $\frac{4x-3z}{12x^2z^2}$
- B. $\sqrt{\frac{4x-3z}{12x^2z^2}}$
- C. $\frac{1}{3xz^3-4x^3z}$
- D. $\sqrt{\frac{1}{3xz^3-4x^3z}}$

Correct Answer: B

The Rule: To solve an equation for a variable that's inside a square (like y^2 , here disguised as xyz where y appears squared once isolated), first clear all fractions by multiplying through by the least common denominator, then isolate the squared variable, and finally take the square root of both sides.

The Tip: When a variable appears in a term like ' xyz ' alongside fraction terms with that SAME variable in the denominator (like $1/4xy$ or $1/3yz$), multiplying the entire equation by that variable (y , in this case) is often the fastest way to consolidate everything into y^2 terms, avoiding messy fraction-within-fraction steps.

Why B is right: Multiply every term in $1/(4xy) + xyz = 1/(3yz)$ by y : this gives $1/(4x) + xy^2z = 1/(3z)$. Subtract $1/(4x)$ from both sides: $xy^2z = 1/(3z) - 1/(4x)$. Combine the right side over a common denominator: $1/(3z) - 1/(4x) = (4x - 3z)/(12xz)$. So $xy^2z = (4x-3z)/(12xz)$. Divide both sides by xz : $y^2 = (4x-3z)/(12x^2z)$. Take the square root of both sides (keeping the positive root since y is positive): $y = \sqrt{[(4x-3z)/(12x^2z)]}$.

Step-by-Step Solution:

Step 1: Start with the given equation: $1/(4xy) + xyz = 1/(3yz)$.

Step 2: Multiply every single term in the equation by y to begin clearing y from the denominators: $y/(4xy) + xy(yz) = y/(3yz)$.

Step 3: Simplify each term: $y/(4xy) = 1/(4x)$; $xy(yz) = xy^2z$; $y/(3yz) = 1/(3z)$. The equation becomes: $1/(4x) + xy^2z = 1/(3z)$.

Step 4: Subtract $1/(4x)$ from both sides to isolate the xy^2z term: $xy^2z = 1/(3z) - 1/(4x)$.

Step 5: Combine the right side into a single fraction using the common denominator $12xz$: $1/(3z) = 4x/(12xz)$, and $1/(4x) = 3z/(12xz)$, so $1/(3z) - 1/(4x) = (4x - 3z)/(12xz)$.

Step 6: Now the equation reads: $xy^2z = (4x - 3z)/(12xz)$.

Step 7: Divide both sides by xz to isolate y^2 : $y^2 = (4x - 3z)/(12xz \times xz) = (4x - 3z)/(12x^2z^2)$.

Step 8: Take the square root of both sides. Since y is stated to be a positive number, keep only the positive root: $y = \sqrt{[(4x - 3z)/(12x^2z^2)]}$.

Step 9: This matches choice B.

The Trap: Choice A is the most commonly picked wrong answer — it correctly derives the expression for y^2 , $(4x-3z)/(12x^2z^2)$, but forgets to take the final square root step, mistakenly presenting y^2 as if it were y itself.

POE — Eliminate the other three:

✗ A — This expression, $(4x-3z)/(12x^2z^2)$, is actually equal to y^2 (as derived in Step 7), NOT y itself — the crucial final square root step was skipped.

✗ C — This has a completely different structure (1 over a difference of terms) that doesn't match the algebra needed to isolate y from the original equation — likely from an error in combining fractions.

✗ D — This applies a square root to the correct 'shape' of expression (C) but that underlying expression itself is already incorrect, compounding the earlier error rather than fixing it.

Solving with Desmos:

Step 1: Since this equation has three variables (x, y, z), pick simple test values for two of them to verify your algebra — let's use $x = 1$ and $z = 1$. Step 2: Open Desmos and type the original equation with these values substituted, using y as the remaining variable: $1/(4*1*y)+1*y*1=1/(3*y*1)$, or simplified: $1/(4y)+y=1/(3y)$

Step 3: Press Enter — Desmos will solve this one-variable equation for y and plot the positive solution as a point.

Step 4: Now test your derived formula with the same values ($x=1, z=1$): type $\text{sqrt}((4*1-3*1)/(12*1^2*1^2))$ — this should evaluate to $\text{sqrt}(1/12)$, and you can compare this decimal value to the y -value Desmos found by solving the original equation directly — if they match, your formula (and choice B) is confirmed correct.

Nonlinear equations in one variable and systems of equations in two variables — Q8 [Hard] (ID: 9a1f1941)

Question ID: 9a1f1941

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$|4x - 3| = -9$$

How many solutions does the given equation have?

Answer

- A. Zero
- B. One
- C. Two
- D. Infinitely many

Correct Answer: A

The Rule: The absolute value of any real number is always greater than or equal to zero — an absolute value expression can NEVER equal a negative number. If an equation sets an absolute value expression equal to a negative number, that equation has NO solutions, no matter what expression is inside the absolute value bars.

The Tip: Before doing any algebra on an absolute value equation, always glance at what's on the OTHER side of the equals sign first. If that side is a negative number, you can immediately conclude there are zero solutions without any further work — this is one of the fastest 'free points' available on this type of question.

Why A is right: The equation $|4x - 3| = -9$ sets an absolute value expression equal to -9 , a negative number. Since absolute value always produces a result that is zero or positive, it can never equal a negative number like -9 — so this equation has no solutions at all.

Step-by-Step Solution:

Step 1: Look at the given equation: $|4x - 3| = -9$.

Step 2: Recall the fundamental property of absolute value: $|\text{anything}|$ is always greater than or equal to 0 — it can never be negative, because absolute value measures distance from zero, and distance is never negative.

Step 3: Since the right side of the equation is -9 (a negative number), and the left side (an absolute value expression) can never be negative, there is no possible value of x that could make this equation true.

Step 4: Conclude that the equation has zero solutions, matching choice A.

The Trap: Choice B (One) is the most commonly picked wrong answer — students sometimes rush into solving the equation algebraically (splitting into $4x - 3 = -9$ and $4x - 3 = 9$, or similar), without first checking whether the equation is even solvable in the first place, and may mistakenly think one of these branches gives a valid answer.

POE — Eliminate the other three:

X B — Selecting 'One' suggests attempting to solve algebraically without first recognizing that an absolute value can never equal a negative number — this check should happen BEFORE any algebra, avoiding wasted work entirely.

X C — Selecting 'Two' would apply the standard 'absolute value equations have two solutions' rule, but that rule only applies when the other side of the equation is POSITIVE — with a negative number on the other side, this rule doesn't apply at all.

X D — 'Infinitely many' would apply if the equation were something like $|4x-3| = |4x-3|$ (always true) — but this equation asks for a SPECIFIC negative value, which absolute value expressions can never produce, regardless of x .

Solving with Desmos:

Step 1: Open Desmos and type the left side of the equation as a function: $y=\text{abs}(4x-3)$

Step 2: Press Enter — Desmos graphs this V-shaped absolute value function, which you'll notice never dips below the x -axis (y is always ≥ 0).

Step 3: Type a second line: $y=-9$ — this creates a flat horizontal line far below the x -axis.

Step 4: Observe that the V-shaped graph and the horizontal line never touch or cross anywhere on the graph — no matter how far you scroll left or right, the absolute value graph stays at or above $y = 0$, while the line sits at $y = -9$.

Step 5: This visual confirms there are zero intersection points, meaning zero solutions to the equation, matching choice A.

Question ID: 30596346

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\frac{1}{7}(x - k)(16 - x^2) = 0$$

In the given equation, k is a positive constant. The sum of the solutions to the equation is 59. What is the value of k ?

Correct Answer: 59 (Student-Produced Response — no answer choices given)

The Rule: For an equation already written as a product of factors set equal to zero, first find ALL the individual solutions using the Zero Product Property, then use the given information (like the sum of all solutions) to solve for any remaining unknown constant.

The Tip: When one factor is a difference of squares pattern (like $16 - x^2$), remember it factors further into TWO solutions ($x = 4$ and $x = -4$), not just one — don't forget this hidden second solution when counting up all the values that make the equation true.

Why this is the answer: Setting each factor to zero: $(x - k) = 0$ gives $x = k$; $(16 - x^2) = 0$ means $x^2 = 16$, giving $x = 4$ or $x = -4$. So the three solutions are k , 4, and -4 . Since the sum of all solutions is 59: $k + 4 + (-4) = 59$, which simplifies to $k + 0 = 59$, so $k = 59$.

Step-by-Step Solution:

Step 1: Recognize the equation is already factored: $(1/7)(x - k)(16 - x^2) = 0$.

Step 2: Since $1/7$ is just a nonzero constant multiplying everything, it doesn't affect where the equation equals zero — focus on the two remaining factors: $(x - k)$ and $(16 - x^2)$.

Step 3: Set the first factor equal to zero: $x - k = 0$, giving $x = k$.

Step 4: Set the second factor equal to zero: $16 - x^2 = 0$, which means $x^2 = 16$.

Step 5: Take the square root of both sides: $x = 4$ or $x = -4$ (remembering that squaring EITHER a positive or negative 4 gives 16, so BOTH are valid solutions).

Step 6: The complete solution set is $\{k, 4, -4\}$.

Step 7: Add up all the solutions and set this equal to the given sum: $k + 4 + (-4) = 59$.

Step 8: Simplify: $k + 0 = 59$, so $k = 59$.

Step 9: This is a grid-in question — enter 59 as your final answer.

The Trap: A common error on this problem is forgetting that $16 - x^2 = 0$ produces TWO solutions (both $x = 4$ and $x = -4$), not just one — a student who only finds $x = 4$ (forgetting the negative root) would set up the wrong sum equation ($k + 4 = 59$, giving an incorrect $k = 55$).

Solving with Desmos:

Step 1: Open Desmos and type the equation with k as a slider: $y = (1/7)(x - k)(16 - x^2)$

Step 2: Desmos will offer to create a slider for k — click to add it.

Step 3: With any slider value, click on each place where the curve crosses the x -axis — you should always see THREE crossing points: two fixed ones at $x = 4$ and $x = -4$ (these don't move as you change k), and one that moves with the slider (at $x = k$).

Step 4: Adjust the k slider and watch the sum of the three x -axis crossings — you're looking for the slider value where the crossings sum to 59. Since the two fixed crossings (4 and -4) always cancel out to 0, the slider position k must itself equal 59.

Step 5: Set the slider to $k = 59$ and confirm this satisfies the problem — enter 59 into the grid-in box.

Equivalent expressions

7 questions · Easy 1 / Medium 2 / Hard 4

Equivalent expressions — Q1 [Easy] (ID: 91aa9aa9)

Question ID: 91aa9aa9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(9x^2 + 6) - (8x^2)$?

Answer

- A. 7
- B. $17x^2 + 6$
- C. $x^2 - 6$
- D. $x^2 + 6$

Correct Answer: D

The Rule: To combine or simplify polynomial expressions, first remove the parentheses (distributing any negative sign across every term inside), then combine LIKE TERMS — terms with the exact same variable raised to the exact same power.

The Tip: When subtracting a parenthetical expression like $-(8x^2)$, remember the negative sign applies to EVERY term inside those parentheses. Even with just one term inside, it's a good habit to explicitly distribute the negative sign as a separate step to avoid sign errors on more complex problems.

Why D is right: Distribute the subtraction: $(9x^2 + 6) - (8x^2) = 9x^2 + 6 - 8x^2$. Combine the like terms (the x^2 terms): $9x^2 - 8x^2 = 1x^2 = x^2$. The equation simplifies to $x^2 + 6$.

Step-by-Step Solution:

- Step 1: Write out the given expression: $(9x^2 + 6) - (8x^2)$.
- Step 2: Distribute the subtraction sign across the second parentheses: $9x^2 + 6 - 8x^2$.
- Step 3: Identify the like terms: $9x^2$ and $-8x^2$ are like terms (both have x^2), while 6 is a constant with no matching term to combine.
- Step 4: Combine the like terms: $9x^2 - 8x^2 = x^2$.
- Step 5: Write the final simplified expression: $x^2 + 6$.
- Step 6: This matches choice D.

The Trap: Choice C ($x^2 - 6$) is the most commonly picked wrong answer — it correctly combines the x^2 terms but incorrectly changes the sign of the constant term, subtracting 6 instead of keeping it positive (the original +6 was never being subtracted — only the $8x^2$ term was in the second set of parentheses).

POE — Eliminate the other three:

- X A** — 7 completely drops the x^2 term, treating the expression as if it were only constants — this ignores that x^2 terms don't cancel out ($9x^2 - 8x^2 = x^2$, not 0).
- X B** — $17x^2 + 6$ adds the x^2 coefficients ($9 + 8 = 17$) instead of subtracting them — this incorrectly treats the operation between the two parentheses as addition rather than subtraction.
- X C** — $x^2 - 6$ correctly combines the x^2 terms but incorrectly applies the subtraction to the constant 6 as well — but the 6 was inside the FIRST parentheses (being added), not affected by the subtraction of the second parentheses.

Solving with Desmos:

- Step 1: Open Desmos and type the original expression as a function: $y=(9x^2+6)-(8x^2)$

Step 2: Press Enter — Desmos graphs this as a parabola.

Step 3: Type your simplified answer as a second function: $y=x^2+6$

Step 4: Press Enter — Desmos graphs this as well. If your simplification is correct, this second curve should land EXACTLY on top of the first one (they'll appear as a single curve, since they're mathematically identical).

Step 5: As an extra check, use the table feature to compare both functions at a specific x-value, like $x = 2$ — both should give the same output (10), confirming the expressions are truly equivalent and verifying choice D.

Equivalent expressions — Q2 [Medium] (ID: 04887c7b)

Question ID: 04887c7b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

$$f(x) = 2x + 3$$

$$g(x) = 7x - 2$$

$$h(x) = 5x + 6$$

The functions f , g , and h are defined as shown. If $f(x) \cdot g(x) - h(x) = ax^2 + bx + c$, where a , b , and c are constants, what is the value of b ?

Answer

- A. -5
- B. 12
- C. 20
- D. 22

Correct Answer: B

The Rule: To find a specific coefficient in a combined polynomial expression (like the 'b' in $ax^2 + bx + c$), you must fully multiply out (FOIL) any products first, then combine ALL like terms — including terms from a subtraction — before identifying which coefficient the question is asking about.

The Tip: When a problem gives you the FINAL form ($ax^2 + bx + c$) and asks for just one specific letter's value, do the ENTIRE multiplication and simplification first, writing out the complete simplified expression — then simply read off the coefficient the question asks for. Don't try to shortcut by guessing which original term becomes 'b.'

Why B is right: First multiply $f(x)$ and $g(x)$: $(2x+3)(7x-2) = 2x(7x) + 2x(-2) + 3(7x) + 3(-2) = 14x^2 - 4x + 21x - 6 = 14x^2 + 17x - 6$. Now subtract $h(x) = 5x + 6$: $(14x^2 + 17x - 6) - (5x + 6) = 14x^2 + 17x - 6 - 5x - 6 = 14x^2 + 12x - 12$. Comparing to $ax^2 + bx + c$, the coefficient of x is $b = 12$.

Step-by-Step Solution:

Step 1: Write out $f(x) \cdot g(x)$ as a multiplication: $(2x + 3)(7x - 2)$.

Step 2: Use FOIL (First, Outer, Inner, Last) to expand this product: First: $2x \times 7x = 14x^2$. Outer: $2x \times (-2) = -4x$. Inner: $3 \times 7x = 21x$. Last: $3 \times (-2) = -6$.

Step 3: Combine these four terms: $14x^2 - 4x + 21x - 6$.

Step 4: Combine the like x -terms: $-4x + 21x = 17x$. So $f(x) \cdot g(x) = 14x^2 + 17x - 6$.

Step 5: Now subtract $h(x) = 5x + 6$ from this result: $(14x^2 + 17x - 6) - (5x + 6)$.

Step 6: Distribute the negative sign across $h(x)$: $14x^2 + 17x - 6 - 5x - 6$.

Step 7: Combine like terms: the x -terms: $17x - 5x = 12x$. The constants: $-6 - 6 = -12$.

Step 8: The fully simplified expression is: $14x^2 + 12x - 12$.

Step 9: Comparing this to the general form $ax^2 + bx + c$, we can see $b = 12$ (the coefficient of the x term).

Step 10: This matches choice B.

The Trap: Choice D (22) is the most commonly picked wrong answer — it typically results from adding the $-4x$ and $21x$ terms correctly to get $17x$, but then making a sign error while subtracting the $5x$ from $h(x)$, perhaps adding instead of subtracting ($17 + 5 = 22$).

POE — Eliminate the other three:

X A — -5 doesn't correspond to a correct simplification path — likely from a significant sign or arithmetic error somewhere in the FOIL expansion or final combination.

X C — 20 could result from correctly finding $17x$ from the FOIL step but then making an error while combining with $h(x)$'s x -term, perhaps adding 5 instead of subtracting it ($17 + 5 = 22$, close to this but not quite matching this specific wrong value — regardless, it doesn't match the correct combination).

X D — 22 typically comes from adding $17 + 5 = 22$ instead of correctly subtracting $17 - 5 = 12$, reversing the sign of the operation when combining with $h(x)$'s x -coefficient.

Solving with Desmos:

Step 1: Open Desmos and type each function separately: $f(x)=2x+3$ then $g(x)=7x-2$ then $h(x)=5x+6$

Step 2: On a new line, type the full combined expression: $y=f(x)*g(x)-h(x)$

Step 3: Press Enter — Desmos graphs this as a parabola (since it now includes an x^2 term from the multiplication).

Step 4: Click the settings/wrench icon or use the table feature to inspect this parabola's equation in standard form, or simply evaluate the expression at a few x -values and compare to $14x^2+12x-12$ to confirm they match.

Step 5: As a direct check, type $14x^2+12x-12$ as a separate function on another line — this should produce a curve that lands EXACTLY on top of the one from Step 2, confirming $b = 12$ is correct and matches choice B.

Equivalent expressions — Q3 [Medium] (ID: f8871ccd)

Question ID: f8871ccd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Medium

Question

Which expression is equivalent to $2\left(x - \frac{9}{2}\right)(x + 8)$?

Answer

- A. $2x^2 - 72$
- B. $2x^2 + 7x - 72$
- C. $2x^2 - x - 36$
- D. $x^2 + 7x - 36$

Correct Answer: B

The Rule: To multiply an expression by a fraction times two binomials, like $2(x - 9/2)(x + 8)$, it's often cleanest to first combine the leading 2 with the fractional term to clear the fraction, THEN use FOIL to expand the two binomials — this avoids carrying an awkward fraction through the entire expansion.

The Tip: When you see a fraction like $9/2$ paired with a leading coefficient like 2 sitting right outside the parentheses, check if multiplying them together first eliminates the fraction entirely — here, $2 \times (9/2) = 9$, a whole number, which makes the rest of the expansion much cleaner.

Why B is right: Distribute the 2 into the first binomial only: $2(x - 9/2) = 2x - 9$. Now the expression is $(2x - 9)(x + 8)$. Use FOIL: First: $2x \times x = 2x^2$. Outer: $2x \times 8 = 16x$. Inner: $-9 \times x = -9x$. Last: $-9 \times 8 = -72$. Combine: $2x^2 + 16x - 9x - 72 = 2x^2 + 7x - 72$.

Step-by-Step Solution:

Step 1: Start with the given expression: $2(x - 9/2)(x + 8)$.

Step 2: Distribute the leading 2 into the first binomial to clear the fraction: $2 \times x = 2x$, and $2 \times (-9/2) = -9$. This gives $(2x - 9)$ for the first factor.

Step 3: Rewrite the expression with this simplified first factor: $(2x - 9)(x + 8)$.

Step 4: Use FOIL to expand: First: $2x \times x = 2x^2$. Outer: $2x \times 8 = 16x$. Inner: $-9 \times x = -9x$. Last: $-9 \times 8 = -72$.

Step 5: Combine these four terms: $2x^2 + 16x - 9x - 72$.

Step 6: Combine the like x-terms: $16x - 9x = 7x$.

Step 7: The fully expanded expression is: $2x^2 + 7x - 72$.

Step 8: This matches choice B.

The Trap: Choice A ($2x^2 - 72$) is the most commonly picked wrong answer — it correctly computes the 'First' and 'Last' terms of FOIL but completely forgets to include the 'Outer' and 'Inner' cross terms ($16x$ and $-9x$), which combine to give the missing $7x$ term.

POE — Eliminate the other three:

X A — This is missing the middle x-term entirely ($7x$) — it looks like only the 'First' and 'Last' parts of FOIL were computed, skipping the 'Outer' and 'Inner' cross-multiplication steps.

X C — This has the right general shape but an incorrect middle term ($-x$ instead of $+7x$) and an incorrect constant (-36 instead of -72) — likely from distributing the leading 2 unevenly or incompletely.

X D — This drops the leading coefficient entirely (x^2 instead of $2x^2$) and has an incorrect constant (-36 instead of -72) — suggesting the initial distribution of the 2 was mishandled or skipped for some terms.

Solving with Desmos:

Step 1: Open Desmos and type the original expression as a function: $y=2(x-9/2)(x+8)$

Step 2: Press Enter — Desmos graphs this parabola.

Step 3: Type your expanded answer as a second function: $y=2x^2+7x-72$

Step 4: Press Enter — if your expansion is correct, this curve will land EXACTLY on top of the first one.

Step 5: As an extra check, use the table feature to compare both functions at $x = 0$ — both should output -72 , and at $x = 1$, both should output $2+7-72 = -63$, confirming the expressions are equivalent and verifying choice B.

Question ID: 38c90632

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $\frac{29x+102}{x(x+51)}$ is equivalent to $\frac{p}{x} + \frac{w}{x+51}$, where p and w are constants. What is the value of w ?

Correct Answer: 27 (Student-Produced Response — no answer choices given)

The Rule: To split a single fraction into a sum of two simpler fractions with different denominators (called 'partial fraction decomposition'), combine the two simpler fractions back into one over their common denominator, then match the resulting numerator to the original numerator term by term.

The Tip: After combining the right side into a single fraction, line up your combined numerator directly underneath the original numerator, matching the coefficient of x with the coefficient of x , and the constant term with the constant term — this creates a small system of equations you can solve directly.

Why this is the answer: Combine $p/x + w/(x+51)$ into a single fraction using $x(x+51)$ as the common denominator: $[p(x+51) + wx] / [x(x+51)]$. Expand the numerator: $px + 51p + wx = (p+w)x + 51p$. Set this equal to the original numerator: $(p+w)x + 51p = 29x + 102$. Matching coefficients: $p + w = 29$ (coefficient of x) and $51p = 102$ (constant term). Solving the second equation: $p = 102/51 = 2$. Substituting into the first equation: $2 + w = 29$, so $w = 27$.

Step-by-Step Solution:

Step 1: Write the right side of the given equivalence, $p/x + w/(x+51)$, as a single combined fraction using the common denominator $x(x+51)$: $[p(x+51) + w(x)] / [x(x+51)]$.

Step 2: Expand the numerator: $p(x+51) = px + 51p$, so the full numerator is $px + 51p + wx$.

Step 3: Combine the like terms in the numerator (the x -terms): $px + wx = (p+w)x$. The numerator becomes $(p+w)x + 51p$.

Step 4: Since this combined fraction must equal the original expression $(29x+102)/[x(x+51)]$, and the denominators already match, set the numerators equal to each other: $(p+w)x + 51p = 29x + 102$.

Step 5: Match the coefficients of x on both sides: $p + w = 29$.

Step 6: Match the constant terms on both sides: $51p = 102$.

Step 7: Solve the constant-term equation for p : $p = 102/51 = 2$.

Step 8: Substitute $p = 2$ into the x -coefficient equation: $2 + w = 29$.

Step 9: Solve for w : $w = 29 - 2 = 27$.

Step 10: This is a grid-in question — enter 27 as your final answer.

The Trap: A common error on this problem is matching the coefficients incorrectly — for example, accidentally setting $51p$ equal to 29 (the x -coefficient) instead of 102 (the constant term), which comes from confusing which part of the combined numerator corresponds to the x -term versus the constant term.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: $y=(29x+102)/(x*(x+51))$

Step 2: Press Enter — Desmos graphs this rational function.

Step 3: Type your derived answer using $p = 2$ and $w = 27$: $y=2/x+27/(x+51)$

Step 4: Press Enter — if your values for p and w are correct, this second curve will land EXACTLY on top of the first one.

Step 5: As an extra check, use the table feature to compare both functions at a specific x -value, like $x = 1$ — both should give the same output, confirming $w = 27$ is correct.

Question ID: 00165291

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

The expression $6x^4 + 31x^2 + 35$ can be rewritten as $(3x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(3x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + c$?

Correct Answer: 31/2 or 15.5 (Student-Produced Response — no answer choices given)

The Rule: When a quadratic-type expression (like $6x^4 + 31x^2 + 35$, which behaves like a quadratic in x^2) needs to be factored TWO different ways, first find its unique factorization by treating x^2 as a single variable and factoring normally, then figure out the second factorization by redistributing the SAME two roots between the two binomial factors in a different way.

The Tip: A quadratic-type expression like this has exactly two 'roots' when treated as a quadratic in x^2 — but there are multiple ways to distribute those two root-values between two binomial factors $(3x^2 + __)(2x^2 + __)$, depending on which root goes with which leading coefficient. This is why the problem can have BOTH an integer factorization AND a non-integer one from the very same expression.

Why this is the answer: Treating the expression as a quadratic in $y = x^2$, we have $6y^2 + 31y + 35$, which has roots $y = -5/3$ and $y = -7/2$ (found using the quadratic formula or by testing that $6(-5/3)^2 + 31(-5/3) + 35 = 0$). The integer factorization $(3x^2 + a)(2x^2 + b)$ requires matching $a/3 = 5/3$ (giving $a = 5$) and $b/2 = 7/2$ (giving $b = 7$) — both integers. The non-integer factorization $(3x^2 + c)(2x^2 + d)$ instead matches $c/3 = 7/2$ (giving $c = 21/2$) and $d/2 = 5/3$ (giving $d = 10/3$) — both non-integers, as required. So $a = 5$ and $c = 21/2$, giving $a + c = 5 + 21/2 = 10/2 + 21/2 = 31/2$.

Step-by-Step Solution:

Step 1: Notice that $6x^4 + 31x^2 + 35$ is a 'quadratic in disguise' — if you let $y = x^2$, the expression becomes $6y^2 + 31y + 35$, a standard quadratic.

Step 2: Find the roots of $6y^2 + 31y + 35 = 0$ using the quadratic formula: $y = [-31 \pm \sqrt{31^2 - 4(6)(35)}] / (2 \times 6) = [-31 \pm \sqrt{961 - 840}] / 12 = [-31 \pm \sqrt{121}] / 12 = [-31 \pm 11] / 12$.

Step 3: Calculate both roots: $y = (-31 + 11) / 12 = -20 / 12 = -5/3$, and $y = (-31 - 11) / 12 = -42 / 12 = -7/2$.

Step 4: This means $6y^2 + 31y + 35 = 6(y + 5/3)(y + 7/2)$, which can be rewritten (redistributing the 6) as $(3y + 5)(2y + 7)$ — substituting back $y = x^2$ gives the INTEGER factorization $(3x^2 + 5)(2x^2 + 7)$, so $a = 5$ and $b = 7$.

Step 5: For the SECOND (non-integer) factorization, redistribute the SAME two roots differently between the two binomials: instead of matching $5/3$ with the '3' coefficient, match $7/2$ with the '3' coefficient instead: $(3x^2 + 21/2)(2x^2 + 10/3)$ — here, $c/3 = 7/2$ gives $c = 21/2$, and $d/2 = 5/3$ gives $d = 10/3$, both non-integers as required.

Step 6: Now calculate the requested value: $a + c = 5 + 21/2$.

Step 7: Convert 5 to a fraction with denominator 2: $5 = 10/2$. So $a + c = 10/2 + 21/2 = 31/2$.

Step 8: This is a grid-in question — enter $31/2$ or its decimal equivalent, 15.5, as your final answer.

The Trap: A common error on this problem is only finding the integer factorization ($a = 5$, $b = 7$) and stopping there, without realizing the SAME expression has a completely different second factorization using the non-integer constants c and d — this requires recognizing that the two roots of the underlying quadratic can be distributed between the binomial factors in more than one way.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: $y = 6x^4 + 31x^2 + 35$

Step 2: Press Enter — Desmos graphs this quartic (degree-4) curve.

Step 3: Type the integer factorization to verify it: $y = (3x^2 + 5)(2x^2 + 7)$

Step 4: Press Enter — this curve should land exactly on top of the first one, confirming $a = 5$ and $b = 7$.

Step 5: Type the non-integer factorization using fractions: $y = (3x^2 + 21/2)(2x^2 + 10/3)$

Step 6: Press Enter — this curve should ALSO land exactly on top of the original, confirming $c = 21/2$ and $d = 10/3$ are correct.

Step 7: Finally, type $5+21/2$ directly into Desmos — it will evaluate to 15.5, confirming $a + c = 31/2$ and matching your grid-in answer.

Equivalent expressions — Q6 [Hard] (ID: efc469c4)

Question ID: efc469c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$-14(5x-2)^2 + 6(5x-3)^2$$

The given expression can be rewritten as $\frac{a}{6}x^2 + \frac{b}{6}x + \frac{c}{6}$, where a , b , and c are constants. What is the value of $a + b + c$?

Answer

- A. -612
- B. -306
- C. -102
- D. -17

Correct Answer: A

The Rule: To expand and combine an expression with multiple squared binomials (like $-14(5x-2)^2 + 6(5x-3)^2$), first expand EACH squared binomial separately using the pattern $(p-q)^2 = p^2 - 2pq + q^2$, then distribute the leading coefficients, and finally combine all like terms together.

The Tip: Expand each squared term completely as its own separate step before multiplying by the leading coefficient (-14 or 6) — trying to combine the squaring and the coefficient distribution in one step is where most sign errors happen on problems like this.

Why A is right: Expand $(5x-2)^2 = 25x^2 - 20x + 4$, and $(5x-3)^2 = 25x^2 - 30x + 9$. Distribute: $-14(25x^2 - 20x + 4) = -350x^2 + 280x - 56$, and $6(25x^2 - 30x + 9) = 150x^2 - 180x + 54$. Add these together: $(-350+150)x^2 + (280-180)x + (-56+54) = -200x^2 + 100x - 2$. Setting this equal to $(a/6)x^2 + (b/6)x + c/6$: $a/6 = -200$ gives $a = -1200$; $b/6 = 100$ gives $b = 600$; $c/6 = -2$ gives $c = -12$. Sum: $a+b+c = -1200+600-12 = -612$.

Step-by-Step Solution:

Step 1: Expand the first squared binomial: $(5x-2)^2 = (5x)^2 - 2(5x)(2) + 2^2 = 25x^2 - 20x + 4$.

Step 2: Expand the second squared binomial: $(5x-3)^2 = (5x)^2 - 2(5x)(3) + 3^2 = 25x^2 - 30x + 9$.

Step 3: Distribute the leading coefficient -14 across the first expansion: $-14(25x^2 - 20x + 4) = -350x^2 + 280x - 56$.

Step 4: Distribute the leading coefficient 6 across the second expansion: $6(25x^2 - 30x + 9) = 150x^2 - 180x + 54$.

Step 5: Add these two results together, combining like terms: x^2 terms: $-350x^2 + 150x^2 = -200x^2$. x terms: $280x - 180x = 100x$. Constants: $-56 + 54 = -2$.

Step 6: The combined expression is: $-200x^2 + 100x - 2$.

Step 7: Set this equal to the target form $(a/6)x^2 + (b/6)x + c/6$, meaning $a/6 = -200$, $b/6 = 100$, and $c/6 = -2$.

Step 8: Solve for each constant by multiplying both sides by 6: $a = -200 \times 6 = -1200$; $b = 100 \times 6 = 600$; $c = -2 \times 6 = -12$.

Step 9: Calculate the requested sum: $a + b + c = -1200 + 600 + (-12) = -612$.

Step 10: This matches choice A.

The Trap: Choice B (-306) is the most commonly picked wrong answer — it's exactly HALF of the correct answer (-612), which typically results from forgetting to multiply by 6 when solving for a , b , and c individually (i.e., using $a = -200$, $b = 100$, $c = -2$ directly as if they were already the final values, then only partially adjusting).

POE — Eliminate the other three:

- X B** — -306 is exactly half of the correct total — this often results from a computational shortcut gone wrong, such as only multiplying some terms by 6 instead of all three constants consistently.
- X C** — -102 doesn't correspond to a clean derivation from this problem — likely a distractor value or the result of multiple compounding errors.
- X D** — -17 is far too small in magnitude to result from correctly scaling by 6 — this likely comes from forgetting to multiply by 6 at all and instead just adding $-200 + 100 + (-2) = -102$ (close to choice C) or a similar unscaled combination.

Solving with Desmos:

Step 1: Open Desmos and type the original expression: $y = -14(5x-2)^2 + 6(5x-3)^2$

Step 2: Press Enter — Desmos graphs this parabola.

Step 3: Type your simplified expression using the target form with the constants you found: $y = (-1200/6)x^2 + (600/6)x + (-12/6)$

Step 4: Press Enter — if your values for a, b, and c are correct, this curve will land EXACTLY on top of the original one.

Step 5: As a final check, type $-1200+600-12$ directly into Desmos — it will evaluate to -612 , confirming $a+b+c = -612$ and matching choice A.

Equivalent expressions — Q7 [Hard] (ID: 2289b199)

Question ID: 2289b199

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Hard

Question

$$\frac{\sqrt[3]{n^{17}p^5}}{5n^2(\sqrt[5]{p^8})}$$

For all positive values of n and p , the given expression is equivalent to which of the following?

I. $\frac{(n^{14})(p^{\frac{5}{3}})}{5np}$

II. $\frac{\sqrt[3]{n^{11}p^2}}{5}$

Answer

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

Correct Answer: C

The Rule: To simplify an expression involving roots (radicals) with different index numbers (like a cube root and an eighth root), first rewrite every root as a FRACTIONAL EXPONENT (the n th root of x becomes $x^{(1/n)}$), then use standard exponent rules to combine and simplify everything into one unified expression.

The Tip: Whenever you see mixed radical types (cube roots, eighth roots, etc.) in the same expression, immediately convert ALL of them to fractional exponent notation — this turns a confusing-looking radical problem into a much more familiar exponent-simplification problem you already know how to handle.

Why C is right: Rewrite the original expression using fractional exponents: the numerator $\sqrt[3]{n^{17}p^5}$ becomes $n^{(17/3)}p^{(5/3)}$, and the denominator's $5\sqrt[5]{p^8}$ becomes $p^{(8/5)} = p^{1.6} = p^{8/5}$. So the full expression becomes $[n^{(17/3)}p^{(5/3)}] / [5n^2p^{(8/5)}] = (1/5) n^{(17/3-2)} p^{(5/3-1.6)} = (1/5) n^{(11/3)} p^{(2/3)}$. Checking option I: $[n^{(14/3)}p^{(5/3)}] / (5np) = (1/5)n^{(14/3-1)}p^{(5/3-1)} = (1/5)n^{(11/3)}p^{(2/3)}$ — matches! Checking option II: $\sqrt[3]{(n^{11}p^2)/5} = (1/5)n^{(11/3)}p^{(2/3)}$ — also matches! Both expressions are equivalent to the original.

Step-by-Step Solution:

Step 1: Rewrite the numerator using fractional exponent notation: $\sqrt[3]{(n^{17}p^5)} = (n^{17}p^5)^{1/3} = n^{17/3}p^{5/3}$.

Step 2: Rewrite the denominator's radical term: $\sqrt[8]{(p^8)} = p^{8/8} = p^1 = p$ (since 8 divided by 8 equals exactly 1).

Step 3: Write the full original expression using these simplified forms: $[n^{17/3}p^{5/3}] / [5n^2p]$.

Step 4: Simplify by subtracting exponents for matching bases (dividing powers with the same base means subtracting exponents): for n: $17/3 - 2 = 17/3 - 6/3 = 11/3$. For p: $5/3 - 1 = 5/3 - 3/3 = 2/3$.

Step 5: The fully simplified original expression is: $(1/5) n^{11/3} p^{2/3}$.

Step 6: Check option I: $[n^{14/3}p^{5/3}] / (5np)$. Simplify exponents: for n: $14/3 - 1 = 14/3 - 3/3 = 11/3$. For p: $5/3 - 1 = 5/3 - 3/3 = 2/3$. This gives $(1/5)n^{11/3}p^{2/3}$ — an EXACT match to Step 5.

Step 7: Check option II: $\sqrt[3]{(n^{11}p^2)}/5 = (n^{11}p^2)^{1/3}/5 = n^{11/3}p^{2/3}/5$ — also an EXACT match to Step 5.

Step 8: Since BOTH option I and option II are algebraically equivalent to the original expression, the correct answer is C: I and II.

The Trap: Choice A (I only) is the most commonly picked wrong answer — students sometimes correctly verify option I matches but assume option II (written with a cube root instead of separated exponents) must be a 'different' or 'trick' expression, without actually converting it to fractional exponent form to check.

POE — Eliminate the other three:

X A — While option I does correctly match the original expression, option II ALSO matches when converted to fractional exponent form — dismissing option II without checking it carefully is the error here.

X B — While option II does correctly match, option I ALSO matches — both are valid equivalent forms of the same original expression, so 'II only' incorrectly excludes a correct option.

X D — Since BOTH option I and option II simplify to the exact same expression as the original (as shown in Steps 6–7), 'Neither I nor II' is incorrect — both options are actually valid.

Solving with Desmos:

Step 1: Since this involves variables n and p in exponents, pick simple positive test values to verify — let's use n = 8 and p = 4 (chosen because they're powers of 2, which often simplify nicely with fractional exponents). Step 2: Open Desmos and type the original expression with these values: $(8^{17/3} \cdot 4^{5/3}) / (5 \cdot 8^2 \cdot (4^{8/8}))$

Step 3: Press Enter — Desmos will evaluate this to a specific decimal number.

Step 4: Type option I with the same values: $(8^{14/3} \cdot 4^{5/3}) / (5 \cdot 8 \cdot 4)$

Step 5: Press Enter — compare this result to Step 3's result; they should be IDENTICAL, confirming option I matches.

Step 6: Type option II with the same values: $(8^{11} \cdot 4^2)^{1/3} / 5$

Step 7: Press Enter — compare this to Step 3's result as well; it should ALSO be identical, confirming option II matches too, verifying choice C (I and II) is correct.

Problem-Solving and Data Analysis

Math | 19 questions with full Rule / Tip / Trap / POE / Desmos explanations

The Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Ratios, rates, proportional relationships, and units

3 questions · Easy 1 / Medium 0 / Hard 2

Ratios, rates, proportional relationships, and units — Q1 [Easy] (ID: de799680)

Question ID: de799680

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Easy

Question

A cluster fly is traveling at a velocity of 76 centimeters per second. What is this velocity, in millimeters per second?
(1 centimeter = 10 millimeters)

Answer

- A. 7,600
B. 836
C. 760
D. 86

Correct Answer: C

The Rule: Unit conversion problems always use the given conversion factor as a multiplier — set it up so the unit you want to CANCEL is in the denominator and the unit you want to KEEP ends up in the numerator, then multiply straight across.

The Tip: Write the conversion factor as a fraction in two different orientations (10 mm/1 cm, or 1 cm/10 mm) and pick whichever one cancels out centimeters and leaves millimeters — this prevents accidentally dividing when you should multiply, or vice versa.

Why C is right: Since 1 centimeter = 10 millimeters, converting FROM centimeters TO millimeters means MULTIPLYING by 10 (going from a bigger unit to a smaller one always gives a bigger number). So $76 \text{ cm/s} \times 10 \text{ mm/cm} = 760 \text{ mm/s}$.

Step-by-Step Solution:

Step 1: Identify the given rate: 76 centimeters per second.

Step 2: Identify the conversion factor: 1 centimeter = 10 millimeters.

Step 3: Since millimeters are a SMALLER unit than centimeters, converting to millimeters should give a LARGER number — this means multiplying, not dividing.

Step 4: Multiply: $76 \text{ cm/s} \times 10 \text{ mm/cm} = 760 \text{ mm/s}$ (the 'cm' units cancel out, leaving 'mm').

Step 5: This matches choice C.

The Trap: Choice D (86) is the most commonly picked wrong answer — it comes from mistakenly ADDING 10 to 76 ($76 + 10 = 86$) instead of correctly MULTIPLYING ($76 \times 10 = 760$), confusing an additive shift with a multiplicative unit conversion.

POE — Eliminate the other three:

✗ A — 7,600 is off by a factor of 10 from the correct answer — likely from multiplying by 100 instead of 10, perhaps double-applying the conversion factor.

✗ B — 836 doesn't correspond to a natural calculation from this conversion — possibly from adding an unrelated number ($760 + 76 = 836$) rather than performing the correct single multiplication.

✗ D — 86 comes from mistakenly ADDING 10 to 76 ($76 + 10 = 86$) instead of correctly MULTIPLYING ($76 \times 10 = 760$) — confusing an additive shift with a multiplicative unit conversion.

Solving with Desmos:

Step 1: Open Desmos and type the conversion directly as a multiplication: 76×10

Step 2: Press Enter — Desmos will instantly evaluate this to 760, confirming the converted velocity in millimeters per second, matching choice C.

Step 3: As a sanity check on direction, type $76/10$ as well — this gives 7.6, which would be the (incorrect) result of dividing instead of multiplying; comparing both results helps confirm multiplying by 10 is the correct operation when converting from a larger unit (cm) to a smaller one (mm).

Question ID: 4c5ea142

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

On average, a certain plant grows 87 millimeters every m months. At this rate, which expression represents the number of millimeters, on average, the plant grows every k years?

Answer

- A. $\frac{29m}{4k}$
- B. $\frac{29k}{4m}$
- C. $\frac{1,044m}{k}$
- D. $\frac{1,044k}{m}$

Correct Answer: D

The Rule: For rate-conversion word problems with two different unit changes happening at once (here, converting a monthly rate to a yearly rate, AND scaling by an unknown number of years), set up the rate as a fraction and multiply by the necessary conversion factors and variables step by step, keeping careful track of what's in the numerator and denominator.

The Tip: Break multi-step unit conversions into separate, small steps rather than trying to do everything at once — first convert 'per month' to 'per year' using the fixed number 12, and only afterward multiply by the variable representing the number of years.

Why D is right: The plant grows 87 mm every m months, so its rate is $87/m$ mm per month. Converting to a yearly rate: since there are 12 months in a year, multiply by 12: $(87/m) \times 12 = 1044/m$ mm per year. For k years, multiply by k : $(1044/m) \times k = 1044k/m$.

Step-by-Step Solution:

Step 1: Write the given growth rate as a fraction: 87 millimeters per m months, or $87/m$ mm/month.

Step 2: Convert this monthly rate into a YEARLY rate by multiplying by 12 (since there are 12 months in one year): $(87/m) \times 12 = 1044/m$ mm per year.

Step 3: To find the total growth over k years (not just one year), multiply this yearly rate by k : $(1044/m) \times k = 1044k/m$.

Step 4: This matches choice D.

The Trap: Choice C is the most commonly picked wrong answer — it correctly computes 1044 (87×12) but multiplies by m instead of k , and divides by k instead of m , essentially swapping the roles of the two variables in the final expression.

POE — Eliminate the other three:

X A — $29m/4k$ doesn't come from a clean derivation of this problem — the numbers 29 and 4 don't naturally arise from 87 and 12 in this context ($87/3=29$, $12/3=4$, suggesting a reduction error somewhere, but the variable placement is also backwards).

X B — $29k/4m$ has the same reduction issue as choice A, with variables swapped in a way that doesn't match the correct setup.

X C — $1,044m/k$ correctly computes 1,044 but swaps which variable (m or k) belongs in the numerator versus denominator — m should be in the denominator (since a larger m means slower growth per month) and k in the numerator (since more years means more total growth).

Solving with Desmos:

Step 1: Since this problem has two unknown variables (m and k), pick simple test values to verify your formula — let's say $m = 3$ (grows 87mm every 3 months) and $k = 2$ (over 2 years). Step 2: Open Desmos and calculate the rate per month directly: $87/3$ — this gives 29 mm per month.

Step 3: Convert to yearly rate: type 29×12 — this gives 348 mm per year.

Step 4: Multiply by the number of years: type 348×2 — this gives 696 mm total growth over 2 years.

Step 5: Now test your formula $1044k/m$ with the same values: type $1044 \times 2/3$ — this should also give 696, confirming your formula matches the step-by-step calculation and verifying choice D.

Ratios, rates, proportional relationships, and units — Q3 [Hard] (ID: 1b96c116)

Question ID: 1b96c116

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

A rectangle has an area of 3,168 square inches. What is the area, in **square feet**, of this rectangle? (1 foot = 12 inches)

Answer

- A. 22
- B. 56.3
- C. 264
- D. 675.4

Correct Answer: A

The Rule: When converting AREA (not just length) between units, you must apply the conversion factor TWICE — once for each dimension (length and width) — meaning you divide (or multiply) by the SQUARE of the linear conversion factor, not just the factor itself.

The Tip: Never apply a linear conversion factor (like 12 inches = 1 foot) directly to an area measurement — always square the conversion factor first ($12^2 = 144$) before using it to convert square inches to square feet, since area involves two dimensions.

Why A is right: Since 1 foot = 12 inches, 1 square foot = $12 \times 12 = 144$ square inches. To convert 3,168 square inches to square feet, divide by 144: $3,168 \div 144 = 22$.

Step-by-Step Solution:

Step 1: Recall the linear conversion: 1 foot = 12 inches.

Step 2: Since area involves TWO dimensions (length $\times$ width), square the conversion factor to get the area conversion: 1 square foot = $12^2 = 144$ square inches.

Step 3: To convert the given area from square inches to square feet, divide by 144 (since square inches is the smaller unit, converting to the larger unit square feet requires dividing): $3,168 \div 144 = 22$.

Step 4: This matches choice A.

The Trap: Choice C (264) is the most commonly picked wrong answer — it results from dividing by 12 instead of 144, applying only the LINEAR conversion factor instead of correctly squaring it for an AREA conversion.

POE — Eliminate the other three:

X B — 56.3 doesn't correspond to a clean derivation from this problem — likely from an unrelated arithmetic error.

X C — 264 comes from dividing 3,168 by 12 instead of 144 — this uses only the linear (one-dimensional) conversion factor rather than correctly squaring it for the two-dimensional area conversion.

X D — 675.4 doesn't correspond to a clean derivation from this problem either — possibly from multiplying instead of dividing, or a different unrelated calculation error.

Solving with Desmos:

Step 1: Open Desmos and calculate the area conversion factor first: type 12^2 — this confirms 144 square inches equals 1 square foot.

Step 2: On the next line, type 3168/144 — Desmos will evaluate this to 22, confirming the area in square feet.

Step 3: As a sanity check, try the common error calculation as well: type 3168/12 — this gives 264, which is what you'd get if you forgot to square the conversion factor; comparing both results helps you see why squaring is necessary for area conversions.

Percentages

4 questions · Easy 0 / Medium 0 / Hard 4

Percentages — Q1 [Hard] (ID: 28ff3966)

Question ID: 28ff3966

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

A square scale drawing has a side length of 55 inches, and 1 inch on the drawing represents an actual distance of 19 miles. A larger version of the same drawing is printed as a square with the side length 65% longer than the side length of the previous drawing. On the larger drawing, which of the following is closest to the actual distance, in miles, represented by 1 inch?

Answer

- A. 6.65
- B. 11.52
- C. 31.35
- D. 35.75

Correct Answer: B

The Rule: In scale-drawing problems, the ACTUAL physical size being represented stays the same even when the drawing itself is resized — so first calculate the total actual distance represented by the original drawing, then use that same fixed actual distance to find the new scale for the resized drawing.

The Tip: Don't confuse 'the drawing got bigger' with 'the actual place got bigger' — the real-world distance being depicted stays constant; only how much space on paper represents that distance changes. Calculate the fixed real-world total first, before working with the new drawing's dimensions.

Why B is right: The original drawing's side is 55 inches, with each inch representing 19 actual miles, so the total actual distance across is $55 \times 19 = 1,045$ miles. This actual distance doesn't change when the drawing is resized. The new drawing's side length is 65% longer: $55 \times 1.65 = 90.75$ inches. Since this new, larger drawing still represents the same 1,045 actual miles, the new scale is $1,045 \div 90.75 \approx 11.52$ miles per inch.

Step-by-Step Solution:

Step 1: Calculate the total actual distance represented by the original drawing's side: $55 \text{ inches} \times 19 \text{ miles/inch} = 1,045$ miles.

Step 2: Recognize that this actual distance (1,045 miles) stays the SAME even when the drawing is resized — only the drawing's size changes, not the real place it represents.

Step 3: Calculate the new drawing's side length: the new side is 65% LONGER than the original 55 inches, so multiply by 1.65 ($100\% + 65\% = 165\%$, or 1.65 as a decimal): $55 \times 1.65 = 90.75$ inches.

Step 4: Since the new 90.75-inch side still represents the same 1,045 actual miles, calculate the new scale by dividing: $1,045 \div 90.75 \approx 11.5152$ miles per inch.

Step 5: Rounding to match the answer choices, this is closest to 11.52, matching choice B.

The Trap: Choice C (31.35) is the most commonly picked wrong answer — it typically results from incorrectly multiplying the ORIGINAL scale (19 miles/inch) by 1.65 directly, as if the scale itself grows by 65% ($19 \times 1.65 \approx 31.35$),

rather than recognizing that a LARGER drawing of the SAME place actually needs a SMALLER (not larger) miles-per-inch scale.

POE — Eliminate the other three:

- X A** — 6.65 comes from dividing 19 by something close to 1.65 backwards, or another inverted calculation — this doesn't match the correct scale-recalculation approach.
- X C** — 31.35 results from multiplying the original scale (19) by 1.65, treating the SCALE itself as growing — but a bigger drawing of the same place actually needs a SMALLER scale (fewer miles per inch), not a bigger one.
- X D** — 35.75 doesn't correspond to a clean derivation from this problem — likely an arithmetic slip or distractor value.

Solving with Desmos:

- Step 1: Open Desmos and calculate the fixed actual distance first: type 55×19 — this confirms 1,045 miles is the actual distance represented by the original drawing's side.
- Step 2: Calculate the new drawing's side length: type 55×1.65 — this confirms 90.75 inches for the larger drawing.
- Step 3: Calculate the new scale by dividing the FIXED actual distance by the NEW side length: type $1045 / 90.75$ — Desmos will evaluate this to approximately 11.5152, confirming the new scale is closest to 11.52 miles per inch, matching choice B.

Percentages — Q2 [Hard] (ID: 8bb94668)

Question ID: 8bb94668

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

The speeds of objects A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of object A is 5,900% of the speed of object C and the speed of object C is 0.008% of the speed of object B, which expression represents the value of $a + b$ in terms of c ?

Answer

- A. $1,840c$
- B. $5,908c$
- C. $6,025c$
- D. $12,559c$

Correct Answer: D

The Rule: When a problem describes percentages chained together across multiple variables (A is a percent of C, and C is a percent of B), convert EACH percentage into a decimal multiplier first, then use substitution to express everything in terms of a single variable.

The Tip: Convert every percentage to its decimal form immediately (5,900% becomes 59, and 0.008% becomes 0.00008) — working with percentages over 100% or under 1% in their original form is a common source of decimal-point errors; converting early keeps the rest of the algebra clean.

Why D is right: Since a is 5,900% of c , convert to decimal: $a = 59c$. Since c is 0.008% of b , convert to decimal: $c = 0.00008b$, which means $b = c / 0.00008 = 12,500c$. Adding: $a + b = 59c + 12,500c = 12,559c$.

Step-by-Step Solution:

- Step 1: Convert 'a is 5,900% of c' into an equation: 5,900% as a decimal is $5900/100 = 59$, so $a = 59c$.
- Step 2: Convert 'c is 0.008% of b' into an equation: 0.008% as a decimal is $0.008/100 = 0.00008$, so $c = 0.00008b$.
- Step 3: Solve this second equation for b (since we need b in terms of c to add it to a): $b = c / 0.00008$.
- Step 4: Calculate this division: $1/0.00008 = 12,500$, so $b = 12,500c$.

Step 5: Now add a and b, both now expressed in terms of c: $a + b = 59c + 12,500c$.

Step 6: Combine like terms: $59c + 12,500c = 12,559c$.

Step 7: This matches choice D.

The Trap: Choice A (1,840c) is the most commonly picked wrong answer — it typically results from a decimal-point error while converting 0.008% to a decimal, such as using 0.08 instead of the correct 0.00008, which throws off the entire calculation for b.

POE — Eliminate the other three:

X A — 1,840c likely results from a decimal-point misplacement while converting 0.008% — using too large a decimal value for this tiny percentage significantly changes the resulting value of b.

X B — 5,908c doesn't correctly combine both relationships — possibly only accounting for one of the two given percentage relationships instead of both.

X C — 6,025c also doesn't align with the correct chain of calculations — likely from a different decimal or arithmetic slip somewhere in converting one of the two percentages.

Solving with Desmos:

Step 1: Open Desmos and set up the relationships directly. First type: $a=59c$ to represent 'a is 5,900% of c.'

Step 2: Type: $c=0.00008b$ to represent 'c is 0.008% of b.'

Step 3: To find b in terms of c, type: $b=c/0.00008$ — Desmos will show this simplifies to $b = 12500c$ (you can verify by typing $1/0.00008$ separately, which evaluates to 12500).

Step 4: Finally, type $a+b$ using the c-based expressions: $59c+12500c$ — Desmos will confirm this combines to 12559c, matching choice D. You can also plug in a specific test value for c (like $c=1$) to see the numeric results directly: with $c=1$, $a=59$ and $b=12500$, so $a+b=12559$.

Percentages — Q3 [Hard] (ID: 6feae9c3)

Question ID: 6feae9c3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

At a recreation center, visitors used the exercise room for a total of t hours in July. At the same recreation center, visitors used the exercise room for $3.49t$ hours in August. What is the percent increase in the number of hours visitors used the exercise room from July to August?

Answer

- A. 2.49%
- B. 3.49%
- C. 249%
- D. 349%

Correct Answer: C

The Rule: Percent increase is always calculated as (the amount of increase) divided by (the ORIGINAL, starting value), then converted to a percentage by multiplying by 100 — never divide by the new/final value.

The Tip: When both the original and new values are given as multiples of the same variable (like t and $3.49t$ here), the variable will always cancel out completely in the percent-increase calculation — don't be intimidated by having a variable instead of a specific number; just follow the formula exactly as you would with numbers.

Why C is right: The increase in hours is $3.49t - t = 2.49t$. The percent increase is $(\text{increase} / \text{original}) \times 100 = (2.49t / t) \times 100 = 2.49 \times 100 = 249\%$.

Step-by-Step Solution:

- Step 1: Identify the original value (July hours): t .
- Step 2: Identify the new value (August hours): $3.49t$.
- Step 3: Calculate the amount of increase: new value – original value = $3.49t - t = 2.49t$.
- Step 4: Apply the percent increase formula: $(\text{increase} / \text{original value}) \times 100\%$.
- Step 5: Substitute the values: $(2.49t / t) \times 100\%$.
- Step 6: The t in the numerator and denominator cancels out completely: $2.49 \times 100\% = 249\%$.
- Step 7: This matches choice C.

The Trap: Choice B (3.49%) is the most commonly picked wrong answer — it mistakes the MULTIPLIER (3.49, representing the NEW total as a multiple of the original) for the percent INCREASE itself, without first subtracting 1 (or subtracting the original amount) to isolate just the increase.

POE — Eliminate the other three:

- X A** — 2.49% correctly identifies the coefficient of the increase (2.49) but forgets to convert this decimal into a percentage by multiplying by 100 — leaving it 100 times too small.
- X B** — 3.49% confuses the total NEW amount's multiplier (3.49, meaning August is 3.49 times July) with the percent INCREASE — but a multiplier of 3.49 actually represents a 249% increase (since $3.49 = 1 + 2.49$, meaning 100% of the original PLUS a 249% increase), not a 3.49% increase.
- X D** — 349% incorrectly uses the full multiplier 3.49 converted directly to a percentage (349%) without first subtracting the original 100% (representing the unchanged portion) to isolate just the increase.

Solving with Desmos:

- Step 1: Since this problem involves a variable (t) that cancels out, pick a simple test value to make it concrete — let's use $t = 100$ (100 hours in July). Step 2: Open Desmos and calculate August's hours: type 3.49×100 — this gives 349 hours in August.
- Step 3: Calculate the increase: type $349 - 100$ — this gives 249 hours of increase.
- Step 4: Calculate the percent increase using the original (July) value as the base: type $(249/100) \times 100$ — Desmos will show this evaluates to 249, confirming a 249% increase and matching choice C.
- Step 5: Try this with a different test value for t (like $t = 50$) to confirm the percentage stays the same regardless of which specific value you pick — this confirms the variable genuinely cancels out in the percent-increase formula.

Percentages — Q4 [Hard] [Grid-In] (ID: 98818acf)

Question ID: 98818acf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard ■■■■■

Question

The number of audiobooks in a library increased by 131% from last year to this year. The number of audiobooks in the library last year was n , and the number of audiobooks in the library this year is bn , where b and n are constants. What is the value of b ?

Correct Answer: 2.31 (Student-Produced Response — no answer choices given)

The Rule: When a quantity increases by $p\%$, the new total is always $(100\% + p\%)$ of the original — expressed as a decimal multiplier, this is $(1 + p/100)$. Never use just $p/100$ alone as the multiplier, since that would only represent the AMOUNT of increase, not the new TOTAL.

The Tip: Whenever you see 'increased by $X\%$ ' and need to find a multiplier that gives you the NEW TOTAL directly (not just the increase amount), always add 1 to $X/100$ — this combines the original 100% (which is 1, in decimal form) with the $X\%$ increase.

Why this is the answer: Since the number of audiobooks increased by 131% from last year (n) to this year, the new total is the original 100% PLUS the 131% increase, totaling 231% of the original, or 2.31 as a decimal multiplier. Since this year's count is given as bn , this means $b = 2.31$.

Step-by-Step Solution:

Step 1: Understand what 'increased by 131%' means: the new amount equals the original amount (100%) PLUS an additional 131% increase, for a total of $100\% + 131\% = 231\%$.

Step 2: Convert 231% into a decimal multiplier: $231/100 = 2.31$.

Step 3: Since the new amount is bn (b times the original n), and we just determined the new amount is 2.31 times the original, this means $b = 2.31$.

Step 4: This is a grid-in question — enter 2.31 as your final answer.

The Trap: A common error on this problem is using 1.31 instead of 2.31 — this happens when a student converts 131% directly to a decimal (1.31) without recognizing that this decimal already represents the FULL new total's relationship to the original as an increase multiplier, but forgetting that 'increased BY 131%' means adding this to the original 100%, or alternatively, using just 0.31 or 1.31 by misunderstanding which part represents the total versus just the increase.

Solving with Desmos:

Step 1: Since this problem involves the variable n , pick a simple test value to make it concrete — let's use $n = 100$ (100 audiobooks last year). Step 2: Open Desmos and calculate the increase amount: type $100 \cdot 1.31$ — this gives 131, representing the SIZE of the increase (131% of 100 is 131 additional audiobooks).

Step 3: Calculate this year's total by adding the increase to the original: type $100 + 131$ — this gives 231, the total number of audiobooks this year.

Step 4: Since this year's count is bn , and $n = 100$, solve for b : type $231/100$ — Desmos will show this equals 2.31, confirming $b = 2.31$ and matching your grid-in answer.

One-variable data: distributions and measures of center and spread

4 questions · Easy 1 / Medium 1 / Hard 2

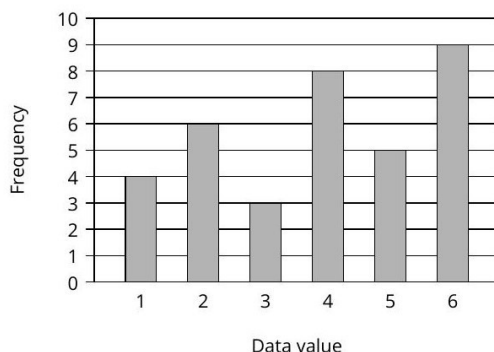
One-variable data: distributions and measures of center and spread — Q1 [Easy] [Grid-In] (ID: 559476f4)

Question ID: 559476f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The bar graph shows the frequency of each data value in a data set.



What is the frequency of data value 2 in this data set?

Correct Answer: 6 (Student-Produced Response — no answer choices given)

The Rule: On a bar graph, the frequency of any specific data value is read directly from the HEIGHT of that value's bar, measured against the vertical (frequency) axis.

The Tip: Trace straight up from the data value you're asked about on the horizontal axis until you hit the top of its bar, then trace straight across (left) to read the exact height on the vertical axis — this two-step 'trace up, then trace across' method prevents misreading a neighboring bar.

Why this is the answer: Looking at the bar directly above the data value '2' on the horizontal axis, its height reaches exactly 6 on the frequency (vertical) axis.

Step-by-Step Solution:

- Step 1: Locate '2' on the horizontal (data value) axis.
- Step 2: Trace straight up from that point until you reach the top of the bar directly above it.
- Step 3: From the top of that bar, trace straight across (left) to the vertical (frequency) axis to read its height.
- Step 4: The bar for data value 2 reaches a height of 6.
- Step 5: This is a grid-in question — enter 6 as your final answer.

The Trap: A common error on this type of problem is misreading an adjacent bar (like the bar for data value 1, which has height 4, or data value 3, which has height 3) instead of carefully tracing up from exactly '2' on the horizontal axis.

Solving with Desmos:

This is a straightforward graph-reading question that doesn't require Desmos to solve — the answer (6) is read directly from the bar graph provided in the question. However, if you want to double check your reading, you could recreate the bar heights as a list in Desmos (typing the data values 1 through 6 and their corresponding frequencies 4, 6, 3, 8, 5, 9 as a table) and confirm that the frequency listed for data value 2 is indeed 6.

Question ID: d73a67d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question

Which of the following lists represents a data set with the smallest standard deviation?

Answer

- A. 51, 52, 54, 56, 57
- B. 52, 52, 54, 56, 56
- C. 52, 53, 54, 55, 56
- D. 53, 54, 54, 54, 55

Correct Answer: D

The Rule: Standard deviation measures how SPREAD OUT a data set is from its mean — a data set with values tightly clustered close together (especially with repeated identical values) will always have a SMALLER standard deviation than one with values spread far apart, even if both sets share the same mean.

The Tip: You don't need to calculate the exact standard deviation for problems like this — just visually compare how 'spread out' or 'clustered together' each list's values are. The list with values closest together (and most repeated identical values) will have the smallest standard deviation.

Why D is right: List D (53, 54, 54, 54, 55) has values clustered extremely tightly around 54, with three of the five values being EXACTLY 54 — this tight clustering around the mean gives it the smallest spread, and therefore the smallest standard deviation, compared to the other lists.

Step-by-Step Solution:

- Step 1: Calculate (or estimate) the mean of each list — notice all four lists happen to have a mean of 54.
- Step 2: Since all means are equal, compare how far each list's individual values deviate from 54 — the list with the SMALLEST deviations overall will have the smallest standard deviation.
- Step 3: List A (51,52,54,56,57): deviations of -3,-2,0,2,3 — fairly spread out.
- Step 4: List B (52,52,54,56,56): deviations of -2,-2,0,2,2 — moderately spread out.
- Step 5: List C (52,53,54,55,56): deviations of -2,-1,0,1,2 — more tightly clustered.
- Step 6: List D (53,54,54,54,55): deviations of -1,0,0,0,1 — extremely tightly clustered, with three values exactly AT the mean.
- Step 7: List D has by far the smallest deviations from the mean, meaning it has the smallest standard deviation.
- Step 8: This matches choice D.

The Trap: Choice C is the most commonly picked wrong answer — its values are evenly and closely spaced (52 through 56, each one apart), which LOOKS tightly clustered, but list D's heavy repetition of the exact mean value (54 appearing three times) actually creates even less spread than C's evenly-spaced-but-still-varying values.

POE — Eliminate the other three:

- X A** — List A has the widest range of values (51 to 57, a spread of 6) among all four lists, giving it the LARGEST standard deviation, not the smallest.
- X B** — List B has a moderate spread (52 to 56), larger than both C and D — its values aren't as tightly clustered around the mean as list D's.
- X C** — List C has evenly spaced values with no repeats, giving it a real but moderate spread — list D's heavy repetition of the exact mean (54 three times) creates even less variability than C's evenly distributed values.

Solving with Desmos:

- Step 1: Open Desmos and enter each list as a set of points to calculate statistics. For list D, type: `stdev({53,54,54,54,55})`

Step 2: Press Enter — Desmos will calculate and display the exact standard deviation for this list.

Step 3: Repeat this same process for lists A, B, and C: $\text{stdev}(\{51,52,54,56,57\})$, $\text{stdev}(\{52,52,54,56,56\})$, and $\text{stdev}(\{52,53,54,55,56\})$

Step 4: Compare all four calculated standard deviations directly — you'll see list D's value is clearly the smallest of the four, confirming choice D without needing to estimate visually.

One-variable data: distributions and measures of center and spread — Q3 [Hard] (ID: f6c18a66)

Question ID: f6c18a66

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of 11 days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 10 data values.

Which statement best compares the median of data set A and the median of data set B?

Answer

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

Correct Answer: C

The Rule: The median of a data set is the MIDDLE value when all values are arranged in order (for an odd number of values), or the AVERAGE of the two middle values (for an even number of values). Removing a value that's far from the middle (like an outlier or error) often doesn't change the median at all.

The Tip: When a data set is described using a frequency table (value paired with how many times it occurs), first 'unroll' it into a full ordered list before trying to find the median — this makes it much easier to correctly count to the middle position(s), especially for larger data sets.

Why C is right: Data set A (11 values, unrolled from the table): 0, 0, 3, 3, 4, 4, 5, 5, 6, 7, 13. With 11 values, the median is the 6th value: counting in order, this is 4. After removing the erroneous 13, data set B (10 values): 0, 0, 3, 3, 4, 4, 5, 5, 6, 7. With 10 values (even), the median is the average of the 5th and 6th values: both are 4, so the median is $(4+4)/2 = 4$. Since both medians equal 4, they are equal.

Step-by-Step Solution:

Step 1: 'Unroll' the frequency table into a full ordered list for data set A: two 0's, two 3's, two 4's, two 5's, one 6, one 7, one 13 — giving: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7, 13 (11 values total).

Step 2: Find the median of data set A: with 11 values (an odd number), the median is the middle (6th) value. Counting through the ordered list: 1st=0, 2nd=0, 3rd=3, 4th=3, 5th=4, 6th=4. The median of set A is 4.

Step 3: Create data set B by removing the erroneous value 13: 0, 0, 3, 3, 4, 4, 5, 5, 6, 7 (10 values total).

Step 4: Find the median of data set B: with 10 values (an even number), the median is the average of the two middle values (the 5th and 6th). Counting through: 5th=4, 6th=4. The median of set B is $(4+4)/2 = 4$.

Step 5: Compare the two medians: both data set A and data set B have a median of 4 — they are EQUAL.

Step 6: This matches choice C.

The Trap: Choice A is the most commonly picked wrong answer — students sometimes assume that removing the largest value (13) from a data set must automatically lower the median, without actually counting through the ordered list to check whether 13 was anywhere near the middle position in the first place (it wasn't — it was the very last, most extreme value).

POE — Eliminate the other three:

- X A** — Removing 13 (the LARGEST value, at the very end of the ordered list) doesn't affect which values sit in the MIDDLE of the data set — the median stays the same, not lower.
- X B** — Similarly, there's no reason removing the largest, most extreme value would cause the median to increase — the middle values of the data set are unaffected by removing an extreme outlier.
- X D** — Since the data set is fully known and given in the table (not an estimate or a sample), there IS enough information to precisely calculate and compare both medians — this choice incorrectly claims otherwise.

Solving with Desmos:

Step 1: Open Desmos and enter data set A as a full list based on the frequency table: `median({0,0,3,3,4,4,5,5,6,7,13})`

Step 2: Press Enter — Desmos will calculate and display the median, confirming it equals 4.

Step 3: On the next line, enter data set B (with 13 removed): `median({0,0,3,3,4,4,5,5,6,7})`

Step 4: Press Enter — Desmos will show this median is also 4, confirming both medians are equal and verifying choice C.

One-variable data: distributions and measures of center and spread — Q4 **[Hard]** (ID: cf2be18d)

Question ID: cf2be18d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$$a, 26, 29, b, 31, 47, c$$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is 36, the median is 29, and the range is 72. What is the value of c ?

Answer

- A. 54
- B. 72
- C. 81
- D. 98

Correct Answer: C

The Rule: When a data set has unknown values mixed with known ones, use the GIVEN statistics (mean, median, range) as separate equations — the mean gives you a SUM equation, the median tells you which position (or unknown) sits in the middle, and the range gives you a MAX-MINUS-MIN equation. Solve this small system of equations together.

The Tip: Since the data set is already given in ascending order, use that order to immediately identify which position corresponds to the median (the exact middle value for an odd-count list) — this often directly tells you the value of one of the unknowns without any calculation.

Why C is right: With 7 values in ascending order (a, 26, 29, b, 31, 47, c), the median is the 4th value, which is b. Since the median is given as 29, this means $b = 29$. The mean is 36, so the sum of all 7 values is $7 \times 36 = 252$. Substituting known values: $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which simplifies to $a + c + 162 = 252$, so $a + c = 90$. The range is 72, and since the list is in ascending order, the range equals the largest value minus the smallest: $c - a = 72$. Now solve the system: $a + c = 90$ and $c - a = 72$. Adding these two equations: $2c = 162$, so $c = 81$.

Step-by-Step Solution:

Step 1: Since the 7 values are listed in ascending order (a, 26, 29, b, 31, 47, c), identify the median's position: for 7 values, the median is the 4th (middle) value, which is b.

Step 2: Since the median is given as 29, set $b = 29$.

Step 3: Use the mean to find the sum of all values: $\text{mean} \times \text{count} = \text{sum}$, so $36 \times 7 = 252$.

Step 4: Substitute all known values into the sum equation: $a + 26 + 29 + 29 + 31 + 47 + c = 252$.

Step 5: Add up the known numbers: $26 + 29 + 29 + 31 + 47 = 162$. The equation becomes: $a + c + 162 = 252$.

Step 6: Solve for $a + c$: $a + c = 252 - 162 = 90$.

Step 7: Use the range (largest minus smallest): since the list is ascending, the range is $c - a = 72$.

Step 8: Now solve this system of two equations: $a + c = 90$ and $c - a = 72$. Add both equations together: $(a+c) + (c-a) = 90 + 72$, which simplifies to $2c = 162$.

Step 9: Divide both sides by 2: $c = 81$.

Step 10: This matches choice C.

The Trap: Choice A (54) is the most commonly picked wrong answer — it typically results from solving only one of the two equations (like using just the range equation with an incorrect assumed value for a) without properly setting up and solving the full system using both the mean AND range information together.

POE — Eliminate the other three:

✗ A — 54 doesn't satisfy both the sum equation ($a+c=90$) and the range equation ($c-a=72$) simultaneously — it likely comes from an incomplete calculation using only one piece of given information.

✗ B — 72 is actually the value of the RANGE itself (given in the problem), not the value of c — a student might mistakenly select the given range number instead of solving for c.

✗ D — 98 doesn't satisfy the system of equations either — if $c = 98$, then from $c - a = 72$, a would be 26, but then $a + c = 26 + 98 = 124$, which doesn't match the required sum of 90.

Solving with Desmos:

Step 1: Open Desmos and set up the two equations from the problem as a system. First type: $a+c=90$

Step 2: Type the second equation: $c-a=72$

Step 3: Desmos will treat these as two lines if you use x and y instead — replace a with x and c with y: $x+y=90$ and $y-x=72$

Step 4: Press Enter after each — Desmos graphs both lines and shows their intersection point when you click on it.

Step 5: The intersection point will show $x = 9$ (the value of a) and $y = 81$ (the value of c), confirming $c = 81$ and matching choice C.

Two-variable data: models and scatterplots

3 questions · Easy 1 / Medium 2 / Hard 0

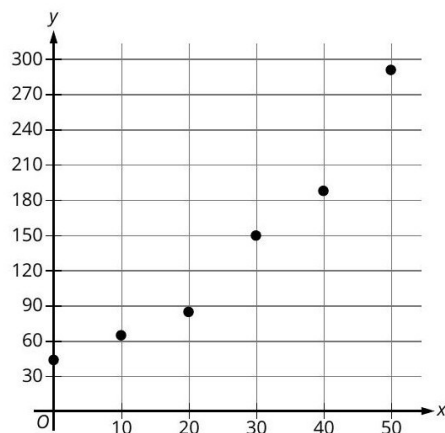
Two-variable data: models and scatterplots — Q1 [Easy] (ID: dbd89c59)

Question ID: dbd89c59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question

The scatterplot shows the relationship between the number of swans, y , in a certain population and the number of years since 1970, x , where $0 \leq x \leq 50$.



Which of the following equations is the most appropriate exponential model for the data shown in the scatterplot?

Answer

A. $y = 43(1.04)^{-x}$

B. $y = 43(1.04)^x$

C. $y = 288(1.04)^{-x}$

D. $y = 288(1.04)^x$

Correct Answer: B

The Rule: For an exponential model $y = a(b)^x$, the constant 'a' (the coefficient out front) represents the STARTING value (the y-intercept, where $x = 0$), and the base 'b' tells you whether the data is growing ($b > 1$) or decaying ($b < 1$). Match both features to the scatterplot's visual pattern.

The Tip: Always check the y-intercept FIRST on an exponential scatterplot — it should closely match the leading coefficient 'a' in the correct equation. Then check whether the data is rising or falling overall to determine if the base should be greater than or less than 1.

Why B is right: The scatterplot shows data starting at approximately $y = 43$ when $x = 0$, and clearly INCREASING (growing) as x increases up to 50. This matches $a = 43$ (the starting value) and a base greater than 1 with a POSITIVE exponent $(1.04)^x$, giving $y = 43(1.04)^x$.

Step-by-Step Solution:

Step 1: Read the y-intercept directly from the scatterplot: the leftmost point (at $x = 0$) sits at approximately $y = 43$.

Step 2: Observe the overall trend of the data: as x increases from 0 to 50, y clearly increases (grows) from about 43 up to nearly 290 — this is GROWTH, not decay.

Step 3: Match these observations to the exponential model form $y = a(b)^x$: the starting value a should be approximately 43, and since the data is growing, the exponent should be POSITIVE x (not negative x , which would represent decay).

Step 4: This matches choice B: $y = 43(1.04)^x$.

The Trap: Choice A is the most commonly picked wrong answer — it uses the correct starting value (43) but a NEGATIVE exponent ($-x$), which would actually make the function DECREASE as x increases, contradicting the clearly increasing pattern shown in the scatterplot.

POE — Eliminate the other three:

- ✗ **A** — The negative exponent ($-x$) would make this function DECREASE as x increases, but the scatterplot clearly shows the data values GROWING over time — this doesn't match the visual trend.
- ✗ **C** — Starting at 288 doesn't match the scatterplot's y -intercept, which is clearly close to 43 (the point at $x=0$), not near 288 — this starting value belongs to a very different curve.
- ✗ **D** — Similarly, this uses 288 as the starting value, which doesn't match the scatterplot's actual y -intercept of approximately 43 — the growth direction (positive exponent) is correct, but the starting point is wrong.

Solving with Desmos:

- Step 1: Open Desmos and type the candidate equation: $y=43(1.04)^x$
- Step 2: Press Enter — Desmos graphs this curve.
- Step 3: Compare this graph to the scatterplot given in the question — check that the curve passes close to the plotted points, starting near $(0, 43)$ and rising to approximately $(50, 290)$.
- Step 4: If the curve fits the scattered points well, you've confirmed choice B. You can also try graphing the other choices (like $y=288(1.04)^x$) to see how poorly they fit the actual data points, reinforcing why B is the best model.

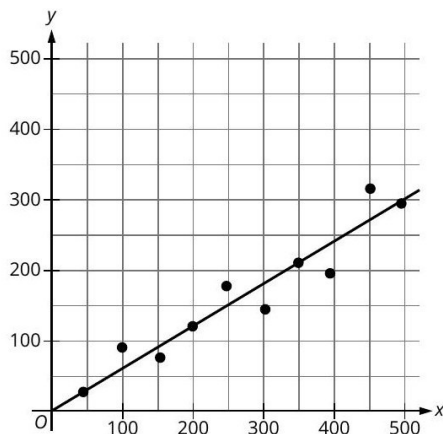
Two-variable data: models and scatterplots — Q2 [Medium] (ID: b3547a10)

Question ID: b3547a10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

A group of 10 botanists recorded data on the germination rates of a certain type of seed for one growing season. The scatterplot shows the relationship between the number of seeds planted, x , and the number of seeds that germinated, y , for each of the botanists. A line of best fit is also shown.



Which of the following is the best interpretation of the slope of the line of best fit in this context?

Answer

- A. The number of seeds planted is predicted to increase by 60 seeds every 100 days.
- B. The number of seeds planted is predicted to increase by 300 seeds every 100 days.
- C. The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.
- D. The number of seeds that germinate is predicted to increase by 300 seeds for every additional 100 seeds that are planted.

Correct Answer: C

The Rule: The slope of a line of best fit represents the RATE OF CHANGE between the two variables graphed — specifically, how much the DEPENDENT variable (on the y -axis) changes for every one-unit increase in the

INDEPENDENT variable (on the x-axis). Always identify which variable is on which axis before interpreting the slope in context.

The Tip: Read the axis labels carefully before interpreting slope — the correct interpretation must have the y-axis variable as the one that 'increases' and the x-axis variable as the one changing 'per' some amount. Answer choices that swap these two roles are testing whether you're reading the graph correctly.

Why C is right: On this scatterplot, x represents the number of seeds PLANTED, and y represents the number of seeds that GERMINATED. The slope of the line of best fit, estimated from the graph (rising from about (0,30) to (500,310), giving a slope of about $280/500 = 0.56$, closest to $60/100 = 0.6$), means that for every 100 additional seeds planted, approximately 60 more seeds are predicted to germinate.

Step-by-Step Solution:

Step 1: Identify which variable is on the x-axis (independent/input variable) and which is on the y-axis (dependent/output variable): x = number of seeds planted, y = number of seeds that germinated.

Step 2: Estimate two points on the line of best fit from the graph: approximately (0, 30) and (500, 310).

Step 3: Calculate the slope using these two points: $\text{slope} = (310 - 30) / (500 - 0) = 280/500 = 0.56$.

Step 4: Compare this slope to the answer choices, which are given as 'X seeds per 100 seeds': 0.56 is closest to $60/100 = 0.60$.

Step 5: Since the slope relates y (germinated) to x (planted), the correct interpretation must describe germinated seeds increasing per planted seeds — not the reverse.

Step 6: This matches choice C: 'The number of seeds that germinate is predicted to increase by 60 seeds for every additional 100 seeds that are planted.'

The Trap: Choice A is the most commonly picked wrong answer — it has the correct approximate slope value (60 per 100) but describes the WRONG variable as increasing (seeds planted, the independent/x-axis variable, rather than seeds germinated, the dependent/y-axis variable) — this reverses the cause-and-effect direction the slope actually represents.

POE — Eliminate the other three:

X A — This describes 'seeds planted' as the variable that increases — but seeds planted is the INDEPENDENT variable (x-axis), which the researchers choose directly; it's seeds GERMINATED (y-axis) that responds to and increases with planting, not the reverse.

X B — This has the same variable-reversal issue as A, plus uses the wrong slope magnitude (300 instead of approximately 60) — doesn't match the graph's actual slope.

X D — This correctly identifies germinated seeds as the increasing variable but uses the wrong slope magnitude (300 per 100, which would be a much steeper line than what's shown) — the actual slope is closer to 60 per 100, not 300.

Solving with Desmos:

Step 1: Open Desmos and estimate several points from the scatterplot to check the line of best fit's slope. Type a few points you can read from the graph, like: (100,90), (200,120), (350,215), (450,315)

Step 2: Desmos will plot these points. To find a best-fit line, click the '+' button and look for a regression feature, or manually type: $y \sim mx + b$ and Desmos will calculate the best-fit slope (m) and intercept (b) automatically using linear regression.

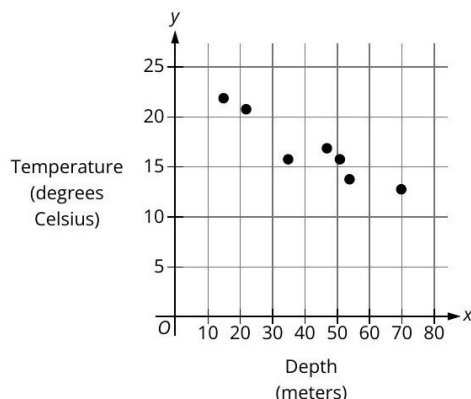
Step 3: Check the calculated slope value — it should be close to 0.56 to 0.6, confirming that for every 100 seeds planted (x), about 60 seeds are predicted to germinate (y), matching choice C's interpretation.

Question ID: 70a2776f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

As part of a study of lake conditions, the water temperature was recorded at different depths below the surface of a certain lake. The data shown in the scatterplot give the recorded temperature, in degrees Celsius, for 7 depths, in meters, below the surface of the lake.



Which of the following is closest to the slope of a line of best fit for the data shown?

Answer

- A. -0.17
- B. -2.17
- C. -6.02
- D. -8.02

Correct Answer: A

The Rule: To estimate the slope of a line of best fit from a scatterplot, choose two points that appear to represent the OVERALL trend of the data (not just two random points), calculate the slope between them using (change in y) / (change in x), and match this to the closest answer choice.

The Tip: When estimating slope from a scatterplot with a downward trend, pick one point from the far LEFT side of the data and one from the far RIGHT side — using points that are far apart minimizes the impact of any single point's imprecise reading, giving a more reliable slope estimate.

Why A is right: Using an early point around (15, 22) and a later point around (70, 13), the slope is estimated as $(13 - 22) / (70 - 15) = -9/55 \approx -0.16$, which is closest to -0.17 among the answer choices.

Step-by-Step Solution:

- Step 1: Choose two points that represent the overall trend of the scattered data — one from early depths and one from later depths. Reading the graph, approximate points are (15, 22) and (70, 13).
- Step 2: Calculate the change in y (temperature): $13 - 22 = -9$.
- Step 3: Calculate the change in x (depth): $70 - 15 = 55$.
- Step 4: Calculate the slope: $-9 / 55 \approx -0.1636$.
- Step 5: Compare this estimated slope to the answer choices: -0.1636 is closest to -0.17 .
- Step 6: This matches choice A.

The Trap: Choice B (-2.17) is the most commonly picked wrong answer — it's off by roughly a factor of 10-13 from the correct slope, which can happen if a student misreads the axis scale (for example, confusing gridline spacing) while estimating the change in x or y .

POE — Eliminate the other three:

- X B** — -2.17 is far too steep for this data — this magnitude of slope would mean temperature drops over 150 degrees across the full depth range shown, which doesn't match the actual data (which only spans about 22 to 13 degrees).
- X C** — -6.02 is even more drastically too steep — this doesn't match the gentle downward trend visible in the scattered points.
- X D** — -8.02 is also far too steep for the gentle decline shown in the data — likely from a significant misreading of the graph's scale or points.

Solving with Desmos:

- Step 1: Open Desmos and enter several points estimated from the scatterplot: (15,22), (20,21), (35,16), (45,17), (50,16), (52,14), (70,13)
- Step 2: Desmos will plot these points. To find the actual best-fit slope, type: $y \sim mx + b$ — Desmos will use linear regression to calculate the best-fit values for m (slope) and b (intercept) automatically.
- Step 3: Check the calculated value of m — it should be close to -0.17 , confirming your estimate and matching choice A.

Probability and conditional probability

2 questions · Easy 0 / Medium 0 / Hard 2

Probability and conditional probability — Q1 [Hard] [Grid-In] (ID: 89ff6a0a)

Question ID: 89ff6a0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

For 100 buttons, the table summarizes the distribution by group and diameter.

Group	Diameter (millimeters)		
	Less than 20	20 to 30	Greater than 30
Group 1	12	8	3
Group 2	0	17	18
Group 3	10	32	0

One of these buttons will be selected at random. What is the probability of selecting a button with a diameter that is less than or equal to 30 millimeters, given that it is **not** in group 2? (Express your answer as a decimal or fraction, not as a percent.)

Correct Answer: 62/65 (Student-Produced Response — no answer choices given)

The Rule: For conditional probability problems presented in a table ('probability of X given Y'), first identify and total ONLY the rows/columns that satisfy the GIVEN condition (the 'given that' part), then find what fraction of that reduced total also satisfies the main event being asked about.

The Tip: Physically cross out or ignore any table cells that DON'T meet the 'given that' condition before calculating anything — this prevents accidentally including irrelevant data in either your numerator or denominator.

Why this is the answer: 'Given that it is NOT in group 2' means we only consider Group 1 and Group 3 (ignoring Group 2 entirely). Group 1 total: $12+8+3 = 23$. Group 3 total: $10+32+0 = 42$. Combined total (not group 2): $23+42 = 65$. Within this reduced group, 'diameter ≤ 30 mm' means the 'less than 20' and '20 to 30' columns: Group 1: $12+8=20$; Group 3: $10+32=42$. Combined: $20+42 = 62$. The probability is $62/65$.

Step-by-Step Solution:

- Step 1: Identify the condition ('given that it is NOT in group 2'): this means we only look at Group 1 and Group 3 rows, completely ignoring Group 2.

- Step 2: Calculate the total number of buttons in Group 1: $12 + 8 + 3 = 23$.
- Step 3: Calculate the total number of buttons in Group 3: $10 + 32 + 0 = 42$.
- Step 4: Calculate the combined total for 'not in group 2': $23 + 42 = 65$. This is our new denominator.
- Step 5: Now find how many of these 65 buttons have a diameter $\leq 30\text{mm}$ — this means the 'Less than 20' AND '20 to 30' columns (both are $\leq 30\text{mm}$), but NOT the 'Greater than 30' column.
- Step 6: From Group 1: 'Less than 20' = 12, '20 to 30' = 8. Sum: $12 + 8 = 20$.
- Step 7: From Group 3: 'Less than 20' = 10, '20 to 30' = 32. Sum: $10 + 32 = 42$.
- Step 8: Combine these: $20 + 42 = 62$. This is our numerator.
- Step 9: The probability is $62/65$ (this fraction cannot be simplified further, since $62 = 2 \times 31$ and $65 = 5 \times 13$ share no common factors).
- Step 10: This is a grid-in question — enter $62/65$ (or its decimal equivalent, approximately 0.954) as your final answer.

The Trap: A common error on this problem is using the TOTAL of all 100 buttons as the denominator (instead of just the 65 buttons that are 'not in group 2'), which incorrectly calculates a regular probability instead of the required **CONDITIONAL** probability.

Solving with Desmos:

- Step 1: Open Desmos as a calculator to organize this conditional probability calculation. Type: $12+8+3$ to confirm Group 1's total is 23.
- Step 2: Type: $10+32+0$ to confirm Group 3's total is 42.
- Step 3: Type: $23+42$ to confirm the 'not group 2' total is 65 (your denominator).
- Step 4: Type: $12+8+10+32$ to confirm the count with diameter $\leq 30\text{mm}$ within 'not group 2' is 62 (your numerator).
- Step 5: Type: $62/65$ — Desmos will show this as approximately 0.9538, confirming your fraction and allowing you to enter either $62/65$ or the decimal into the grid-in box.

Probability and conditional probability — Q2 **[Hard]** (ID: f496d2c0)

Question ID: f496d2c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	Hard

Question

The table shows the distribution of rooms in a certain facility by seating capacity.

Seating capacity	Proportion
Less than 18 seats	26%
18–40 seats	21%
41–65 seats	29%
Greater than 65 seats	24%

If a room in this facility is selected at random, which of the following is closest to the probability of selecting a room that has a seating capacity greater than 65 seats, given that the room has a seating capacity of at least 18 seats?

Answer

- A. 0.24
- B. 0.32
- C. 0.50
- D. 0.92

Correct Answer: B

The Rule: For conditional probability problems with percentages, first add up ONLY the percentages that satisfy the GIVEN condition to find the new total, then divide the specific percentage you're looking for by this new (reduced) total — not by 100%.

The Tip: Rewrite the 'given that' condition as its own smaller, self-contained group first — add up just the relevant percentages to get a new 'whole' for that group, then calculate the requested probability as a fraction of THAT smaller whole, not the original 100%.

Why B is right: 'Given that the room has a seating capacity of at least 18 seats' means combining the '18–40 seats,' '41–65 seats,' and 'greater than 65 seats' categories: $21\% + 29\% + 24\% = 74\%$. Within this 74% group, the probability of 'greater than 65 seats' is $24\% \div 74\% \approx 0.3243$, which rounds to approximately 0.32.

Step-by-Step Solution:

Step 1: Identify the given condition: 'seating capacity of at least 18 seats' — this includes the '18–40 seats,' '41–65 seats,' and 'greater than 65 seats' categories (everything except 'less than 18 seats').

Step 2: Add up the percentages for these three categories to find the new total: $21\% + 29\% + 24\% = 74\%$.

Step 3: Identify what we're looking for within this reduced group: rooms with 'greater than 65 seats,' which is 24%.

Step 4: Calculate the conditional probability by dividing the specific percentage by the new total: $24\% / 74\% \approx 0.3243$.

Step 5: Rounding to match the answer choices, this is closest to 0.32.

Step 6: This matches choice B.

The Trap: Choice A (0.24) is the most commonly picked wrong answer — it uses the original 24% directly as the answer, forgetting to divide by the reduced total (74%) that accounts for the 'given that' condition — this would be the correct answer only if the question asked for the UNCONDITIONAL probability across all rooms, not the conditional probability.

POE — Eliminate the other three:

X A — 0.24 is simply the original percentage for 'greater than 65 seats' without adjusting for the 'given that' condition — this ignores the conditional nature of the question entirely.

X C — 0.50 doesn't correspond to a correct calculation from this table — likely a distractor or a significant miscalculation.

X D — 0.92 might come from dividing the wrong two percentages together, or from a completely different (incorrect) combination of the given values.

Solving with Desmos:

Step 1: Open Desmos and calculate the new total for the 'given' condition: type $21+29+24$ — this confirms 74% is the combined total for rooms with at least 18 seats.

Step 2: Calculate the conditional probability: type $24/74$ — Desmos will evaluate this to approximately 0.3243.

Step 3: Compare this decimal to the answer choices — 0.3243 rounds to 0.32, confirming choice B.

Inference from sample statistics and margin of error

3 questions · Easy 0 / Medium 2 / Hard 1

Inference from sample statistics and margin of error — Q1 [Medium] (ID: 3e1bd4e2)

Question ID: 3e1bd4e2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

Researchers studied a sample of a species of fish from a certain location. The researchers found that the mean length of fish in the sample was 16.59 millimeters. Based on the sample mean and margin of error, the researchers estimated that the mean length of all fish of this species from this location was between 15.86 and 17.32 millimeters. What is the margin of error, in millimeters?

Answer

- A. 0.73
- B. 2.19
- C. 15.86
- D. 16.59

Correct Answer: A

The Rule: A confidence interval (the range given for an estimate) is always constructed as (sample estimate) $\pm$ (margin of error). This means the margin of error is exactly HALF the total width of the interval, or equivalently, the distance from the sample mean to either endpoint of the interval.

The Tip: To find the margin of error from a given interval, you don't need the full formula — simply subtract the sample mean from the UPPER endpoint (or subtract the LOWER endpoint from the sample mean); both calculations give you the same margin of error value.

Why A is right: The sample mean is 16.59mm, and the interval's upper bound is 17.32mm. The margin of error is the difference: $17.32 - 16.59 = 0.73$. (You can double check: the lower bound is $16.59 - 0.73 = 15.86$, which matches the given lower bound exactly.)

Step-by-Step Solution:

- Step 1: Identify the sample mean (the central estimate): 16.59 millimeters.
- Step 2: Identify one endpoint of the confidence interval — let's use the upper bound: 17.32 millimeters.
- Step 3: Calculate the margin of error by finding the difference between the upper bound and the sample mean: $17.32 - 16.59 = 0.73$.
- Step 4: Double-check using the lower bound: sample mean – margin of error should equal the lower bound: $16.59 - 0.73 = 15.86$, which matches the given lower bound exactly, confirming the calculation.
- Step 5: This matches choice A.

The Trap: Choice C (15.86) is the most commonly picked wrong answer — it directly uses the LOWER BOUND of the interval as if it were the margin of error itself, rather than correctly calculating the DIFFERENCE between the sample mean and that bound.

POE — Eliminate the other three:

- X B** — 2.19 doesn't correspond to a direct calculation from the given numbers — possibly from adding instead of subtracting, or another arithmetic slip.
- X C** — 15.86 is simply the LOWER BOUND of the confidence interval itself, not the margin of error — the margin of error is the DISTANCE between the mean and this bound, not the bound's value directly.
- X D** — 16.59 is simply the SAMPLE MEAN itself, not the margin of error — selecting this confuses the central estimate with the amount of uncertainty around it.

Solving with Desmos:

- Step 1: Open Desmos and calculate the margin of error directly: type $17.32 - 16.59$
- Step 2: Press Enter — Desmos will evaluate this to 0.73, confirming the margin of error and matching choice A.
- Step 3: As a double check, verify using the lower bound instead: type $16.59 - 15.86$ — this should ALSO give 0.73, confirming your answer is consistent from both directions of the interval.

Inference from sample statistics and margin of error — Q2 [Medium] (ID: 2bab1399)

Question ID: 2bab1399

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Medium

Question

A researcher is designing a study to investigate the average number of hours participants in a training course spend on a particular activity per day. The researcher will report an estimated average number of hours participants in the training course spend on this activity per day with an associated margin of error. The researcher is considering using a random sample of either 115 or 230 participants from the training course. Which of the following would be the most likely effect of using the larger random sample compared to the smaller random sample?

Answer

- A. The reported margin of error would be lower.
- B. The reported margin of error would be higher.
- C. The reported average number of hours would be lower.
- D. The reported average number of hours would be higher.

Correct Answer: A

The Rule: In statistical sampling, INCREASING the sample size always tends to DECREASE the margin of error (making the estimate more precise), while decreasing the sample size increases the margin of error. This is because larger samples give a more reliable, representative picture of the full population.

The Tip: Remember this general relationship as an inverse one: sample size UP means margin of error DOWN, and sample size DOWN means margin of error UP. This relationship holds true across virtually all standard statistical sampling methods, so you can answer these conceptual questions without any calculation.

Why A is right: Since 230 participants is a LARGER sample than 115 participants (twice as large), using this larger sample would be expected to produce a MORE precise (narrower) estimate — meaning a LOWER margin of error compared to the smaller sample.

Step-by-Step Solution:

Step 1: Compare the two sample sizes being considered: 115 participants versus 230 participants.

Step 2: Recognize that 230 is the LARGER of the two sample sizes (twice as large as 115).

Step 3: Apply the general statistical principle: larger sample sizes lead to more precise estimates, which means a SMALLER (lower) margin of error.

Step 4: Since the question asks about the effect of using the LARGER sample (230) compared to the smaller one (115), the expected effect is a LOWER margin of error.

Step 5: This matches choice A.

The Trap: Choice B is the most commonly picked wrong answer — it reverses the correct relationship, assuming that MORE data somehow leads to a HIGHER margin of error (more 'noise' or uncertainty), when in fact more data generally REDUCES uncertainty and produces a tighter, more reliable estimate.

POE — Eliminate the other three:

X B — This reverses the true relationship between sample size and margin of error — larger samples produce LOWER margins of error (more precision), not higher ones.

X C — The sample size affects the PRECISION of the estimate (the margin of error), not necessarily the reported average value itself — a larger sample doesn't systematically make the average number of hours lower.

X D — Similarly, sample size doesn't systematically make the reported average HIGHER either — it affects how precise (narrow) the margin of error is around whatever the true average happens to be.

Solving with Desmos:

This is a conceptual statistics question about the relationship between sample size and margin of error, which doesn't require a specific calculation in Desmos. However, if you want to explore this relationship numerically, margin of error in many standard formulas is proportional to $1/\sqrt{\text{sample size}}$ — you could type $\sqrt{115}$ and $\sqrt{230}$ into Desmos to compare: you'll see $\sqrt{230}$ is larger than $\sqrt{115}$, and since margin of error is roughly proportional to 1 divided by this square root, a LARGER sample size (230) corresponds to a SMALLER margin of error, confirming choice A conceptually.

Inference from sample statistics and margin of error — Q3 [Hard] (ID: da978ee6)

Question ID: da978ee6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

A researcher surveyed a random sample of 1,020 people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that 86% of people from the city support this proposal, with an associated margin of error of 2.13%. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

Answer

- A. The margin of error is between 2.13% and 3.13%.
- B. The margin of error is between 4.13% and 5.13%.
- C. The margin of error is greater than 5.13%.
- D. The margin of error is less than 2.13%.

Correct Answer: D

The Rule: For simple random sampling, the margin of error is inversely proportional to the SQUARE ROOT of the sample size — meaning if you increase the sample size by some factor, the margin of error decreases by the square root of that same factor, not by the factor itself.

The Tip: When a sample size DOUBLES, don't assume the margin of error simply halves — because of the square-root relationship, doubling the sample size only shrinks the margin of error by a factor of $\sqrt{2}$ (about 1.41), not by a full factor of 2. Recognizing this square-root relationship (rather than a direct proportional one) is the key insight for these problems.

Why D is right: Doubling the sample size from 1,020 to 2,040 people means the margin of error decreases by a factor of $\sqrt{2} \approx 1.414$. So the new margin of error is approximately $2.13\% \div 1.414 \approx 1.506\%$, which is LESS than the original 2.13%.

Step-by-Step Solution:

Step 1: Identify the relationship between sample size and margin of error: margin of error is inversely proportional to the square root of the sample size.

Step 2: Identify the change in sample size: the new sample is DOUBLE the original ($2 \times 1,020 = 2,040$ people).

Step 3: Calculate the factor by which the margin of error changes: since sample size increased by a factor of 2, margin of error decreases by a factor of $\sqrt{2} \approx 1.414$.

Step 4: Calculate the new (approximate) margin of error: $2.13\% \div 1.414 \approx 1.506\%$.

Step 5: Compare this new margin of error (approximately 1.51%) to the original (2.13%) and to the answer choices — 1.51% is LESS than 2.13%, matching the description in choice D.

Step 6: This matches choice D.

The Trap: Choice A is the most commonly picked wrong answer — it assumes the margin of error will still be reasonably close to the original value (between 2.13% and 3.13%, even higher than before), missing that a LARGER sample size should DECREASE the margin of error, not increase or maintain it.

POE — Eliminate the other three:

✗ A — This suggests the margin of error would be HIGHER than or similar to the original (2.13% to 3.13%) — but doubling the sample size should DECREASE the margin of error, not increase it.

✗ B — This suggests an even larger margin of error (4.13% to 5.13%) — this strongly contradicts the fundamental principle that larger samples produce smaller, more precise margins of error.

✗ C — This suggests the margin of error would be extremely high (greater than 5.13%) — this is the opposite of what should happen with a larger, more representative sample size.

Solving with Desmos:

Step 1: Open Desmos and calculate the square-root scaling factor: type $\sqrt{2}$ — this confirms the factor is approximately 1.414 (since the sample size doubled).

Step 2: Calculate the new margin of error by dividing the original by this factor: type $2.13/\sqrt{2}$

Step 3: Press Enter — Desmos will evaluate this to approximately 1.506, confirming the new margin of error is about 1.51%, which is LESS than the original 2.13%, matching choice D.

Step 4: As a broader check, try a few other sample size ratios (like tripling the sample, using $\sqrt{3}$ instead of $\sqrt{2}$) to build intuition for how margin of error shrinks more slowly than the sample size grows — this square-root relationship is a recurring theme in AP Statistics and college-level statistics as well.

Geometry and Trigonometry

Math | 51 questions with full Rule / Tip / Trap / POE / Desmos explanations

The Method

Every question in this file is broken down the same way, so you build one consistent habit instead of guessing question to question:

The Rule — the underlying math principle or formula being tested. Learn the rule, not just this one answer, and you'll be able to answer every future question of this type.

The Tip — a concrete move you can make before diving into algebra, so you solve efficiently instead of the long way.

Step-by-Step Solution — every single algebra step written out in full, with nothing skipped — so you can follow exactly how the answer was reached, not just see the final number.

The Trap — the single wrong answer most students actually pick, and the specific arithmetic or logical slip that produces it.

POE (multiple-choice questions only) — a full walkthrough of the other wrong choices, showing exactly what error produces each one. Some Math questions are Student-Produced Response ('grid-in') questions with no answer choices at all — for those, there's no POE section, just the full solution.

Solving with Desmos — exact, specific instructions for using the built-in Desmos graphing calculator (available on every Digital SAT Math question) to solve or verify the question — including the exact expressions to type in, not just a general suggestion to 'graph it.'

Area and volume

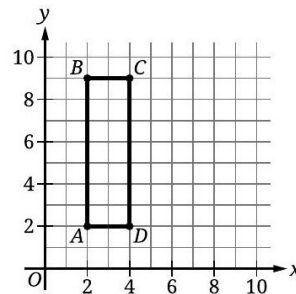
16 questions · Easy 5 / Medium 3 / Hard 8

Area and volume — Q1 [Easy] (ID: 124bc42b)

Question ID: 124bc42b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Rectangle $ABCD$ shown has a length of 7 units and a width of 2 units. What is the area, in square units, of rectangle $ABCD$?

Answer

- A. 5
- B. 9
- C. 14
- D. 28

Correct Answer: C

The Rule: The area of a rectangle is always calculated by multiplying its length by its width: $\text{Area} = \text{length} \times \text{width}$. This formula works regardless of where the rectangle is positioned on a coordinate grid.

The Tip: When a rectangle is shown on a coordinate grid, you don't need to use the coordinates at all if the length and width are already given directly in the problem text — the picture is just there to help you visualize the shape, not to require you to calculate distances from coordinates.

Why C is right: The problem directly states the rectangle has a length of 7 units and a width of 2 units. Using the area formula: $\text{Area} = \text{length} \times \text{width} = 7 \times 2 = 14$.

Step-by-Step Solution:

- Step 1: Identify the given dimensions: length = 7 units, width = 2 units.
- Step 2: Apply the area formula for a rectangle: $\text{Area} = \text{length} \times \text{width}$.
- Step 3: Substitute the values: $\text{Area} = 7 \times 2 = 14$.
- Step 4: This matches choice C.

The Trap: Choice D (28) is the most commonly picked wrong answer — it typically comes from mistakenly using the PERIMETER formula ($2 \times \text{length} + 2 \times \text{width} = 2 \times 7 + 2 \times 2 = 18$, not quite 28 either) or from doubling the correct area by mistake, rather than the simple area formula.

POE — Eliminate the other three:

✗ A — 5 doesn't correspond to any calculation using the given dimensions — likely confused with the difference between length and width ($7 - 2 = 5$) rather than their product.

X B — 9 comes from ADDING the length and width ($7+2=9$) instead of multiplying them — this would be relevant for perimeter-related calculations, not area.

X D — 28 doesn't match a direct area calculation — possibly from doubling the correct area ($14 \times 2 = 28$) or another multiplication error.

Solving with Desmos:

Step 1: Open Desmos and calculate the area directly: type 7×2

Step 2: Press Enter — Desmos will evaluate this to 14, confirming the area and matching choice C.

Step 3: If you want to verify using the coordinate points shown (A, B, C, D), you can also calculate the distance between two adjacent points directly in Desmos using the distance formula, though since the length and width are given directly in the problem, this extra step isn't necessary here.

Area and volume — Q2 [Easy] (ID: c8ab85ad)

Question ID: c8ab85ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Which expression represents the area, in square inches, of a square with a side length of 35 inches?

Answer

- A. $\sqrt{35}$
- B. $35 + 35$
- C. $(4)(35)$
- D. 35^2

Correct Answer: D

The Rule: The area of a square is always calculated by squaring the side length: $\text{Area} = (\text{side length})^2$. This is a special case of the rectangle area formula, since a square is a rectangle where length and width are equal.

The Tip: For 'which expression represents' questions, you don't need to calculate a final numeric answer — just identify the correct FORMULA setup and match it to the choices. Here, recognizing 'area of a square = side²' immediately points you to the answer without any arithmetic.

Why D is right: For a square with side length 35 inches, the area formula gives $\text{Area} = 35^2$, which is exactly what choice D shows.

Step-by-Step Solution:

Step 1: Recall the area formula for a square: $\text{Area} = (\text{side length})^2$.

Step 2: Substitute the given side length: $\text{Area} = 35^2$.

Step 3: This matches choice D exactly — no further calculation is needed since the question asks for the EXPRESSION, not a numeric value.

The Trap: Choice C, $(4)(35)$, is the most commonly picked wrong answer — it correctly uses the number 4 (associated with a square's four equal sides) but incorrectly applies it as a multiplication factor, which would actually calculate the PERIMETER of a square with side 35, not its area.

POE — Eliminate the other three:

X A — $\sqrt{35}$ represents a square ROOT operation, which would be used to find a side length FROM a given area — but here we're going the opposite direction (finding area FROM a given side length), so this is backwards.

X B — $35 + 35$ represents adding two sides together, which doesn't correspond to any standard area or perimeter formula for a square in this simple form.

X C — $(4)(35)$ represents multiplying the side length by 4 — this is actually the PERIMETER formula for a square ($4 \times$ side), not the area formula.

Solving with Desmos:

Step 1: This question asks for an expression, not a numeric value, so Desmos verification is optional — but you can still check that your chosen expression gives a sensible area. Open Desmos and type: 35^2

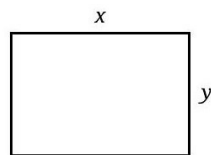
Step 2: Press Enter — Desmos evaluates this to 1225, which is the actual numeric area in square inches, confirming that 35^2 (choice D) is the correct way to represent this square's area.

Area and volume — Q3 [Easy] (ID: f3f06c3a)

Question ID: f3f06c3a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

In the rectangle shown, $x = 7$ centimeters and $y = 5$ centimeters. What is the area, in square centimeters, of the rectangle?

Answer

- A. 7
- B. 12
- C. 35
- D. 40

Correct Answer: C

The Rule: The area of a rectangle is always calculated by multiplying its length by its width: $\text{Area} = \text{length} \times \text{width}$. This holds true regardless of how the rectangle is labeled or oriented in a figure.

The Tip: When a figure labels its dimensions with variables (like x and y here) and then gives you specific numeric values for those variables in the text, simply substitute the numbers directly into the area formula — don't worry about matching the exact pixel proportions shown in the (often not-to-scale) figure.

Why C is right: The rectangle's dimensions are given as $x = 7$ centimeters and $y = 5$ centimeters. Using the area formula: $\text{Area} = x \times y = 7 \times 5 = 35$.

Step-by-Step Solution:

Step 1: Identify the given dimensions: $x = 7$ centimeters, $y = 5$ centimeters.

Step 2: Apply the area formula for a rectangle: $\text{Area} = \text{length} \times \text{width} = x \times y$.

Step 3: Substitute the values: $\text{Area} = 7 \times 5 = 35$.

Step 4: This matches choice C.

The Trap: Choice B (12) is the most commonly picked wrong answer — it results from ADDING the two dimensions ($7 + 5 = 12$) instead of multiplying them, confusing the area formula with the process for finding half the perimeter.

POE — Eliminate the other three:

- ✗ **A** — 7 is simply the value of x alone — this ignores the y dimension entirely.
- ✗ **B** — 12 comes from adding x and y ($7+5=12$) instead of multiplying them — this is a perimeter-related calculation, not area.
- ✗ **D** — 40 doesn't correspond to a direct calculation using $x=7$ and $y=5$ — possibly confused with different dimension values.

Solving with Desmos:

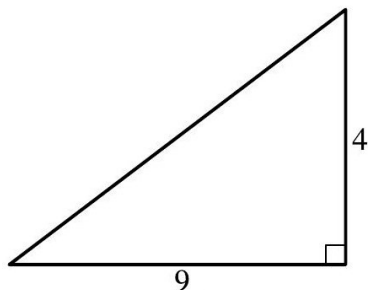
- Step 1: Open Desmos and calculate the area directly: type $7*5$
- Step 2: Press Enter — Desmos will evaluate this to 35, confirming the area and matching choice C.

Area and volume — Q4 [Easy] (ID: cab8f907)

Question ID: cab8f907

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Note: Figure not drawn to scale.

The figure shows the lengths, in units, of two sides of a triangle. What is the area, in square units, of the triangle?

Answer

- A. 18
- B. 13
- C. 9
- D. 4

Correct Answer: A

The Rule: The area of a triangle is always calculated using the formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$, where the base and height must be PERPENDICULAR to each other (meeting at a right angle, as shown by the small square symbol in the figure).

The Tip: Look carefully for the right-angle marker (small square) in a triangle figure — it tells you exactly which two sides are the 'base' and 'height' to use in the area formula. Not every pair of given side lengths in a triangle problem are automatically the base and height; they must be the two sides meeting at that right angle.

Why A is right: The right-angle marker in the figure shows that the sides of length 9 and 4 meet at a right angle, making them the base and height. Using the formula: $\text{Area} = (1/2) \times 9 \times 4 = (1/2) \times 36 = 18$.

Step-by-Step Solution:

- Step 1: Identify the base and height from the figure: the right-angle marker confirms the sides labeled 9 and 4 are perpendicular to each other, making them the base and height.
- Step 2: Apply the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$.
- Step 3: Substitute the values: $\text{Area} = (1/2) \times 9 \times 4$.
- Step 4: Calculate: $9 \times 4 = 36$, then $(1/2) \times 36 = 18$.
- Step 5: This matches choice A.

The Trap: Choice B (13) is the most commonly picked wrong answer — it results from simply ADDING the two given side lengths ($9 + 4 = 13$) instead of correctly applying the triangle area formula, which requires multiplying and then halving.

POE — Eliminate the other three:

- X B** — 13 comes from adding the two given lengths ($9+4=13$) — this doesn't correspond to any part of the correct area formula.
- X C** — 9 is simply the base length alone, without multiplying by the height or applying the 1/2 factor.
- X D** — 4 is simply the height alone, without multiplying by the base or applying the 1/2 factor.

Solving with Desmos:

- Step 1: Open Desmos and calculate the area directly using the triangle area formula: type $(1/2)*9*4$
- Step 2: Press Enter — Desmos will evaluate this to 18, confirming the area and matching choice A.

Area and volume — Q5 [Easy] (ID: 17049054)

Question ID: 17049054

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Easy

Question
 The length of a side of square X is 9 centimeters. The area of rectangle Y is 32 square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

- Answer**
- A. 145
 - B. 113
 - C. 82
 - D. 81

Correct Answer: B

The Rule: To find the TOTAL area of two separate shapes, first calculate each shape's area individually using its own appropriate formula, then ADD the two areas together.

The Tip: When a problem describes two different shapes (like a square and a separately-defined rectangle), resist the urge to combine their given measurements before calculating — find each area completely on its own first, then add the two final results together.

Why B is right: The area of square X is $(\text{side length})^2 = 9^2 = 81$ square centimeters. The area of rectangle Y is given directly as 32 square centimeters. The total combined area is $81 + 32 = 113$.

Step-by-Step Solution:

- Step 1: Calculate the area of square X using the given side length: $\text{Area} = \text{side}^2 = 9^2 = 81$ square centimeters.
- Step 2: Note the area of rectangle Y, which is given directly in the problem: 32 square centimeters.
- Step 3: Add the two areas together to find the total: $81 + 32 = 113$.

Step 4: This matches choice B.

The Trap: Choice D (81) is the most commonly picked wrong answer — it's the area of square X ALONE, calculated correctly but then the student forgets to add rectangle Y's area (32) to get the actual TOTAL area the question asks for.

POE — Eliminate the other three:

✗ A — 145 doesn't correspond to a correct combination of 81 and 32 — likely from a different, incorrect calculation of the square's area before adding.

✗ C — 82 is very close to the square's correct area (81) but off by one — possibly from a minor arithmetic slip, and still missing rectangle Y's area entirely.

✗ D — 81 is the area of square X ALONE ($9^2 = 81$) — this is a correct partial calculation, but it forgets to ADD rectangle Y's area (32) to find the TOTAL area the question asks for.

Solving with Desmos:

Step 1: Open Desmos and calculate the square's area first: type 9^2

Step 2: Press Enter — this confirms 81 square centimeters for square X.

Step 3: Add rectangle Y's given area: type $81+32$

Step 4: Press Enter — Desmos will evaluate this to 113, confirming the total combined area and matching choice B.

Area and volume — Q6 [Medium] [Grid-In] (ID: 25cc5327)

Question ID: 25cc5327

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The base of a right rectangular pyramid has a length of 27 centimeters and a width of 2 centimeters. The height of this pyramid is 7 centimeters. What is the volume, in cubic centimeters, of this pyramid?

Correct Answer: 126 (Student-Produced Response — no answer choices given)

The Rule: The volume of any pyramid is calculated using the formula: $\text{Volume} = (1/3) \times (\text{base area}) \times \text{height}$. For a RECTANGULAR pyramid, the base area is simply $\text{length} \times \text{width}$.

The Tip: Always calculate the base area FIRST as its own separate step, then plug that single number into the pyramid volume formula — trying to do everything in one big calculation increases the chance of a multiplication error.

Why this is the answer: The rectangular base has $\text{area} = \text{length} \times \text{width} = 27 \times 2 = 54$ square centimeters. Using the pyramid volume formula: $\text{Volume} = (1/3) \times 54 \times 7 = (1/3) \times 378 = 126$.

Step-by-Step Solution:

Step 1: Calculate the base area of the rectangular pyramid: $\text{base area} = \text{length} \times \text{width} = 27 \times 2 = 54$ square centimeters.

Step 2: Apply the pyramid volume formula: $\text{Volume} = (1/3) \times \text{base area} \times \text{height}$.

Step 3: Substitute the values: $\text{Volume} = (1/3) \times 54 \times 7$.

Step 4: Calculate $54 \times 7 = 378$, then $(1/3) \times 378 = 126$.

Step 5: This is a grid-in question — enter 126 as your final answer.

The Trap: A common error on this problem is forgetting the $(1/3)$ factor entirely, calculating just $\text{base area} \times \text{height}$ ($54 \times 7 = 378$) as if it were a rectangular PRISM instead of a pyramid — pyramids always have exactly one-third the volume of a prism with the same base and height.

Solving with Desmos:

Step 1: Open Desmos and calculate the base area first: type 27×2 — this confirms 54 square centimeters.

Step 2: Calculate the full volume using the pyramid formula: type $(1/3) \times 54 \times 7$

Step 3: Press Enter — Desmos will evaluate this to 126, confirming your grid-in answer.

Area and volume — Q7 [Medium] (ID: 42155bac)

Question ID: 42155bac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of one of the bases of a right circular cylinder is 81 square centimeters, and the volume of the cylinder is 2,187 cubic centimeters. What is the height, in centimeters, of the cylinder?

Answer

- A. 27
- B. 2,025
- C. 2,106
- D. 177,147

Correct Answer: A

The Rule: The volume of a cylinder is calculated using the formula: $\text{Volume} = (\text{base area}) \times \text{height}$. If you're given the volume and the base area, you can find the height by dividing: $\text{height} = \text{Volume} \div \text{base area}$.

The Tip: Recognize that a cylinder's volume formula is structurally the same as a rectangular prism's ($\text{base area} \times \text{height}$) — this makes it easy to rearrange and solve for height using simple division once you have the volume and base area.

Why A is right: Using the formula $\text{Volume} = \text{base area} \times \text{height}$, and solving for height: $\text{height} = \text{Volume} \div \text{base area} = 2,187 \div 81 = 27$.

Step-by-Step Solution:

Step 1: Recall the cylinder volume formula: $\text{Volume} = \text{base area} \times \text{height}$.

Step 2: Rearrange this formula to solve for height: $\text{height} = \text{Volume} \div \text{base area}$.

Step 3: Substitute the given values: $\text{height} = 2,187 \div 81$.

Step 4: Calculate: $2,187 \div 81 = 27$.

Step 5: This matches choice A.

The Trap: Choice B (2,025) is the most commonly picked wrong answer — it likely results from SUBTRACTING ($2,187 - 81 = 2,106$, close to choice C actually) or another incorrect operation instead of correctly DIVIDING the volume by the base area.

POE — Eliminate the other three:

X B — 2,025 doesn't correspond to a correct division of 2,187 by 81 — likely from an unrelated arithmetic error or confusing which operation to use.

X C — 2,106 comes from SUBTRACTING the base area from the volume ($2,187 - 81 = 2,106$) instead of correctly dividing to solve for height.

X D — 177,147 comes from MULTIPLYING the volume by the base area ($2,187 \times 81$) instead of dividing — the exact opposite, incorrect operation.

Solving with Desmos:

Step 1: Open Desmos and calculate the height directly: type $2187/81$

Step 2: Press Enter — Desmos will evaluate this to 27, confirming the height and matching choice A.

Area and volume — Q8 [Medium] [Grid-In] (ID: 1b687ae3)

Question ID: 1b687ae3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of a triangle is 36 square units. If the height of this triangle is 2 units, what is the length, in units, of the base of this triangle?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

The Rule: The area of a triangle is always calculated using the formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$. If you're given the area and the height, you can find the base by rearranging: $\text{base} = (2 \times \text{Area}) \div \text{height}$.

The Tip: When solving backwards from a given area to find a missing base or height, always multiply the area by 2 FIRST to cancel out the $(1/2)$ in the formula, then divide by whichever dimension you already know.

Why this is the answer: Using the triangle area formula $\text{Area} = (1/2) \times \text{base} \times \text{height}$, and solving for base: $\text{base} = (2 \times \text{Area}) \div \text{height} = (2 \times 36) \div 2 = 72 \div 2 = 36$.

Step-by-Step Solution:

Step 1: Recall the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$.

Step 2: Rearrange this formula to solve for the base: $\text{base} = (2 \times \text{Area}) \div \text{height}$.

Step 3: Substitute the given values: $\text{base} = (2 \times 36) \div 2$.

Step 4: Calculate the numerator first: $2 \times 36 = 72$.

Step 5: Divide: $72 \div 2 = 36$.

Step 6: This is a grid-in question — enter 36 as your final answer.

The Trap: A common error on this problem is forgetting to multiply the area by 2 before dividing by the height, which would incorrectly give $\text{base} = 36 \div 2 = 18$ instead of the correct 36 — always remember the factor of 2 needed to 'undo' the $(1/2)$ in the area formula.

Solving with Desmos:

Step 1: Open Desmos and calculate the base directly using the rearranged formula: type $(2*36)/2$

Step 2: Press Enter — Desmos will evaluate this to 36, confirming your grid-in answer.

Area and volume — Q9 [Hard] (ID: 1fe41c6b)

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

Correct Answer: D

The Rule: For SIMILAR figures, the ratio of their PERIMETERS equals the ratio of their corresponding LINEAR dimensions (the 'scale factor'), while the ratio of their AREAS equals the SQUARE of that same scale factor. Always find the scale factor from the area ratio FIRST (by taking a square root), then apply that same scale factor directly to the perimeter.

The Tip: When given both area and perimeter information for similar figures, use the AREA ratio to find the scale factor (by taking its square root), since area ratios reveal the TRUE linear scale factor more reliably than trying to reason from perimeter alone.

Why D is right: The area ratio between Rectangle B and Rectangle A is $2,520 \div 630 = 4$. Since area ratio equals the SQUARE of the scale factor, the actual linear scale factor is $\sqrt{4} = 2$. Since perimeter scales linearly (not squared), Rectangle B's perimeter is 2 times Rectangle A's perimeter: $n = 210 \times 2 = 420$.

Step-by-Step Solution:

Step 1: Calculate the area ratio between the two similar rectangles: Rectangle B's area $\div$ Rectangle A's area = $2,520 \div 630 = 4$.

Step 2: Since area scales as the SQUARE of the linear scale factor, find the actual scale factor by taking the square root of the area ratio: $\sqrt{4} = 2$.

Step 3: Since perimeter scales LINEARLY (directly proportional to the scale factor, not squared), multiply Rectangle A's perimeter by this same scale factor: $n = 210 \times 2$.

Step 4: Calculate: $210 \times 2 = 420$.

Step 5: This matches choice D.

The Trap: Choice A (2,100) is the most commonly picked wrong answer — it results from directly multiplying Rectangle A's perimeter by the AREA ratio (4) instead of correctly using the LINEAR scale factor (2) — perimeter should only be multiplied by the linear scale factor, not the area ratio.

POE — Eliminate the other three:

- X A** — 2,100 comes from multiplying 210 by the area ratio (4) directly — but perimeter scales with the LINEAR scale factor (2), not the area ratio itself.
- X B** — 1,680 doesn't correspond to a clean derivation from this problem — possibly from an unrelated calculation error.
- X C** — 840 comes from multiplying 210 by 4 and then dividing by something, or another miscalculation that doesn't correctly apply the square-root relationship between area ratio and linear scale factor.

Solving with Desmos:

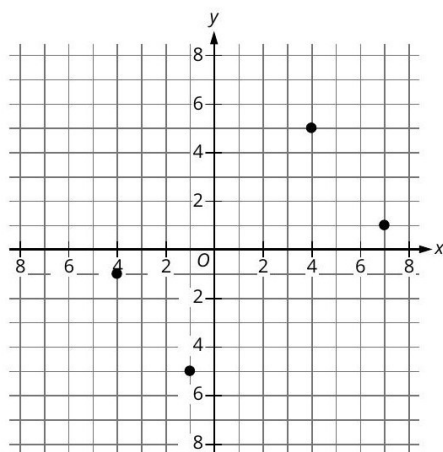
Step 1: Open Desmos and calculate the area ratio first: type $2520/630$ — this confirms the ratio is 4.
 Step 2: Find the linear scale factor by taking the square root: type $\sqrt{4}$ — this confirms the scale factor is 2.
 Step 3: Apply this scale factor to Rectangle A's perimeter: type 210×2
 Step 4: Press Enter — Desmos will evaluate this to 420, confirming $n = 420$ and matching choice D.

Area and volume — Q10 [Hard] [Grid-In] (ID: 30aa51b7)

Question ID: 30aa51b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Correct Answer: 50 (Student-Produced Response — no answer choices given)

The Rule: When a rectangle's vertices are given as coordinate points (not aligned with the x and y axes), find the side lengths using the DISTANCE FORMULA between adjacent vertices, then multiply those two side lengths together for the area — just like any other rectangle.

The Tip: For a 'tilted' rectangle on a coordinate grid, identify which pairs of points are adjacent (connected by a side) versus diagonal (opposite corners) — adjacent sides should be PERPENDICULAR to each other, which you can verify by checking that their slopes multiply to -1 .

Why this is the answer: Reading the four points from the graph: approximately $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$. Using the distance formula, one side has length $\sqrt{[(4 - (-4))]^2 + [(5 - (-1))]^2} = \sqrt{(8)^2 + (6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$, and the adjacent side has length $\sqrt{[(7 - 4)]^2 + [(1 - 5)]^2} = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. The area is $10 \times 5 = 50$.

Step-by-Step Solution:

Step 1: Read the four vertex coordinates from the graph: approximately $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$.

Step 2: Calculate the length of one side using the distance formula between two adjacent vertices, $(-4, -1)$ and $(4, 5)$:
 distance = $\sqrt{[(4 - (-4))]^2 + [(5 - (-1))]^2} = \sqrt{(8)^2 + (6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$.

Step 3: Calculate the length of the adjacent side between $(4, 5)$ and $(7, 1)$: distance = $\sqrt{[(7 - 4)]^2 + [(1 - 5)]^2} = \sqrt{(3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Step 4: Verify these two sides are perpendicular (confirming it's truly a rectangle) by checking their slopes multiply to -1 : slope of first side = $6/8 = 3/4$; slope of second side = $-4/3$. Product: $(3/4) \times (-4/3) = -1$. Confirmed perpendicular.

Step 5: Calculate the area by multiplying the two side lengths: Area = $10 \times 5 = 50$.

Step 6: This is a grid-in question — enter 50 as your final answer.

The Trap: A common error on this problem is misreading the exact coordinates of the points from the graph (since they're not always sitting exactly on grid intersections), which throws off the entire distance calculation — always double check each point's position carefully against the grid lines.

Solving with Desmos:

- Step 1: Open Desmos and plot the four points you read from the graph: (-4,-1), (4,5), (7,1), (-1,-5)
- Step 2: To calculate the distance between two adjacent points, type: $\sqrt{(4-(-4))^2+(5-(-1))^2}$ — this confirms one side length is 10.
- Step 3: Type: $\sqrt{(7-4)^2+(1-5)^2}$ — this confirms the adjacent side length is 5.
- Step 4: Calculate the area: type 10×5 — Desmos will evaluate this to 50, confirming your grid-in answer.

Area and volume — Q11 [Hard] [Grid-In] (ID: 179edb12)

Question ID: 179edb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of 22 centimeters (cm) and a base with a diameter of 18 cm. The volume of this cone is $n\pi$ cm³. What is the value of n ?

Correct Answer: 594 (Student-Produced Response — no answer choices given)

The Rule: The volume of a cone is calculated using the formula: $\text{Volume} = (1/3) \times \pi \times r^2 \times \text{height}$, where r is the RADIUS of the circular base (not the diameter). Always remember to divide the given diameter by 2 to get the radius before plugging into the formula.

The Tip: When a cone problem gives you the DIAMETER instead of the radius, immediately divide by 2 as your very first step, before touching the volume formula — using the diameter directly in place of the radius is one of the most common errors on cone and cylinder problems.

Why this is the answer: The diameter is 18 cm, so the radius is $18 \div 2 = 9$ cm. Using the cone volume formula: $\text{Volume} = (1/3) \times \pi \times 9^2 \times 22 = (1/3) \times \pi \times 81 \times 22 = (1/3) \times 1,782 \times \pi = 594\pi$. Since the volume is given as $n\pi$, this means $n = 594$.

Step-by-Step Solution:

- Step 1: Convert the given diameter to radius: $\text{radius} = \text{diameter} \div 2 = 18 \div 2 = 9$ cm.
- Step 2: Recall the cone volume formula: $\text{Volume} = (1/3) \times \pi \times r^2 \times \text{height}$.
- Step 3: Substitute the values: $\text{Volume} = (1/3) \times \pi \times 9^2 \times 22$.
- Step 4: Calculate $9^2 = 81$.
- Step 5: Multiply: $81 \times 22 = 1,782$.
- Step 6: Multiply by $(1/3)$: $(1/3) \times 1,782 = 594$.
- Step 7: The volume is 594π cubic centimeters, so comparing to the given form $n\pi$, we get $n = 594$.
- Step 8: This is a grid-in question — enter 594 as your final answer.

The Trap: A common error on this problem is using the given diameter (18) directly as the radius in the formula, instead of first dividing by 2 to get the correct radius (9) — this would lead to an answer four times too large, since the radius is squared in the formula.

Solving with Desmos:

- Step 1: Open Desmos and calculate the radius first: type $18/2$ — this confirms the radius is 9 cm.
- Step 2: Calculate the volume coefficient (the value of n): type $(1/3) \times 9^2 \times 22$

Step 3: Press Enter — Desmos will evaluate this to 594, confirming $n = 594$ for your grid-in answer. (Desmos will show 594 as a plain number since you're calculating the coefficient of π , not the full numeric volume.)

Area and volume — Q12 [Hard] (ID: e6c557f9)

Question ID: e6c557f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

Answer

- A. $\frac{9}{7,744}$
B. $\frac{27}{3,872}$
C. $\frac{3}{88}$
D. $\frac{9}{44}$

Correct Answer: A

The Rule: For SIMILAR three-dimensional shapes, the ratio of their SURFACE AREAS equals the SQUARE of the ratio of their corresponding linear dimensions (like edge length) — this is the 3D analog of how area scales for similar 2D shapes.

The Tip: Whenever you see a ratio comparing similar shapes' surface areas (a 2D measurement), square whatever linear (1D) ratio you're given — don't use the linear ratio directly, and don't cube it either (cubing is for VOLUME ratios, not surface area).

Why A is right: The edge length ratio of cube Y to cube X is $\frac{3}{88}$. Since surface area scales as the SQUARE of the linear ratio, the surface area ratio $n = (\frac{3}{88})^2 = \frac{9}{7,744}$.

Step-by-Step Solution:

- Step 1: Identify the given linear (edge length) ratio: cube Y's edge is $\frac{3}{88}$ of cube X's edge.
Step 2: Recall that surface area scales as the SQUARE of the linear scale factor for similar 3D shapes.
Step 3: Square the given ratio: $n = (\frac{3}{88})^2 = \frac{(3^2)}{(88^2)} = \frac{9}{7,744}$.
Step 4: This matches choice A.

The Trap: Choice C ($\frac{3}{88}$) is the most commonly picked wrong answer — it uses the LINEAR edge-length ratio directly as the answer, without squaring it to correctly account for the fact that surface area is a two-dimensional measurement.

POE — Eliminate the other three:

- X B** — $\frac{27}{3,872}$ comes from CUBING the ratio ($\frac{3^3}{88^2}$ or a similar mismatched operation) instead of correctly squaring both the numerator and denominator — cubing would be appropriate for VOLUME ratios, not surface area.
X C — $\frac{3}{88}$ is simply the original linear edge-length ratio, used directly without squaring — this doesn't account for surface area being a two-dimensional measurement.
X D — $\frac{9}{44}$ doesn't correctly square the denominator ($88^2 = 7,744$, not 44) — this likely results from only partially squaring the ratio.

Solving with Desmos:

- Step 1: Open Desmos and calculate the surface area ratio by squaring the given edge-length ratio: type $(\frac{3}{88})^2$
Step 2: Press Enter — Desmos will evaluate this to a decimal (approximately 0.000388...), and you can compare it to $\frac{9}{7,744}$ by typing that as well to confirm they match exactly, verifying choice A.

Area and volume — Q13 [Hard] [Grid-In] (ID: 73e1b793)

Question ID: 73e1b793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is 972 cubic centimeters (cm^3), and the volume of prism Q is 36 cm^3 . One edge of prism P has a length of 12 cm. What is the length, in cm, of the corresponding edge of prism Q?

Correct Answer: 4 (Student-Produced Response — no answer choices given)

The Rule: For SIMILAR three-dimensional shapes, the ratio of their VOLUMES equals the CUBE of the ratio of their corresponding linear dimensions — this is different from area, which uses the SQUARE of the ratio.

The Tip: When given a volume ratio for similar solids and asked to find a linear dimension, take the CUBE ROOT of the volume ratio to find the actual linear scale factor — don't take the square root (that's for area problems) or use the volume ratio directly.

Why this is the answer: The volume ratio of prism P to prism Q is $972/36 = 27$. Since volume scales as the CUBE of the linear scale factor, the actual linear scale factor is the cube root of 27, which is 3. This means every edge of prism P is 3 times the corresponding edge of prism Q. Since one edge of P is 12 cm, the corresponding edge of Q is $12 \div 3 = 4$ cm.

Step-by-Step Solution:

Step 1: Calculate the volume ratio between the two similar prisms: prism P's volume $\div$ prism Q's volume = $972 \div 36 = 27$.

Step 2: Since volume scales as the CUBE of the linear scale factor, find the actual scale factor by taking the cube root of the volume ratio: the cube root of 27 is 3 (since $3 \times 3 \times 3 = 27$).

Step 3: This means every linear dimension of prism P is 3 times the corresponding dimension of prism Q.

Step 4: Since the given edge of prism P is 12 cm, find the corresponding edge of prism Q by dividing by the scale factor: $12 \div 3 = 4$.

Step 5: This is a grid-in question — enter 4 as your final answer.

The Trap: A common error on this problem is taking the SQUARE ROOT of the volume ratio ($\sqrt{27} \approx 5.2$) instead of the correct CUBE ROOT (3) — remembering that volume is a three-dimensional measurement (requiring a cube root to find the linear scale) is the key insight here.

Solving with Desmos:

Step 1: Open Desmos and calculate the volume ratio first: type $972/36$ — this confirms the ratio is 27.

Step 2: Find the linear scale factor by taking the cube root: type $27^{(1/3)}$ — Desmos will show this equals 3.

Step 3: Calculate the corresponding edge of prism Q: type $12/3$

Step 4: Press Enter — Desmos will evaluate this to 4, confirming your grid-in answer.

Area and volume — Q14 [Hard] (ID: 9fdc4eaa)

Question ID: 9fdc4eaa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of 70,560 square inches, and the combined surface area of the four lateral faces of this pyramid is 38,160 square inches. What is the height, in inches, of this pyramid?

Answer

- A. 56
- B. 90
- C. 106
- D. 180

Correct Answer: A

The Rule: For a right square pyramid, the total surface area equals the base area PLUS the combined lateral (side) face area. The slant height relates to the pyramid's actual height through the Pythagorean theorem, using half the base's side length as the other leg of a right triangle.

The Tip: Break this multi-step problem into clear stages: (1) find the base area by subtracting lateral area from total surface area, (2) find the base's side length by taking the square root, (3) find the slant height using the lateral area formula, and (4) use the Pythagorean theorem to find the actual height. Tackling one stage at a time prevents getting overwhelmed.

Why A is right: The base area is $70,560 - 38,160 = 32,400$ square inches, so the base's side length is $\sqrt{32,400} = 180$ inches. The lateral area formula for a square pyramid is $2 \times (\text{side length}) \times (\text{slant height}) = 38,160$, so slant height = $38,160 \div (2 \times 180) = 38,160 \div 360 = 106$. Using the Pythagorean theorem with half the base (90) and the slant height (106): $\text{height}^2 = 106^2 - 90^2 = 11,236 - 8,100 = 3,136$, so height = $\sqrt{3,136} = 56$.

Step-by-Step Solution:

Step 1: Find the base area by subtracting the lateral surface area from the total surface area: base area = $70,560 - 38,160 = 32,400$ square inches.

Step 2: Since the base is a square, find its side length by taking the square root: side length = $\sqrt{32,400} = 180$ inches.

Step 3: Use the lateral surface area formula for a square pyramid (4 triangular faces, each with area $(1/2) \times \text{base side} \times \text{slant height}$, combined = $2 \times \text{base side} \times \text{slant height}$): $38,160 = 2 \times 180 \times \text{slant height}$.

Step 4: Solve for slant height: slant height = $38,160 \div 360 = 106$ inches.

Step 5: Set up the Pythagorean theorem relating the pyramid's height, half the base side length, and the slant height (these three form a right triangle): $\text{height}^2 + (\text{base side}/2)^2 = \text{slant height}^2$.

Step 6: Substitute the values: $\text{height}^2 + 90^2 = 106^2$.

Step 7: Calculate: $\text{height}^2 + 8,100 = 11,236$.

Step 8: Solve for height^2 : $\text{height}^2 = 11,236 - 8,100 = 3,136$.

Step 9: Take the square root: height = $\sqrt{3,136} = 56$.

Step 10: This matches choice A.

The Trap: Choice B (90) is the most commonly picked wrong answer — it's actually HALF the base side length (used as an intermediate step in the Pythagorean theorem calculation), not the final height — a student who completes most of the steps correctly but stops early might mistakenly select this intermediate value.

POE — Eliminate the other three:

X B — 90 is half the base's side length ($180 \div 2 = 90$), an intermediate value used WITHIN the Pythagorean theorem calculation — it's not the final height the question asks for.

X C — 106 is the SLANT HEIGHT (a different measurement along the triangular face), not the perpendicular height of the pyramid that the question asks for.

X D — 180 is the full side length of the square base — not the height of the pyramid at all.

Solving with Desmos:

- Step 1: Open Desmos and calculate the base area: type $70560-38160$ — this confirms 32,400 square inches.
- Step 2: Find the base's side length: type $\text{sqrt}(32400)$ — this confirms 180 inches.
- Step 3: Find the slant height: type $38160/(2*180)$ — this confirms 106 inches.
- Step 4: Use the Pythagorean theorem to find the actual height: type $\text{sqrt}(106^2-90^2)$
- Step 5: Press Enter — Desmos will evaluate this to 56, confirming the height and matching choice A.

Area and volume — Q15 [Hard] [Grid-In] (ID: f6ca90cc)

Question ID: f6ca90cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■

Question

A right prism has a square base with side lengths of 10 inches. The height of the prism is 58 inches. What is the volume, in cubic inches, of the prism?

Correct Answer: 5800 (Student-Produced Response — no answer choices given)

The Rule: The volume of any right prism (including one with a square base) is calculated using the formula: $\text{Volume} = (\text{base area}) \times \text{height}$.

The Tip: For a prism with a square base, calculate the base area first (side length squared) as a separate step, then multiply by the height — breaking the calculation into two clean stages reduces the chance of a multiplication error.

Why this is the answer: The square base has $\text{area} = \text{side}^2 = 10^2 = 100$ square inches. Using the prism volume formula: $\text{Volume} = \text{base area} \times \text{height} = 100 \times 58 = 5,800$.

Step-by-Step Solution:

- Step 1: Calculate the area of the square base: $\text{base area} = \text{side}^2 = 10^2 = 100$ square inches.
- Step 2: Apply the prism volume formula: $\text{Volume} = \text{base area} \times \text{height}$.
- Step 3: Substitute the values: $\text{Volume} = 100 \times 58$.
- Step 4: Calculate: $100 \times 58 = 5,800$.
- Step 5: This is a grid-in question — enter 5,800 as your final answer.

The Trap: A common error on this problem is using the PERIMETER of the square base ($4 \times 10 = 40$) instead of its AREA ($10^2 = 100$) when applying the volume formula — always make sure you're using area, not perimeter, for volume calculations.

Solving with Desmos:

- Step 1: Open Desmos and calculate the base area first: type 10^2 — this confirms 100 square inches.
- Step 2: Calculate the full volume: type $100*58$
- Step 3: Press Enter — Desmos will evaluate this to 5800, confirming your grid-in answer.

Area and volume — Q16 [Hard] [Grid-In] (ID: 4757123b)

Question ID: 4757123b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard ■■■■■

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is 41 times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Correct Answer: 1681 (Student-Produced Response — no answer choices given)

The Rule: For SIMILAR two-dimensional figures (including trapezoids), the ratio of their AREAS equals the SQUARE of the ratio of their corresponding linear dimensions (side lengths).

The Tip: When a problem gives you a linear (side-length) ratio between similar figures and asks about the area relationship, always SQUARE that ratio — don't use it directly, since area is a two-dimensional measurement while side length is only one-dimensional.

Why this is the answer: Since each side of trapezoid A is 41 times the corresponding side of trapezoid B, the linear scale factor is 41. Since area scales as the SQUARE of the linear scale factor, trapezoid A's area is $41^2 = 1,681$ times as large as trapezoid B's area.

Step-by-Step Solution:

- Step 1: Identify the given linear scale factor: each side of trapezoid A is 41 times the corresponding side of trapezoid B.
- Step 2: Recall that area scales as the SQUARE of the linear scale factor for similar figures.
- Step 3: Square the given scale factor: $41^2 = 41 \times 41 = 1,681$.
- Step 4: This means trapezoid A's area is 1,681 times as large as trapezoid B's area.
- Step 5: This is a grid-in question — enter 1,681 as your final answer.

The Trap: A common error on this problem is using the linear scale factor (41) directly as the final answer, without squaring it to correctly account for the fact that area is a two-dimensional measurement — remember, side-length ratios must always be squared to find area ratios.

Solving with Desmos:

- Step 1: Open Desmos and calculate the area ratio by squaring the given linear scale factor: type 41^2
- Step 2: Press Enter — Desmos will evaluate this to 1681, confirming your grid-in answer.

Lines, angles, and triangles

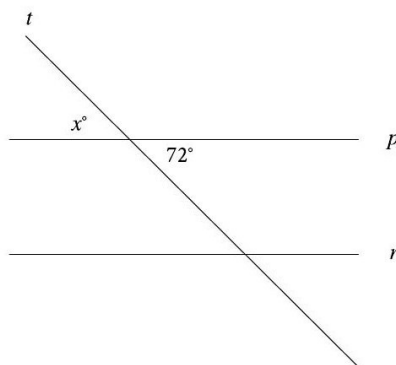
14 questions · Easy 2 / Medium 6 / Hard 6

Lines, angles, and triangles — Q1 [Easy] (ID: 4a141e77)

Question ID: 4a141e77

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line p is parallel to line r , and line t intersects both lines. What is the value of x ?

Answer

- A. 36
- B. 72
- C. 180
- D. 252

Correct Answer: B

The Rule: When two straight lines cross, the two angles directly ACROSS from each other (diagonally opposite, formed by the same two lines) are called vertical angles, and vertical angles are always EQUAL — this is true regardless of whether any other lines in the figure are parallel.

The Tip: Before using any parallel-line facts (corresponding, alternate interior angles), first check if the two labeled angles are simply VERTICAL angles at the SAME intersection point — if so, they're equal by a much more basic rule, and you don't need the parallel-line information at all.

Why B is right: The angles labeled x° and 72° are formed at the same intersection point (where line t crosses line p), positioned diagonally opposite each other — making them vertical angles, which are always equal: $x = 72$.

Step-by-Step Solution:

- Step 1: Identify the two labeled angles' positions: x° and 72° are both located at the single intersection point where line t crosses line p .
- Step 2: Determine their relationship: they are diagonally opposite each other at this intersection, making them vertical angles.
- Step 3: Apply the vertical angles theorem: vertical angles are always equal, regardless of any other properties in the figure (like whether p is parallel to r).
- Step 4: Since 72° is given, $x = 72$.
- Step 5: This matches choice B.

The Trap: Choice A (36) is the most commonly picked wrong answer — it results from incorrectly halving 72° , perhaps confusing this problem with a different angle-bisector scenario that doesn't apply here.

POE — Eliminate the other three:

X A — 36 is half of 72 — this might come from mistakenly assuming some bisecting relationship that doesn't apply to vertical angles, which are simply equal, not related by a factor of 2.

X C — 180 would be the result of treating x and 72 as a LINEAR PAIR (supplementary angles) instead of vertical angles — but based on their diagonal positions in the figure, they are vertical (equal), not supplementary.

X D — 252 comes from adding $180 + 72$ — this doesn't correspond to any standard angle relationship in this figure.

Solving with Desmos:

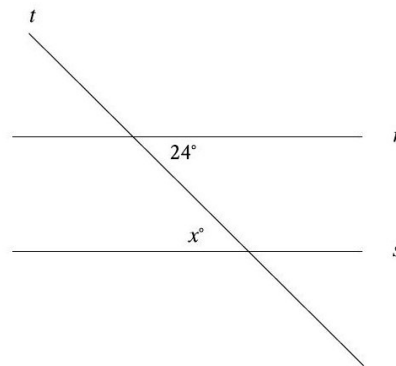
This is a straightforward vertical-angles reading question that doesn't require Desmos calculation — the answer ($x=72$) follows directly from the vertical angles theorem, visible in the figure itself. If you want to visually confirm, you could sketch the intersection using Desmos's geometry tools by plotting two intersecting lines and observing that the opposite angle pairs are always visually equal.

Lines, angles, and triangles — Q2 [Easy] (ID: d21270da)

Question ID: d21270da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines r and s . Which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x < 24$
- B. $x = 24$
- C. $x > 24$
- D. $x - 24 = 360$

Correct Answer: B

The Rule: When a transversal crosses two lines, CORRESPONDING angles (angles in the SAME relative position at each intersection) are equal IF AND ONLY IF the two lines are parallel. This is the Corresponding Angles Postulate, and it works in both directions — equal corresponding angles PROVE the lines are parallel.

The Tip: To identify corresponding angles, imagine 'sliding' one intersection point along the transversal until it lands on top of the other intersection point — whichever two angles end up in the same relative position (same side of the transversal, same side of the crossed lines) are corresponding angles.

Why B is right: The angle 24° (at the r intersection) and the angle x° (at the s intersection) are both positioned in the same relative spot — below their respective horizontal line, on the same side of the transversal t . This makes them corresponding angles. For lines r and s to be parallel, corresponding angles must be equal, so the required condition is $x = 24$.

Step-by-Step Solution:

- Step 1: Identify the positions of the two labeled angles: 24° is at the intersection of t and r ; x° is at the intersection of t and s .
- Step 2: Compare their relative positions: both angles sit in the same location relative to their own intersection (same side of line t , same side of the horizontal line) — this makes them corresponding angles.
- Step 3: Recall the Corresponding Angles Postulate: two lines cut by a transversal are parallel IF AND ONLY IF corresponding angles are equal.
- Step 4: Apply this to find the necessary condition: for r and s to be parallel, x must equal 24.
- Step 5: This matches choice B.

The Trap: Choice D is the most commonly picked wrong answer — it proposes an equation ($x - 24 = 360$) that doesn't correspond to any standard angle relationship; a difference of 360° doesn't make sense for two angles that should logically be somewhere between 0° and 180° .

POE — Eliminate the other three:

- X A** — $x < 24$ doesn't establish a precise EQUALITY needed for the Corresponding Angles Postulate — parallel lines require the angles to be exactly equal, not merely one smaller than the other.
- X C** — $x > 24$ has the same problem as choice A — an inequality doesn't provide the precise equality condition needed to prove the lines are parallel.
- X D** — $x - 24 = 360$ doesn't correspond to any meaningful angle relationship — angle measures in this context are between 0° and 180° , making a 360° difference nonsensical here.

Solving with Desmos:

This is a conceptual angle-relationship question that doesn't require numerical calculation in Desmos — the answer follows directly from correctly identifying corresponding angles and applying the Corresponding Angles Postulate. If you want to visually verify, you could use Desmos's geometry tools to draw two parallel lines cut by a transversal and confirm that the corresponding angle positions always show equal angle measures.

Lines, angles, and triangles — Q3 [Medium] (ID: e89f63bf)

Question ID: e89f63bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

The sum of the measures of two angles in a triangle is 64° . What is the measure of the third angle?

Answer

- A. 26°
- B. 52°
- C. 116°
- D. 128°

Correct Answer: C

The Rule: The three interior angles of ANY triangle always add up to exactly 180° . If you know the sum of two angles, you can always find the third by subtracting from 180.

The Tip: This is one of the most fundamental and frequently-used facts in geometry — memorize it so thoroughly that you can apply it instantly without hesitation: $\text{angle1} + \text{angle2} + \text{angle3} = 180^\circ$ for every triangle, no exceptions.

Why C is right: Since all three angles of a triangle sum to 180° , and two of the angles sum to 64° , the third angle must be $180 - 64 = 116$.

Step-by-Step Solution:

- Step 1: Recall the triangle angle sum theorem: the three interior angles of any triangle always add up to 180° .
- Step 2: The problem states two of the angles sum to 64° .
- Step 3: Find the third angle by subtracting this sum from 180° : $180 - 64 = 116$.
- Step 4: This matches choice C.

The Trap: Choice A (26) is the most commonly picked wrong answer — it results from an unrelated calculation (like $90 - 64 = 26$, perhaps confusing this with a right-triangle complementary-angle problem) rather than correctly using the 180° triangle angle sum.

POE — Eliminate the other three:

- X A** — 26 comes from $90 - 64 = 26$ — this incorrectly applies the COMPLEMENTARY angle rule (which only applies to two acute angles in a RIGHT triangle summing to 90°), not the general triangle angle sum of 180° that applies here.
- X B** — 52 is exactly half of 104, or possibly half of the correct answer — doesn't correspond to a direct calculation from the given 64° sum.
- X D** — 128 is exactly double the correct answer (116×2 ... actually let's check: 128 doesn't have an obvious simple relationship, possibly from a different arithmetic error).

Solving with Desmos:

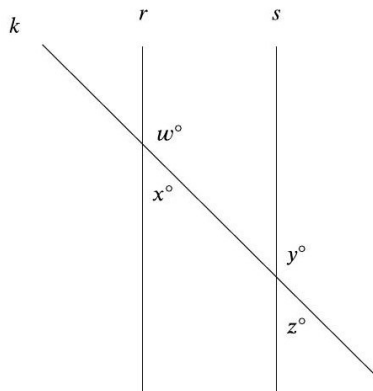
- Step 1: Open Desmos and calculate the third angle directly: type $180-64$
- Step 2: Press Enter — Desmos will evaluate this to 116, confirming the third angle and matching choice C.

Lines, angles, and triangles — Q4 [Medium] (ID: ecc98c87)

Question ID: ecc98c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line k intersects lines r and s . If $w = 147$, which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x = 33$
- B. $y = 147$
- C. $w + y = 180$
- D. $y + z = 180$

Correct Answer: B

The Rule: When a transversal crosses two lines, CORRESPONDING angles (same relative position at each intersection) are equal IF AND ONLY IF the two lines are parallel. Identifying which pair of labeled angles are truly 'corresponding' (not vertical, not alternate) is the key step in these problems.

The Tip: Carefully trace the SAME relative position at each intersection point — the angle that occupies the identical corner (upper-left, upper-right, lower-left, or lower-right) relative to the crossing lines at BOTH intersections is the corresponding angle pair, regardless of how far apart the intersections are.

Why B is right: The angle w° (at the r intersection) and the angle y° (at the s intersection) occupy the same relative position at their respective crossings — both in the 'upper-right' region relative to the transversal k and their vertical line. This makes w and y corresponding angles. Since $w = 147$ (as given), for r and s to be parallel, y must also equal 147.

Step-by-Step Solution:

Step 1: Identify the given angle: $w = 147^\circ$ at the r intersection.

Step 2: Determine which angle at the s intersection occupies the SAME relative position as w — comparing the figure, this is y° (both are in the upper-right region relative to their intersection).

Step 3: Recall the Corresponding Angles Postulate: for two lines cut by a transversal to be parallel, corresponding angles must be equal.

Step 4: Apply this: since $w = 147$, for $r \parallel s$, we need $y = 147$ as well.

Step 5: This matches choice B.

The Trap: Choice C ($w + y = 180$) is the most commonly picked wrong answer — it applies the SAME-SIDE INTERIOR angle rule (which requires supplementary angles for parallel lines), but w and y are actually CORRESPONDING angles in this figure (requiring equality, not supplementary sums), not same-side interior angles.

POE — Eliminate the other three:

X A — $x = 33$ is already automatically true (since w and x form a linear pair at the SAME intersection point, $147+33=180$) regardless of whether r is parallel to s — this doesn't provide NEW information that would help prove the lines are parallel.

X C — $w + y = 180$ applies the co-interior (same-side interior) angle relationship — but w and y are positioned as CORRESPONDING angles in this figure, which requires EQUALITY ($w=y$), not a supplementary sum.

X D — $y + z = 180$ is already automatically true (since y and z form a linear pair at the SAME intersection point at s) regardless of whether r is parallel to s — like choice A, this doesn't provide new proof information.

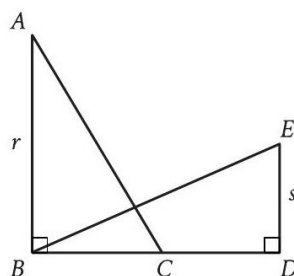
Solving with Desmos:

This is a conceptual angle-relationship question about identifying corresponding angles in a figure, which doesn't require numerical calculation in Desmos. The key insight is recognizing that w and y occupy matching relative positions at their two intersections — you can verify this type of relationship by sketching two parallel lines with a transversal in Desmos's geometry tools and observing which angle positions consistently match.

Question ID: 697b3954

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

Answer

- A. 26
- B. 27
- C. 28
- D. 54

Correct Answer: C

The Rule: When two triangles are stated to be CONGRUENT, the order of the letters in their names tells you EXACTLY which vertices (and therefore which sides and angles) correspond to each other. Always match up the letters in order — the first letter of one triangle corresponds to the first letter of the other, and so on.

The Tip: Write out the correspondence explicitly before doing any calculations: 'triangle $ABC \cong$ triangle BDE ' means $A \leftrightarrow B$, $B \leftrightarrow D$, $C \leftrightarrow E$ — then figure out which given side lengths (r , s) correspond to which sides in each triangle before setting up any equations.

Why C is right: Since triangle $ABC \cong$ triangle BDE (correspondence $A \leftrightarrow B$, $B \leftrightarrow D$, $C \leftrightarrow E$), side AB corresponds to side BD , and side BC corresponds to side DE . Given $AB = r = 54$, this means $BD = 54$. Given $DE = s = 26$, this means $BC = 26$. Since $BD = BC + CD$ (the full horizontal segment from B to D passes through C), we can solve: $54 = 26 + CD$, so $CD = 54 - 26 = 28$.

Step-by-Step Solution:

- Step 1: Write out the vertex correspondence from the congruence statement: triangle $ABC \cong$ triangle BDE means $A \leftrightarrow B$, $B \leftrightarrow D$, $C \leftrightarrow E$.
- Step 2: Match up corresponding sides: side AB (in triangle ABC) corresponds to side BD (in triangle BDE); side BC corresponds to side DE .
- Step 3: Use the given value $r = 54$ for side AB (the vertical leg in the figure): since AB corresponds to BD , this means $BD = 54$.
- Step 4: Use the given value $s = 26$ for side DE (the vertical leg on the right side of the figure): since BC corresponds to DE , this means $BC = 26$.
- Step 5: Notice from the figure that B , C , and D are all on the same horizontal line, with C between B and D — so $BD = BC + CD$.
- Step 6: Substitute the known values: $54 = 26 + CD$.
- Step 7: Solve for CD : $CD = 54 - 26 = 28$.
- Step 8: This matches choice C.

The Trap: Choice A (26) is the most commonly picked wrong answer — it directly uses the value of s (26) as the final answer, mistakenly assuming CD equals BC (or DE) directly, without correctly working through the $BD = BC + CD$ relationship along the shared horizontal line.

POE — Eliminate the other three:

- X A** — 26 is simply the given value of s (which equals BC , not CD) — this skips the necessary step of using $BD = BC + CD$ to solve for the specific segment CD .
- X B** — 27 doesn't correspond to a clean derivation from the given values — likely an arithmetic slip in combining r and s .
- X D** — 54 is simply the given value of r (which equals BD , the ENTIRE segment from B to D) — not the same as CD , which is only part of that full segment.

Solving with Desmos:

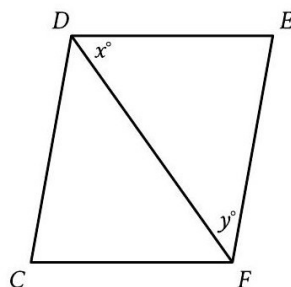
- Step 1: Open Desmos and set up the relationship you derived: $BD (=r=54)$ equals $BC (=s=26)$ plus CD . Type: $54=26+x$
- Step 2: Desmos will treat this as an equation to solve for x — it should plot the solution as $x = 28$, confirming $CD = 28$ and matching choice C.
- Step 3: Alternatively, just calculate directly: type $54-26$ — Desmos will evaluate this to 28.

Lines, angles, and triangles — Q6 [Medium] (ID: ffc39dc5)

Question ID: ffc39dc5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

Answer

- A. 30°
- B. 65°
- C. 95°
- D. 130°

Correct Answer: C

The Rule: When two triangles share a common side and have two pairs of corresponding sides equal (SSS conditions), they are congruent — and congruent triangles have all corresponding angles equal too. Use this congruence to transfer angle relationships between the two triangles.

The Tip: When a diagonal splits a quadrilateral into two triangles that share that diagonal as a common side, check whether the OTHER two pairs of sides are also equal (as stated in the problem) — if so, the shared diagonal PLUS the two given equal pairs gives you SSS congruence between the two triangles.

Why C is right: Since $CD = EF$ and $FC = DE$ (given), plus the shared diagonal $DF = FD$, triangles CDF and EFD are congruent by SSS (with correspondence $C \leftrightarrow E, D \leftrightarrow F, F \leftrightarrow D$). This congruence means angle CDF (in triangle CDF) equals angle EFD (in triangle EFD), which is $y = 30^\circ$. The full angle CDE is the sum of angle CDF and angle FDE (which is $x = 65^\circ$): angle CDE = $30 + 65 = 95$.

Step-by-Step Solution:

Step 1: Identify the given side equalities: $CD = EF$ and $FC = DE$, plus the shared diagonal DF (common to both triangles CDF and EFD).

Step 2: Recognize this establishes SSS congruence between triangle CDF and triangle EFD, with correspondence $C \leftrightarrow E, D \leftrightarrow F, F \leftrightarrow D$.

Step 3: From this congruence, corresponding angles are equal: angle CDF (at vertex D in triangle CDF) corresponds to angle EFD (at vertex F in triangle EFD), which is given as $y = 30^\circ$. So angle CDF = 30° .

Step 4: Identify that the full angle CDE (the angle the question asks about) is made up of two parts: angle CDF (just found to be 30°) plus angle FDE (which is given as $x = 65^\circ$, the angle between DF and DE at vertex D).

Step 5: Add these two parts together: angle CDE = angle CDF + angle FDE = $30 + 65 = 95$.

Step 6: This matches choice C.

The Trap: Choice B (65°) is the most commonly picked wrong answer — it uses only the value of x (the angle FDE alone) as the final answer, forgetting that angle CDE is actually the SUM of two parts (angle CDF PLUS angle FDE), not just the single labeled x° angle.

POE — Eliminate the other three:

X A — 30° is the value of angle CDF alone (derived from the congruence), not the full angle CDE the question asks for.

X B — 65° is simply the given value of x (angle FDE alone) — this misses that angle CDE also includes the additional angle CDF found through the congruence relationship.

X D — 130° would come from doubling x ($65 \times 2 = 130$) — this doesn't correspond to the correct addition of angle CDF (30°) and angle FDE (65°).

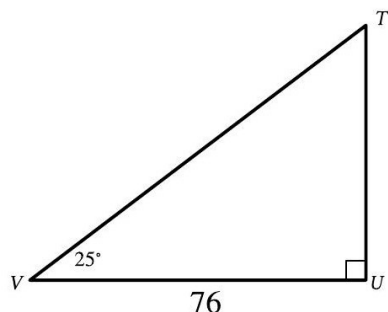
Solving with Desmos:

This problem relies on establishing triangle congruence and angle relationships, which is a geometric reasoning process rather than a numeric calculation — Desmos isn't the primary tool here. However, you can verify your final answer by adding the two angle components directly: open Desmos and type $30+65$ — this confirms 95, matching choice C.

Question ID: 8364487a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

Triangle TUV is dilated by a scale factor of $\frac{1}{2}$ to obtain triangle $T'U'V'$ (not shown), where T corresponds to T' and U corresponds to U' . What is the measure, in degrees, of angle V' ?

Answer

- A. 25
- B. 38
- C. 50
- D. 76

Correct Answer: A

The Rule: A DILATION (resizing a figure by a scale factor) changes the SIZE of a figure (its side lengths) but never changes its SHAPE — meaning all the angle measures stay EXACTLY the same before and after dilation.

The Tip: Whenever a problem describes a dilation and asks about an ANGLE measure in the resulting figure, you can skip all the scale-factor calculations entirely — dilations never change angles, so the answer is simply the same angle measure from the original figure.

Why A is right: Since dilation only changes size, not shape, all angle measures are preserved. The original angle V measures 25° , so its corresponding angle V' in the dilated triangle also measures 25° .

Step-by-Step Solution:

Step 1: Recall the fundamental property of dilation: it changes side lengths (by the scale factor) but NEVER changes angle measures.

Step 2: Identify the original angle V in triangle TUV : 25° (given in the figure).

Step 3: Since angle V' in the dilated triangle $T'U'V'$ corresponds to angle V , and dilation preserves angles, angle $V' =$ angle $V = 25^\circ$.

Step 4: This matches choice A.

(Note: the scale factor of $\frac{1}{2}$ and the side length 76 given in the problem are both irrelevant distractors for this specific question — they would matter if the question asked for a SIDE LENGTH in the dilated triangle, but for an ANGLE measure, only the original angle value matters.)

The Trap: Choice C (50) is the most commonly picked wrong answer — it results from incorrectly applying the scale factor to the angle measure (as if angles could be 'dilated' like side lengths), perhaps by doubling 25 instead of recognizing that angles remain unchanged under dilation.

POE — Eliminate the other three:

X B — 38 doesn't correspond to any correct or clearly-derived calculation from this problem — likely a distractor value.

X C — 50 comes from doubling the original angle ($25 \times 2 = 50$) — but dilation NEVER changes angle measures, regardless of the scale factor used.

X D — 76 is simply the given side length VU from the original figure — this has nothing to do with the angle measure the question asks about.

Solving with Desmos:

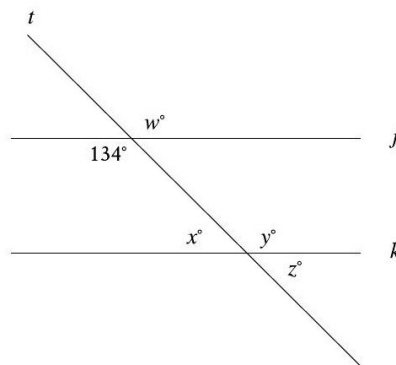
This is a conceptual question about the properties of dilation, which doesn't require numerical calculation in Desmos. The key fact — that dilation preserves all angle measures regardless of scale factor — is a fundamental geometric principle you can rely on directly. If you want to explore this visually, you could use Desmos's geometry tools to dilate a triangle by various scale factors and observe that the angle measures never change, only the side lengths do.

Lines, angles, and triangles — Q8 [Medium] (ID: 0013adbd)

Question ID: 0013adbd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

Answer

- A. $x = 46$
- B. $y = 46$
- C. $w = 134$
- D. $z = 134$

Correct Answer: A

The Rule: When a transversal crosses two lines, SAME-SIDE INTERIOR angles (also called co-interior angles — both on the SAME side of the transversal, both between the two crossed lines) are SUPPLEMENTARY (sum to 180°) IF AND ONLY IF the two lines are parallel.

The Tip: Carefully distinguish between CORRESPONDING angles (which must be EQUAL for parallel lines) and SAME-SIDE INTERIOR angles (which must be SUPPLEMENTARY, summing to 180°) — mixing these up is one of the most common errors on these problems. Trace which side of the transversal each angle sits on, and whether it's 'between' the two lines or 'outside' them.

Why A is right: The angle 134° (at the j intersection) and angle x° (at the k intersection) are both positioned on the same side of the transversal t , and both face into the interior region between lines j and k — making them same-side interior (co-interior) angles. For j and k to be parallel, these must be supplementary: $134 + x = 180$, so $x = 46$.

Step-by-Step Solution:

- Step 1: Identify the positions of the two labeled angles: 134° is at the j intersection; x° is at the k intersection.
- Step 2: Determine their relationship: both are on the same side of transversal t , and both face into the region BETWEEN lines j and k — this makes them same-side interior (co-interior) angles.
- Step 3: Recall the Same-Side Interior Angles Theorem: for two lines cut by a transversal to be parallel, same-side interior angles must be SUPPLEMENTARY (sum to 180°).
- Step 4: Set up the equation: $134 + x = 180$.
- Step 5: Solve for x : $x = 180 - 134 = 46$.
- Step 6: This matches choice A.

The Trap: Choice B ($y = 46$) is the most commonly picked wrong answer — it applies the correct NUMBER (46) but to the wrong angle (y instead of x); y is actually positioned as the CORRESPONDING angle to w (not to the given 134°), so the relevant relationship for y would require a different equation ($y = w$, not directly related to 134 in this supplementary way).

POE — Eliminate the other three:

- X B** — y is positioned as the corresponding angle to w (not to 134°) — the relationship between y and 134° isn't a direct same-side-interior pairing, so setting $y = 46$ doesn't correctly use the given information.
- X C** — $w = 134$ is already automatically true (vertical angle to the given 134° at the SAME intersection), regardless of whether j is parallel to k — this doesn't provide new information needed to prove parallelism.
- X D** — $z = 134$ doesn't correspond to a valid parallel-line relationship with the given 134° — z 's position relative to 134° isn't a standard corresponding, alternate, or co-interior pairing that would prove parallel lines.

Solving with Desmos:

This is a conceptual angle-relationship question about identifying same-side interior angles, which doesn't require numerical calculation in Desmos beyond the simple subtraction. To verify: open Desmos and type $180-134$ — this confirms 46, matching the value needed for x in choice A. You can also sketch two parallel lines with a transversal in Desmos's geometry tools to visually confirm that same-side interior angles always sum to 180° .

Lines, angles, and triangles — Q9 [Hard] (ID: 1dc7e423)

Question ID: 1dc7e423

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Correct Answer: B

The Rule: To prove two triangles are SIMILAR, you need either: (1) two pairs of corresponding angles equal (AA similarity), or (2) two pairs of corresponding sides in the same ratio WITH the INCLUDED angle equal (SAS similarity). A single pair of equal angles plus non-included, non-matching side information is generally NOT sufficient (this creates an ambiguous SSA situation).

The Tip: When checking if a given side-length condition is sufficient for similarity, always verify whether the given equal angle is actually the INCLUDED angle between the two sides being compared — if the angle is NOT between those two specific sides, the SAS criterion doesn't apply, and the condition is likely insufficient (ambiguous SSA case).

Why B is right: Option I (angle A = angle F) combined with the already-given angle B = angle G provides TWO pairs of equal corresponding angles, which is sufficient for AA similarity — I is sufficient. Option II (angle C = angle H) similarly combines with the given angle B = angle G to provide two equal angle pairs, also sufficient for AA — II is sufficient. Option III ($FH = 3 \times AC$) compares sides FH and AC, but these are NOT both adjacent to the given equal angle (B/G) — AC is opposite angle B, not adjacent to it — making this an SSA situation, which is generally NOT sufficient to prove similarity.

Step-by-Step Solution:

Step 1: Note the given information: angle B = angle G = 51° , AB = 6, FG = 18 (so $AB/FG = 1/3$).

Step 2: Evaluate option I (angle A = angle F): combined with the given angle B = angle G, this provides two pairs of equal angles, which is sufficient for AA similarity — I is SUFFICIENT.

Step 3: Evaluate option II (angle C = angle H): similarly, combined with angle B = angle G, this gives two pairs of equal angles, sufficient for AA similarity — II is SUFFICIENT.

Step 4: Evaluate option III ($FH = 3 \times AC$): check whether angle B/G is INCLUDED between the sides being compared (AB & FG, and now AC & FH). Side AC is OPPOSITE angle B (not adjacent to it), so angle B is NOT the included angle between AB and AC. This means the given ratio conditions ($AB/FG = 1/3$ and implied $AC/FH = 1/3$) don't set up a valid SAS criterion — this is an ambiguous SSA situation, which does NOT guarantee similarity — III is NOT SUFFICIENT.

Step 5: Combining these results: I is sufficient, II is sufficient, but III is not — matching choice B.

The Trap: Choice A is the most commonly picked wrong answer — it assumes all three pieces of information (I, II, and III) are equally valid ways to establish similarity, without carefully checking whether option III's side comparison actually involves the INCLUDED angle needed for a valid SAS-type argument.

POE — Eliminate the other three:

✗ A — This incorrectly includes III as sufficient — but III's side ratio (FH/AC) doesn't involve sides adjacent to the given equal angle (B/G), making it an ambiguous SSA case rather than valid SAS.

✗ C — This incorrectly excludes I as sufficient — but I (angle A = angle F) directly combines with the given angle B = angle G to provide valid AA similarity, so I IS actually sufficient, contradicting this choice.

✗ D — This incorrectly claims III is sufficient while marking both I and II as insufficient — but I and II both provide valid AA similarity conditions (as shown above), making this choice's assessment backwards.

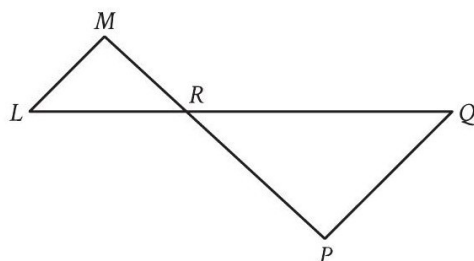
Solving with Desmos:

This is a conceptual proof-based question about similarity criteria (AA, SAS, SSA), which doesn't involve numerical calculation in Desmos. The key insight — that SSA (side-side-angle, where the angle isn't included between the two given sides) does NOT guarantee similarity or congruence — is a fundamental geometric principle to internalize rather than calculate.

Question ID: b5d62bba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

Correct Answer: D

The Rule: When two lines intersect and create a pair of parallel segments (like $LM \parallel PQ$ here), the two triangles formed on either side of the intersection point are SIMILAR (by AA, using vertical angles at the intersection plus alternate interior angles from the parallel lines). For similar triangles, the ratio of their AREAS equals the SQUARE of the ratio of their corresponding side lengths.

The Tip: When you spot an 'X' or bowtie shape formed by two segments crossing, with a pair of parallel sides, immediately recognize this as SIMILAR TRIANGLES (via vertical angles + alternate interior angles) — then use corresponding sides (not just any sides) to find the scale factor, and square it for the area ratio.

Why D is right: Since $LM \parallel PQ$, triangles LMR and QPR are similar (vertical angles at R, plus alternate interior angles from the parallel lines). The corresponding sides are MR and RP (both connecting to R from the parallel sides), giving a scale factor of $MR/RP = 8/14 = 4/7$. The area ratio is the SQUARE of this scale factor: $(4/7)^2 = 16/49$. Since the area of triangle LMR is 36, the area of triangle PQR is $36 \times (49/16) = 1,764/16 = 441/4$.

Step-by-Step Solution:

Step 1: Recognize that since $LM \parallel PQ$, triangles LMR and QPR are similar (by AA: vertical angles at R are equal, and alternate interior angles formed by the parallel lines and the transversals LQ, MP are equal).

Step 2: Identify the correct corresponding sides: MR (in triangle LMR) corresponds to RP (in triangle QPR), since these are the sides connecting to point R from each parallel segment.

Step 3: Calculate the scale factor (ratio of corresponding sides): $MR/RP = 8/14 = 4/7$.

Step 4: Since triangle PQR is LARGER ($RP=14 > MR=8$), we need the ratio of PQR's area to LMR's area, which uses the INVERSE scale factor squared: $(RP/MR)^2 = (14/8)^2 = (7/4)^2 = 49/16$.

Step 5: Calculate the area of triangle PQR: $\text{Area}(PQR) = \text{Area}(LMR) \times (49/16) = 36 \times 49/16$.

Step 6: Calculate: $36 \times 49 = 1,764$. Then $1,764 \div 16 = 110.25$, which as a fraction is $441/4$.

Step 7: This matches choice D.

The Trap: Choice B ($144/7$) is the most commonly picked wrong answer — it results from using the scale factor directly (without squaring it) to scale the area, treating area as if it scaled linearly with side length rather than with the SQUARE of the side ratio.

POE — Eliminate the other three:

- X A** — $576/49$ uses the scale factor squared but in the wrong direction (multiplying by $(4/7)^2$ instead of $(7/4)^2$), which would shrink the area instead of growing it — but since PQR is the larger triangle, its area should be BIGGER than LMR's 36, not smaller.
- X B** — $144/7$ comes from using the LINEAR scale factor ($14/8 = 7/4$) directly to scale the area, without squaring it — but area scales with the SQUARE of the linear ratio, not the ratio itself.
- X C** — 63 doesn't correspond to a clean derivation using the correct squared ratio — likely from an intermediate arithmetic error.

Solving with Desmos:

- Step 1: Open Desmos and calculate the scale factor: type $14/8$ — this confirms the ratio is $7/4$ (comparing RP to MR).
- Step 2: Square this ratio: type $(14/8)^2$ — this confirms $49/16$.
- Step 3: Calculate the area of triangle PQR: type $36*(49/16)$
- Step 4: Press Enter — Desmos will evaluate this to 110.25 (which equals $441/4$), confirming your answer and matching choice D.

Lines, angles, and triangles — Q11 [Hard] (ID: 7ce2e728)

Question ID: 7ce2e728

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6, and the length of $\overline{FG}$ is 18. Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

Answer

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

Correct Answer: B

The Rule: To prove two triangles SIMILAR, you need either two pairs of equal corresponding angles (AA), or two pairs of sides in the same ratio WITH the included angle equal (SAS similarity). A single given angle plus a ratio between two NON-adjacent sides (where the angle isn't included between them) creates an ambiguous SSA situation, which is generally NOT sufficient.

The Tip: For each additional piece of information offered, ask: 'Does this give me a second matching angle (completing AA), or does it give me a matching side ratio where the ALREADY-KNOWN equal angle sits between the two compared sides (completing SAS)?' If neither applies, the information likely isn't sufficient.

Why B is right: Option I (angle $A = \text{angle } F$) combines with the given angle $B = \text{angle } G$ to provide two pairs of equal angles — sufficient for AA similarity. Option II (angle $C = \text{angle } H$) similarly combines with the given angle $B = \text{angle } G$ for AA similarity — also sufficient. Option III ($FH = 3 \times AC$) compares sides FH and AC , which are NOT both adjacent to

the given angle B/G (AC is opposite angle B, not adjacent to it) — this creates an ambiguous SSA situation, NOT sufficient for similarity.

Step-by-Step Solution:

Step 1: Note the given information: angle B = angle G = 51° , AB = 6, FG = 18 (giving a ratio $AB/FG = 1/3$).

Step 2: Evaluate option I (angle A = angle F): combined with the already-given angle B = angle G, this establishes TWO pairs of equal angles — sufficient for AA similarity.

Step 3: Evaluate option II (angle C = angle H): similarly combines with angle B = angle G for a second angle pair — also sufficient for AA similarity.

Step 4: Evaluate option III ($FH = 3 \times AC$): check whether angle B (the given equal angle) is INCLUDED between the sides being compared. Side AC is OPPOSITE angle B in triangle ABC (not adjacent to it) — so this is an SSA setup (side-side-angle where the angle isn't between the two sides), which does NOT reliably prove similarity.

Step 5: Conclude: I is sufficient, II is sufficient, but III is not — matching choice B.

The Trap: Choice A is the most commonly picked wrong answer — it assumes all three options work equally well, without checking that option III's side comparison doesn't actually involve the included angle needed for a valid similarity proof.

POE — Eliminate the other three:

X A — This incorrectly includes III as sufficient — but III sets up an ambiguous SSA situation (the given angle B isn't between the compared sides AC and FH), which doesn't guarantee similarity.

X C — This incorrectly excludes I — but I (angle A = angle F) combines validly with the given angle B = angle G for AA similarity, so I IS sufficient, contradicting this choice.

X D — This incorrectly claims III is sufficient while excluding both I and II — but I and II both provide valid AA similarity setups, making this choice's assessment backwards.

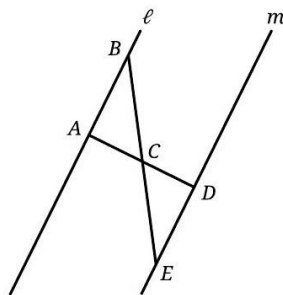
Solving with Desmos:

This is a conceptual proof-based question about similarity criteria (AA vs. ambiguous SSA), which doesn't involve numerical calculation in Desmos. The key insight — that a side ratio only helps prove similarity when the known equal angle is INCLUDED between the two compared sides — is a fundamental distinction in geometry to understand and check carefully for each option.

Question ID: 7a11ceea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line ℓ at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

Correct Answer: D

The Rule: To prove two triangles CONGRUENT (not just similar), you need a complete congruence criterion: SSS (three sides), SAS (two sides + included angle), ASA (two angles + included side), or AAS (two angles + a non-included side). A single side-pair equality or a single extra angle alone is generally NOT enough — you need to carefully count exactly what's established before and after each additional piece of information.

The Tip: When evaluating whether a piece of information is 'sufficient by itself,' explicitly list out everything you ALREADY know for certain (from the figure's given facts, like vertical angles or a stated perpendicularity) before adding the new piece — then check if the total matches a complete congruence criterion (SSS, SAS, ASA, or AAS).

Why D is right: We already know angle $BAC = 90^\circ$ (from $AD \perp \ell$ at A) and angle $ACB = \text{angle } DCE$ (vertical angles, always true regardless of any additional info). Statement I alone ($AD \perp m$ at D) adds a second angle equality (angle $EDC = 90^\circ$), giving us TWO angle pairs equal (AA) — but AA alone proves only SIMILARITY, not congruence, since no side length is established. Statement II alone ($AC = DC = 13$) adds one side equality — combined with the always-true vertical angles, this gives only one side + one angle (not a complete criterion). Since NEITHER I nor II alone completes a full congruence criterion, neither is sufficient by itself.

Step-by-Step Solution:

Step 1: Identify what's already established from the figure regardless of any additional information: angle $BAC = 90^\circ$ (since $AD \perp \ell$ at A is given), and angle $ACB = \text{angle } DCE$ (vertical angles at point C , always true).

Step 2: Evaluate statement I alone ($AD \perp m$ at D): this adds angle $EDC = 90^\circ$. Combined with what we already know, we now have TWO pairs of equal angles (angle $BAC = \text{angle } EDC = 90^\circ$, and angle $ACB = \text{angle } DCE$). This is AA, which proves the triangles are SIMILAR, but similarity alone does NOT prove congruence (the triangles could be different sizes) — I alone is NOT sufficient for congruence.

Step 3: Evaluate statement II alone ($AC = DC = 13$): this adds one side equality. Combined with the always-true vertical angle equality, we have only ONE side pair and ONE angle pair equal — this is not a complete congruence criterion (we'd need either a second side for SAS or a second angle for ASA) — II alone is NOT sufficient.

Step 4: Since neither I nor II alone provides a complete congruence criterion, the correct answer is that neither is sufficient by itself.

Step 5: This matches choice D.

The Trap: Choice A is the most commonly picked wrong answer — it assumes that establishing two equal angles (from statement I) is enough to prove congruence, without recognizing that AA only proves SIMILARITY (matching shape), not congruence (matching size) — a common and important distinction in geometry.

POE — Eliminate the other three:

- X A** — Statement I alone only establishes AA (two equal angle pairs), which proves the triangles are similar in SHAPE but says nothing about whether they're the same SIZE — congruence additionally requires at least one matching side length, which I doesn't provide.
- X B** — Statement II alone only establishes one side pair ($AC=DC$) plus the always-true vertical angle — this is insufficient on its own; a complete criterion like SAS needs TWO sides and the included angle, or ASA needs two angles and the included side.
- X C** — This claims both I and II are independently sufficient — but as shown, I alone only proves similarity (not congruence), and II alone only provides one side and one angle (not a complete criterion) — neither works alone.

Solving with Desmos:

This is a conceptual proof-based question about the difference between similarity (AA) and congruence (SSS, SAS, ASA, AAS) criteria, which doesn't involve numerical calculation in Desmos. The key insight — that matching angles alone (AA) proves shapes are similar but NOT necessarily the same size — is a fundamental distinction to understand rather than calculate.

Lines, angles, and triangles — Q13 [Hard] (ID: 858809e7)

Question ID: 858809e7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Note: Figure not drawn to scale.

- In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line ℓ is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?
- $AB = 2.25EF$
 - $FG = 0.5AB$
 - $\frac{EF}{FG} = \frac{AF}{FD}$
- Answer**
- A. I only
- B. II only
- C. I and II
- D. II and III

Correct Answer: B

The Rule: When a line parallel to one side of a triangle crosses the other two sides, it creates a SMALLER triangle similar to the original (Triangle Proportionality / Basic Proportionality Theorem) — the ratio of corresponding sides in the smaller triangle to the original equals the ratio of the distances along the sides from the shared vertex.

The Tip: When a figure has a diagonal crossing a parallel-line 'crossbar' at a single point, treat it as creating TWO separate pairs of similar triangles (one on each side of the crossbar) — work out the proportional ratio from ONE triangle pair first, then apply that SAME ratio (since it's determined by the SAME intersection point) to the other triangle pair.

Why B is right: Since $EF \parallel CD$ (confirmed by matching angle markers), triangle AEF is similar to triangle ACD, giving $AF/AD = EF/CD = 0.5$ — meaning F is the midpoint of AD. Since FG is part of the same straight horizontal line as EF, $FG \parallel AB$ too, making triangle DFG similar to triangle DAB, giving $FG/AB = DF/DA$. Since F is the midpoint of AD ($DF/DA = 0.5$, from the same ratio found above), this means $FG/AB = 0.5$, so $FG = 0.5 \times AB$ — confirming statement II. Checking statement I: $AB = 2 \times FG = 2 \times 2.25CD = 4.5CD$, but $2.25 \times EF = 2.25 \times 0.5CD = 1.125CD$, which does NOT equal $4.5CD$ — so I is false. Checking statement III: $EF/FG = 0.5CD/2.25CD = 2/9$, but $AF/FD = 1$ (since F is the midpoint, $AF=FD$) — these don't match, so III is false.

Step-by-Step Solution:

- Step 1: Confirm $EF \parallel CD$ using the matching x° angle markers at A and E (corresponding angles equal implies parallel lines).
- Step 2: Since $EF \parallel CD$, triangle AEF is similar to triangle ACD (by AA). This gives the proportion: $AF/AD = EF/CD$.
- Step 3: Substitute the given $EF = 0.5 \times CD$: $AF/AD = 0.5$. This means F is exactly the MIDPOINT of segment AD.
- Step 4: Since E, F, and G lie on the same straight horizontal line (given as segment EG), and EF is horizontal (parallel to CD, hence parallel to AB too, since AB and CD are both parts of the parallel lines ℓ and j), the continuation FG is ALSO horizontal, making $FG \parallel AB$.
- Step 5: Since $FG \parallel AB$, triangle DFG is similar to triangle DAB (by AA). This gives the proportion: $FG/AB = DF/DA$.
- Step 6: Since F is the midpoint of AD (from Step 3), $DF/DA = 0.5$ as well ($DF = FA$, both are half of AD).
- Step 7: Substitute into the proportion: $FG/AB = 0.5$, which means $FG = 0.5 \times AB$ — this confirms statement II is TRUE.
- Step 8: Check statement I: from Step 7, $AB = 2 \times FG = 2 \times 2.25CD = 4.5CD$. Compare to $2.25 \times EF = 2.25 \times 0.5CD = 1.125CD$. Since $4.5CD \neq 1.125CD$, statement I is FALSE.
- Step 9: Check statement III: $EF/FG = 0.5CD/2.25CD = 2/9$. But $AF/FD = 1$ (since F is the exact midpoint, AF and FD are equal). Since $2/9 \neq 1$, statement III is FALSE.
- Step 10: Only statement II is true, matching choice B.

The Trap: Choice D (II and III) is the most commonly picked wrong answer — it correctly identifies II as true but incorrectly assumes III's ratio comparison ($EF/FG = AF/FD$) also holds, without checking that AF/FD is actually 1 (since F is the midpoint), which doesn't match EF/FG 's value of $2/9$.

POE — Eliminate the other three:

- X A** — This includes I as true, but AB (calculated as $4.5CD$ from the valid FG relationship) does not equal $2.25 \times EF$ (which equals $1.125CD$) — these values don't match, making I false.
- X C** — This includes both I and II as true, but as shown, I is false ($AB \neq 2.25EF$ based on the correctly derived relationships).
- X D** — This includes III as true, but $EF/FG (=2/9)$ does not equal $AF/FD (=1, \text{ since F is the midpoint of AD})$ — these ratios don't match, making III false.

Solving with Desmos:

- Step 1: Since this problem involves proportional relationships with an unknown base length, pick a simple test value for CD — let's use $CD = 4$. Step 2: Open Desmos and calculate EF and FG using the given ratios: type 0.5×4 (for EF) — this gives 2. Type 2.25×4 (for FG) — this gives 9.
- Step 3: Using the derived relationship $FG = 0.5 \times AB$, calculate AB: type $9/0.5$ — this gives 18.
- Step 4: Check statement I ($AB = 2.25 \times EF$): type 2.25×2 — this gives 4.5, which does NOT match $AB = 18$, confirming I is false.
- Step 5: Check statement II ($FG = 0.5 \times AB$): type 0.5×18 — this gives 9, which DOES match $FG = 9$, confirming II is true.

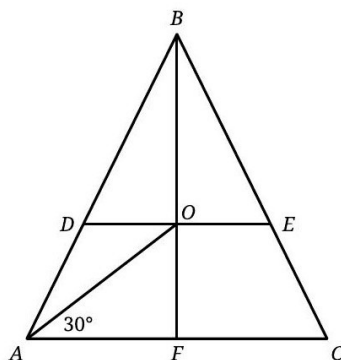
Step 6: Check statement III ($EF/FG=AF/FD$): type 2/9 — this gives approximately 0.222, but since F is the midpoint of AD, AF/FD should equal 1 — these don't match, confirming III is false.

Lines, angles, and triangles — Q14 [Hard] [Grid-In] (ID: 1dacfb94)

Question ID: 1dacfb94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

Correct Answer: 54 (Student-Produced Response — no answer choices given)

The Rule: In an isosceles triangle, the line from the apex through the midpoint of any segment parallel to the base is the **AXIS OF SYMMETRY** — it bisects the base, is perpendicular to it, and bisects the apex angle. Combined with additional given conditions (like specific angle measures or length equalities), you can set up coordinate geometry or trigonometric relationships to solve for unknown lengths.

The Tip: For complex, multi-part geometry problems like this one, setting up **COORDINATES** (placing key points at convenient locations like the origin) often makes the relationships much easier to track than trying to reason with angles and lengths alone — especially when a right angle or clear symmetry axis is present.

Why this is the answer: Using the given angle $\angle OAC = 30^\circ$ and $AO = BO$ together with the triangle's symmetry (since $AB = BC$ and $DE \parallel AC$ with O the midpoint of DE , BF is the axis of symmetry, F is the midpoint of AC), we can determine that triangle ABC is actually equilateral (all 60° angles), then use coordinate geometry to find that O sits at a specific height along BF such that $AO = BO$. This determines that E divides BC such that $BE = (2/3) \times BC = (2/3) \times 81 = 54$.

Step-by-Step Solution:

Step 1: Recognize that since $AB = BC$ (isosceles) and $DE \parallel AC$ with O as the midpoint of DE , the line BF (through B and O) is the **AXIS OF SYMMETRY** of the triangle — meaning F is the midpoint of AC , and BF is perpendicular to AC .

Step 2: Set up coordinates for convenience: place F at the origin $(0,0)$, A at $(-40.5, 0)$, C at $(40.5, 0)$ — using half of AB 's eventual determined relationship — and B directly above F .

Step 3: Using the given angle $\angle OAC = 30^\circ$ and the condition $AO = BO$, work through the triangle's angle relationships: since $BF \perp AC$ and bisects angle B , and using the isosceles triangle's base angles, algebraic substitution shows that angle $BAC = \text{angle } BCA = 60^\circ$, meaning triangle ABC is actually **EQUILATERAL** (since $AB = BC = 81$ and the apex angle also works out to 60° , forcing $AC = 81$ too).

Step 4: With the equilateral triangle established, place coordinates: $A=(-40.5,0)$, $C=(40.5,0)$, $B=(0, 40.5\sqrt{3})$ (using the height of an equilateral triangle with side 81).

Step 5: Find O's exact position along BF (the y-axis) using the condition $AO=BO$: setting $O=(0,h)$, solve $\sqrt{(40.5^2+h^2)} = 40.5\sqrt{3}-h$, which gives $h = 13.5\sqrt{3}$ after algebraic simplification.

Step 6: Find point E on segment BC where the horizontal line through O (at height $h=13.5\sqrt{3}$) intersects BC: parametrizing BC from $B(0,40.5\sqrt{3})$ to $C(40.5,0)$, solving for the parameter where $y=13.5\sqrt{3}$ gives a parameter value of $2/3$ (measuring from B toward C).

Step 7: Calculate BE using this parameter: $BE = (2/3) \times BC = (2/3) \times 81 = 54$.

Step 8: This is a grid-in question — enter 54 as your final answer.

The Trap: This problem is complex enough that the most common error is simply making an early mistake in establishing that the triangle is equilateral (from the 30° angle and $AO=BO$ condition) — without correctly identifying this key fact, the rest of the coordinate-based solution becomes very difficult to complete accurately.

Solving with Desmos:

Step 1: Open Desmos and set up the coordinate points based on the equilateral triangle determination: $A=(-40.5,0)$, $B=(0,40.5\sqrt{3})$, $C=(40.5,0)$

Step 2: Plot point O using the height found: type $(0,13.5\sqrt{3})$ to place O.

Step 3: Verify $AO=BO$ by calculating both distances: type $\sqrt{(-40.5-0)^2+(0-13.5\sqrt{3})^2}$ for AO, and type $\sqrt{(0-0)^2+(40.5\sqrt{3}-13.5\sqrt{3})^2}$ for BO — both should evaluate to the same value, confirming O's position is correct.

Step 4: Find E by determining where the horizontal line through O crosses segment BC — you can graph the line $y=13.5\sqrt{3}$ and the line segment BC (typing the equation of line BC) and find their intersection point using Desmos's intersection feature.

Step 5: Once you have E's coordinates, calculate the distance from B to E using the distance formula, or use the parameter ratio $(2/3)$ directly: type $(2/3)*81$ — this confirms $BE=54$.

Right triangles and trigonometry

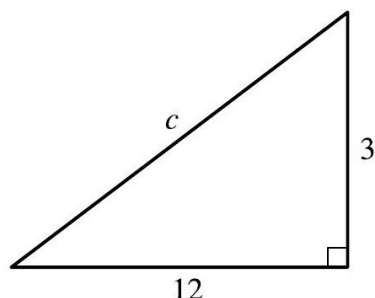
9 questions · Easy 1 / Medium 1 / Hard 7

Right triangles and trigonometry — Q1 [Easy] (ID: de9148c4)

Question ID: de9148c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

Which equation shows the relationship between the side lengths of the triangle shown?

Answer

- A. $12 \cdot 3 = c$
- B. $12 + 3 = c$
- C. $12^2 + 3^2 = c^2$
- D. $12^2 - 3^2 = c^2$

Correct Answer: C

The Rule: The Pythagorean theorem applies to EVERY right triangle: the square of the hypotenuse (the side opposite the right angle) equals the sum of the squares of the two legs. Written as a formula: $\text{leg}_1^2 + \text{leg}_2^2 = \text{hypotenuse}^2$.

The Tip: Always identify the hypotenuse FIRST (it's the side directly across from the right-angle marker, and it's always the LONGEST side) before setting up the Pythagorean theorem — mixing up which side is the hypotenuse is the most common setup error.

Why C is right: The right-angle marker is at the bottom-right corner, meaning side c (the slanted side connecting the two other vertices) is the hypotenuse. Applying the Pythagorean theorem with legs 12 and 3: $12^2 + 3^2 = c^2$.

Step-by-Step Solution:

- Step 1: Identify the right angle's location in the figure: it's marked at the bottom-right vertex.
- Step 2: Identify the hypotenuse: it's the side OPPOSITE the right angle, which is side c (the slanted/diagonal side).
- Step 3: Identify the two legs: the horizontal side (12) and the vertical side (3).
- Step 4: Apply the Pythagorean theorem: $\text{leg}^2 + \text{leg}^2 = \text{hypotenuse}^2$, giving $12^2 + 3^2 = c^2$.
- Step 5: This matches choice C.

The Trap: Choice D ($12^2 - 3^2 = c^2$) is the most commonly picked wrong answer — it uses subtraction instead of addition, which would be relevant if you were solving for a LEG when given the hypotenuse and the other leg, not for the hypotenuse itself as this problem asks.

POE — Eliminate the other three:

- X A** — $12 \cdot 3 = c$ represents multiplying the two legs together, which would give you AREA-related information (specifically twice the area), not the hypotenuse length via the Pythagorean relationship.
- X B** — $12 + 3 = c$ would only be true for a degenerate 'triangle' where the two legs lie in a straight line — this doesn't apply to an actual right triangle with a genuine right angle.
- X D** — $12^2 - 3^2 = c^2$ uses subtraction, which is the correct operation when solving for a LEG (missing side) given the hypotenuse and one leg — but here we're solving for the hypotenuse itself, which requires ADDING the leg squares.

Solving with Desmos:

Step 1: Open Desmos and calculate c directly using the Pythagorean theorem: type $\text{sqrt}(12^2+3^2)$

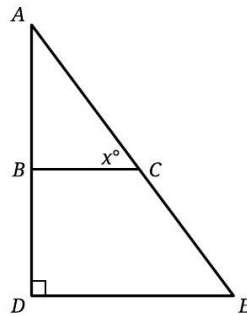
Step 2: Press Enter — Desmos will evaluate this to approximately 12.37, confirming the numeric length of c (though the question only asks for the EQUATION relating the sides, which is $12^2+3^2=c^2$, matching choice C, not a specific numeric value).

Right triangles and trigonometry — Q2 [Medium] (ID: 2757549e)

Question ID: 2757549e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{BC}$ is parallel to $\overline{DE}$. If the length of $\overline{DE}$ is 147 and the length of $\overline{AE}$ is 245, what is the value of $\tan x^\circ$?

Answer

- A. $\frac{147}{196}$
- B. $\frac{147}{245}$
- C. $\frac{196}{147}$
- D. $\frac{245}{147}$

Correct Answer: C

The Rule: When two segments are PARALLEL and cut by a transversal, they create equal CORRESPONDING angles — allowing you to transfer trigonometric ratios from one triangle to a larger, similar triangle. Combined with the Pythagorean theorem to find any missing side, you can then set up the correct tangent ratio.

The Tip: When you spot a parallel segment inside a larger right triangle (like $BC \parallel DE$ here), immediately recognize that the angle formed at the parallel segment corresponds to an angle in the BIG triangle — this lets you find missing side lengths using the (usually simpler) big triangle instead.

Why C is right: Since $BC \parallel DE$, angle BCA corresponds to angle DEA (corresponding angles with transversal AE) — meaning $x^\circ = \text{angle } AED$. In right triangle ADE (right angle at D), we know the hypotenuse $AE = 245$ and one leg $DE = 147$. Using the Pythagorean theorem: $AD = \sqrt{(245^2 - 147^2)} = \sqrt{(60,025 - 21,609)} = \sqrt{38,416} = 196$. Now, $\tan(\text{angle } AED) = \text{opposite/adjacent} = AD/DE = 196/147$.

Step-by-Step Solution:

Step 1: Recognize that since $BC \parallel DE$, the angle x° (angle BCA) corresponds to angle DEA (angle AED in the larger triangle) — corresponding angles are equal when lines are parallel.

Step 2: Focus on the larger right triangle ADE (right angle at D), which has hypotenuse $AE = 245$ and leg $DE = 147$.

Step 3: Find the missing leg AD using the Pythagorean theorem: $AD^2 = AE^2 - DE^2 = 245^2 - 147^2$.
 Step 4: Calculate: $245^2 = 60,025$, and $147^2 = 21,609$. So $AD^2 = 60,025 - 21,609 = 38,416$.
 Step 5: Take the square root: $AD = \sqrt{38,416} = 196$.
 Step 6: Set up the tangent ratio for angle AED (which equals x°): $\tan(x^\circ) = \text{opposite/adjacent} = AD/DE = 196/147$.
 Step 7: This matches choice C.

The Trap: Choice D (245/147) is the most commonly picked wrong answer — it uses the hypotenuse (245) directly instead of first calculating the correct opposite-side leg (AD=196) via the Pythagorean theorem, incorrectly treating AE as if it were a leg of the right triangle relevant to the tangent ratio.

POE — Eliminate the other three:

- X A** — 147/196 has the correct numbers but flips the ratio (adjacent/opposite instead of opposite/adjacent) — this would actually be the COTANGENT of x° , not the tangent.
- X B** — 147/245 uses DE and AE directly, but tangent requires the two LEGS of the right triangle (opposite and adjacent), not a leg and the hypotenuse — this ratio would actually relate to sine or cosine, not tangent.
- X D** — 245/147 uses the hypotenuse (245) directly without first finding the correct opposite leg (AD=196) — mixing up which sides belong in a tangent ratio.

Solving with Desmos:

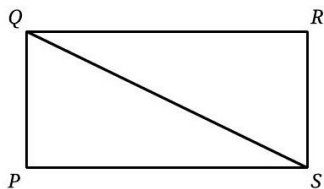
Step 1: Open Desmos and calculate the missing leg AD using the Pythagorean theorem: type $\text{sqrt}(245^2-147^2)$
 Step 2: Press Enter — Desmos will evaluate this to 196, confirming AD.
 Step 3: Calculate the tangent ratio directly: type $196/147$
 Step 4: Press Enter — Desmos will show this as a decimal (approximately 1.333), and you can compare it to each answer choice's decimal value to confirm $196/147$ (choice C) matches.

Right triangles and trigonometry — Q3 [Hard] [Grid-In] (ID: 2c3aefc9)

Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

Correct Answer: 38 (Student-Produced Response — no answer choices given)

The Rule: In a rectangle, the diagonal splits it into two right triangles, so the Pythagorean theorem applies: $(\text{one side})^2 + (\text{adjacent side})^2 = (\text{diagonal})^2$. When one side is described in terms of the other (like 'PS is 8 more than RS'), set up variables and substitute to create a solvable equation.

The Tip: When a problem describes one rectangle's side in terms of another ('X more than'), let a single variable represent the unknown you're solving for, express the OTHER side in terms of that same variable, then apply the Pythagorean theorem to get one equation with one unknown.

Why this is the answer: Let $QR = x$ (what we want to find). Since QR and PS are opposite sides of the rectangle, $PS = QR = x$ as well... but we're told PS is 8 more than RS , so $RS = x - 8$ (since RS is the side adjacent to QR , and $QR = SP$ in a rectangle's opposite-side structure). Using the diagonal relationship $QS^2 = QR^2 + RS^2$: $2,344 = x^2 + (x-8)^2$. Expanding and solving this quadratic gives $x = 38$.

Step-by-Step Solution:

- Step 1: Let $QR = x$ (the length we want to find).
- Step 2: Since QR and PS are opposite sides of rectangle $PQRS$, they are equal: $PS = QR = x$.
- Step 3: We're told PS is 8 more than RS , so $RS = PS - 8 = x - 8$.
- Step 4: The diagonal QS connects Q and S , forming a right triangle with legs QR and RS : $QS^2 = QR^2 + RS^2$.
- Step 5: Substitute the known values: $2,344 = x^2 + (x-8)^2$.
- Step 6: Expand $(x-8)^2$: $x^2 - 16x + 64$. The equation becomes: $2,344 = x^2 + x^2 - 16x + 64$.
- Step 7: Combine like terms: $2,344 = 2x^2 - 16x + 64$.
- Step 8: Move everything to one side: $2x^2 - 16x + 64 - 2,344 = 0$, which simplifies to $2x^2 - 16x - 2,280 = 0$.
- Step 9: Divide the entire equation by 2: $x^2 - 8x - 1,140 = 0$.
- Step 10: Solve using the quadratic formula: $x = [8 \pm \sqrt{(64 + 4,560)}] / 2 = [8 \pm \sqrt{4,624}] / 2 = [8 \pm 68] / 2$.
- Step 11: Taking the positive root (since a length can't be negative): $x = (8+68)/2 = 76/2 = 38$.
- Step 12: This is a grid-in question — enter 38 as your final answer.

The Trap: A common error on this problem is mixing up which side (QR or RS) is described as '8 more than' the other, or incorrectly assuming QR and RS are the two DIFFERENT variables directly, without correctly using the rectangle's opposite-side-equality ($QR=PS$) to set up the relationship properly.

Solving with Desmos:

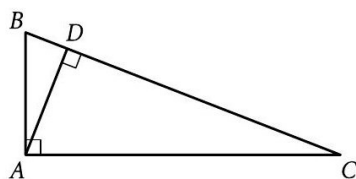
- Step 1: Open Desmos and set up the equation directly: type $x^2+(x-8)^2=2344$
- Step 2: Press Enter — since this is a one-variable equation, Desmos will solve it and plot the solution(s) as labeled points.
- Step 3: Look for the POSITIVE solution (since a rectangle's side length must be positive) — it should show $x = 38$.
- Step 4: Enter 38 into the grid-in answer box. (As a check, you can also verify: $38^2 + 30^2 = 1444+900 = 2344$, confirming this matches the given diagonal length squared.)

Right triangles and trigonometry — Q4 [Hard] [Grid-In] (ID: c4a9c94d)

Question ID: c4a9c94d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

Correct Answer: 6.5 (Student-Produced Response — no answer choices given)

The Rule: When an altitude is drawn from the right angle of a right triangle down to the hypotenuse, it creates a special GEOMETRIC MEAN relationship: the altitude's length squared equals the product of the two segments it divides the hypotenuse into. Formula: $(\text{altitude})^2 = (\text{segment 1}) \times (\text{segment 2})$.

The Tip: Whenever you see a right triangle with a SECOND right angle formed by an altitude dropped to the hypotenuse (creating two smaller right triangles inside), immediately think 'geometric mean relationship' — this often provides a much faster path to the answer than working through multiple separate triangle equations.

Why this is the answer: Since angle $BAC = 90^\circ$ and $AD \perp BC$ (right angle at D), AD is the altitude from the right angle to the hypotenuse BC, creating the geometric mean relationship $AD^2 = BD \times DC$. We're given $\tan(\text{angle CAD}) = 1.3$, and in right triangle ACD (right angle at D), this means $DC/AD = 1.3$, so $DC = 1.3 \times AD$. Substituting into the geometric mean formula: $AD^2 = 5 \times (1.3 \times AD) = 6.5 \times AD$. Dividing both sides by AD: $AD = 6.5$.

Step-by-Step Solution:

- Step 1: Identify the key structure: angle $BAC = 90^\circ$ (right angle at A) and $AD \perp BC$ (right angle at D) means AD is the altitude from the right angle to the hypotenuse in right triangle ABC.
- Step 2: Recall the geometric mean relationship for this configuration: $AD^2 = BD \times DC$.
- Step 3: Use the given tangent value to find DC in terms of AD: in right triangle ACD (right angle at D), $\tan(\text{angle CAD}) = \text{opposite/adjacent} = DC/AD$. Given $\tan(CAD) = 1.3$, this means $DC = 1.3 \times AD$.
- Step 4: Substitute $BD = 5$ (given) and $DC = 1.3 \times AD$ into the geometric mean formula: $AD^2 = 5 \times (1.3 \times AD)$.
- Step 5: Simplify the right side: $AD^2 = 6.5 \times AD$.
- Step 6: Divide both sides by AD (valid since $AD \neq 0$): $AD = 6.5$.
- Step 7: This is a grid-in question — enter 6.5 as your final answer.

The Trap: A common error on this problem is not recognizing the geometric mean relationship at all, and instead trying to set up multiple separate trigonometric equations across both smaller triangles without a clear connecting strategy — the geometric mean shortcut avoids this complexity entirely.

Solving with Desmos:

- Step 1: Open Desmos and set up the relationship you derived: $AD^2 = 5 \times (1.3 \times AD)$. Type this as an equation using x for AD: $x^2=5*1.3*x$
- Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution(s).
- Step 3: Look for the positive, non-zero solution — it should show $x = 6.5$.
- Step 4: Enter 6.5 into the grid-in answer box.

Right triangles and trigonometry — Q5 [Hard] (ID: f389569d)

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Correct Answer: C

The Rule: In any equilateral triangle, the height relates to the side length through the specific ratio: height = (side length) $\times (\sqrt{3}/2)$. This comes from splitting the equilateral triangle into two 30-60-90 right triangles using the altitude.

The Tip: Memorize the equilateral triangle height formula (height = side $\times \sqrt{3}/2$) as a direct shortcut — when a problem gives you the height and asks for the side (or perimeter), just rearrange this formula to solve for the side directly, avoiding a full 30-60-90 triangle setup each time.

Why C is right: Using the formula height = side $\times (\sqrt{3}/2)$, and solving for side: side = height $\div (\sqrt{3}/2)$ = height $\times (2/\sqrt{3})$. Substituting the given height of $39\sqrt{3}$: side = $39\sqrt{3} \times 2/\sqrt{3} = 39 \times 2 = 78$ (the $\sqrt{3}$ terms cancel out). Since the perimeter of an equilateral triangle is $3 \times$ side: perimeter = $3 \times 78 = 234$.

Step-by-Step Solution:

Step 1: Recall the equilateral triangle height formula: height = side $\times (\sqrt{3}/2)$.

Step 2: Rearrange to solve for side: side = height $\div (\sqrt{3}/2)$ = height $\times (2/\sqrt{3})$.

Step 3: Substitute the given height: side = $39\sqrt{3} \times (2/\sqrt{3})$.

Step 4: Notice the $\sqrt{3}$ in the numerator and denominator cancel out: side = $39 \times 2 = 78$.

Step 5: Calculate the perimeter using the formula perimeter = $3 \times$ side: perimeter = $3 \times 78 = 234$.

Step 6: This matches choice C.

The Trap: Choice B ($117\sqrt{3}$) is the most commonly picked wrong answer — it results from forgetting to cancel the $\sqrt{3}$ terms correctly, perhaps by multiplying the height directly by 3 ($39\sqrt{3} \times 3 = 117\sqrt{3}$) instead of first solving for the side length and THEN multiplying by 3.

POE — Eliminate the other three:

X A — 117 comes from $39 \times 3 = 117$, treating the given height as if it were directly the side length (ignoring the $\sqrt{3}$ and the height formula entirely) — this skips the necessary conversion step.

X B — $117\sqrt{3}$ results from multiplying the given height ($39\sqrt{3}$) directly by 3 (for perimeter) without first solving for the actual side length — this conflates the height with the side length.

X D — $1,521\sqrt{3}$ likely results from squaring some value incorrectly or a significant misapplication of the height formula — this doesn't correspond to a clean derivation.

Solving with Desmos:

Step 1: Open Desmos and solve for the side length directly using the rearranged formula: type $39 \cdot \sqrt{3} \cdot 2 / \sqrt{3}$

Step 2: Press Enter — Desmos will simplify this and evaluate to 78, confirming the side length.

Step 3: Calculate the perimeter: type $78 \cdot 3$

Step 4: Press Enter — Desmos will evaluate this to 234, confirming the perimeter and matching choice C.

Right triangles and trigonometry — Q6 [Hard] (ID: e3158182)

Question ID: e3158182

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

Answer

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

Correct Answer: D

The Rule: For any two ACUTE angles in a right triangle (or any two angles that are complementary, summing to 90°), sine of one angle always equals cosine of the other: $\sin(\theta) = \cos(90^\circ - \theta)$. This is called the 'cofunction identity.' If $\sin(p) = \cos(t)$, this means p and t are complementary angles: $p + t = 90^\circ$ (in radians, $p + t = \pi/2$).

The Tip: Whenever a problem states 'sin of one angle equals cos of another,' immediately translate this into 'these two angles are complementary' (they add up to 90° , or $\pi/2$ radians) — this is a direct shortcut that avoids needing to work with the trig functions themselves.

Why D is right: Since $\sin(p) = \cos(t)$, angles P and T are complementary, meaning $p + t = \pi/2$ (in radians). Solving for p : $p = \pi/2 - t$. Now substitute $x = 0$ into the function $g(x) = p(8)^x + 6t$: $g(0) = p(8)^0 + 6t = p(1) + 6t = p + 6t$ (since anything to the power of 0 equals 1). Substitute $p = \pi/2 - t$: $g(0) = (\pi/2 - t) + 6t = \pi/2 + 5t$, which can be rewritten as $5t + \pi/2$.

Step-by-Step Solution:

Step 1: Use the given relationship $\sin(p) = \cos(t)$ to establish that angles P and T are complementary: $p + t = \pi/2$ (since this cofunction identity holds specifically for complementary angles, measured in radians here since p and t are defined in radians).

Step 2: Solve this relationship for p : $p = \pi/2 - t$.

Step 3: Evaluate the function $g(x) = p(8)^x + 6t$ at $x = 0$: $g(0) = p(8)^0 + 6t$.

Step 4: Simplify using the zero exponent rule: $(8)^0 = 1$, so $g(0) = p(1) + 6t = p + 6t$.

Step 5: Substitute the expression for p from Step 2: $g(0) = (\pi/2 - t) + 6t$.

Step 6: Combine like terms: $-t + 6t = 5t$, so $g(0) = \pi/2 + 5t$, which can be written as $5t + \pi/2$.

Step 7: This matches choice D.

The Trap: Choice C ($5t + \pi$) is the most commonly picked wrong answer — it correctly combines the t -terms to get $5t$, but uses π instead of $\pi/2$ for the constant term, likely from forgetting that complementary angles sum to $\pi/2$ (90°) specifically, not a full π (180°).

POE — Eliminate the other three:

X A — $6t + \pi$ incorrectly keeps the original $6t$ coefficient (forgetting to subtract t after substituting $p = \pi/2 - t$) and uses the wrong constant (π instead of $\pi/2$).

X B — $6t$ keeps the original coefficient without substituting for p at all, and drops the constant term entirely — this skips the substitution step.

X C — $5t + \pi$ correctly combines the t -terms but uses the wrong constant (π instead of $\pi/2$), likely confusing the complementary angle relationship (sum to $\pi/2$) with a supplementary one (sum to π).

Solving with Desmos:

Step 1: Since this problem involves the relationship between p and t through radians, pick a simple test value for t to verify — let's use $t = \pi/6$ (30° , a common angle). Step 2: Calculate the corresponding p value using $p = \pi/2 - t$: type $\pi/2 - \pi/6$ — this gives $p = \pi/3$ (60°), confirming $p + t = \pi/2$ (complementary).

Step 3: Calculate $g(0)$ using the original function with these values: type $p(8)^0 + 6t$, substituting $p = \pi/3$ and $t = \pi/6$: $(\pi/3) * 1 + 6 * (\pi/6)$

Step 4: Press Enter — Desmos will evaluate this numerically; compare it to evaluating $5t + \pi/2$ with the same $t = \pi/6$: $5 * (\pi/6) + \pi/2$ — both expressions should give the SAME numeric value, confirming the formula $5t + \pi/2$ (choice D) correctly represents $g(0)$.

Question ID: 98e02472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle XYZ , angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is a° , the measure of $\angle XYR$ is b° , and the length of $\overline{XY}$ is 19. If $\sin a^\circ = \frac{12}{13}$, what is the value of $\tan b^\circ$?

Correct Answer: 12/5 (Student-Produced Response — no answer choices given)

The Rule: When an altitude is drawn from the right angle of a right triangle to the hypotenuse, it creates TWO smaller right triangles, each SIMILAR to the original triangle AND to each other. This means corresponding angles across all three triangles (the original and the two smaller ones) are equal.

The Tip: When you see an altitude drawn from a right angle to the hypotenuse, immediately identify which smaller triangle is similar to which — the angle in the smaller triangle 'nearest' to a specific vertex of the original triangle typically corresponds to that vertex's original angle.

Why this is the answer: Since $YR \perp XZ$ (creating a right angle at R), triangle XYR is similar to triangle XZY (sharing angle X , both containing a right angle). This similarity means angle XYR (labeled b°) corresponds to angle XZY (labeled a°) — so $b = a$. Given $\sin(a^\circ) = 12/13$, this corresponds to a right triangle with opposite side 12, hypotenuse 13, and (by the Pythagorean theorem) adjacent side $\sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} = 5$. So $\tan(a^\circ) = 12/5$. Since $b = a$, $\tan(b^\circ) = 12/5$ as well.

Step-by-Step Solution:

- Step 1: Identify the right angle in triangle XYZ : since X and Z are both acute angles (given), the right angle must be at Y .
- Step 2: Since $YR \perp XZ$, this creates two smaller right triangles (XYR and YZR), each similar to the original triangle XYZ .
- Step 3: Identify the similarity: triangle XYR shares angle X with triangle XZY , and both have a right angle (at R and at Y respectively) — this makes them similar, with angle XYR corresponding to angle XZY .
- Step 4: Since angle $XYR = b^\circ$ and angle $XZY = a^\circ$, and these are corresponding angles in similar triangles, $b = a$.
- Step 5: Use the given $\sin(a^\circ) = 12/13$ to find the full right triangle ratio: if the opposite side is 12 and the hypotenuse is 13, use the Pythagorean theorem to find the adjacent side: adjacent $= \sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = \sqrt{25} = 5$.
- Step 6: Calculate $\tan(a^\circ) = \text{opposite/adjacent} = 12/5$.
- Step 7: Since $b = a$, $\tan(b^\circ) = \tan(a^\circ) = 12/5$.
- Step 8: This is a grid-in question — enter 12/5 (or its decimal equivalent, 2.4) as your final answer.

The Trap: A common error on this problem is not recognizing that $b = a$ (from the similar triangles relationship), and instead trying to work out angle b using unrelated information like the given side length $XY=19$, which is actually a distractor not needed for this particular calculation.

Solving with Desmos:

- Step 1: Open Desmos and verify the sine-to-tangent conversion: type $\text{asin}(12/13)$ — this gives the angle a in radians (or you can convert to degrees).
- Step 2: Calculate the tangent of this same angle: type $\tan(\text{asin}(12/13))$
- Step 3: Press Enter — Desmos will evaluate this to exactly 2.4, confirming $\tan(a^\circ) = 12/5 = 2.4$.
- Step 4: Since $b = a$ (from the similar triangles relationship established in the solution), this same value (12/5 or 2.4) is your answer for $\tan(b^\circ)$ as well.

Question ID: 4fb972f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Correct Answer: 36 (Student-Produced Response — no answer choices given)

The Rule: The Pythagorean theorem applies to every right triangle: $(\text{leg } 1)^2 + (\text{leg } 2)^2 = (\text{hypotenuse})^2$. When given the hypotenuse and one leg, rearrange the formula to solve for the missing leg by SUBTRACTING (not adding) the known leg's square from the hypotenuse's square.

The Tip: When solving for a MISSING LEG (not the hypotenuse), remember the formula rearranges to: missing leg = $\sqrt{(\text{hypotenuse}^2 - \text{known leg}^2)}$ — this is subtraction, the opposite operation from when you're solving for the hypotenuse itself.

Why this is the answer: Using the Pythagorean theorem with hypotenuse $SQ = 4\sqrt{82}$ and leg $QR = 4$: $RS^2 = SQ^2 - QR^2 = (4\sqrt{82})^2 - 4^2$. Calculating $(4\sqrt{82})^2 = 16 \times 82 = 1,312$. And $4^2 = 16$. So $RS^2 = 1,312 - 16 = 1,296$. Taking the square root: $RS = \sqrt{1,296} = 36$.

Step-by-Step Solution:

- Step 1: Identify the given information: hypotenuse $SQ = 4\sqrt{82}$, one leg $QR = 4$.
- Step 2: Set up the Pythagorean theorem to solve for the missing leg RS : $QR^2 + RS^2 = SQ^2$.
- Step 3: Rearrange to isolate RS^2 : $RS^2 = SQ^2 - QR^2$.
- Step 4: Calculate SQ^2 : $(4\sqrt{82})^2 = 4^2 \times (\sqrt{82})^2 = 16 \times 82 = 1,312$.
- Step 5: Calculate QR^2 : $4^2 = 16$.
- Step 6: Substitute these values: $RS^2 = 1,312 - 16 = 1,296$.
- Step 7: Take the square root of both sides: $RS = \sqrt{1,296} = 36$.
- Step 8: This is a grid-in question — enter 36 as your final answer.

The Trap: A common error on this problem is forgetting to square the ENTIRE expression $4\sqrt{82}$ correctly — some students might only square the 82 (getting 4×82^2 instead of the correct $(4^2) \times 82$), leading to a significantly incorrect intermediate value.

Solving with Desmos:

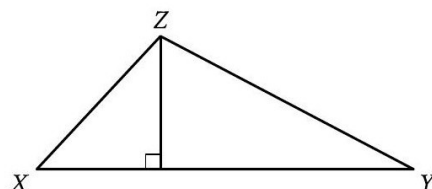
- Step 1: Open Desmos and calculate the hypotenuse squared first: type $(4*\text{sqrt}(82))^2$ — this confirms 1,312.
- Step 2: Calculate the missing leg using the Pythagorean theorem: type $\text{sqrt}(1312-4^2)$
- Step 3: Press Enter — Desmos will evaluate this to 36, confirming $RS = 36$ for your grid-in answer.

Right triangles and trigonometry — Q9 [Hard] (ID: 6c67dd47)

Question ID: 6c67dd47

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

Answer

- A. 247
- B. 494
- C. $247 \sin 54^\circ$
- D. $494 \sin 54^\circ$

Correct Answer: C

The Rule: To find the area of a triangle when you're given two sides and the INCLUDED angle (or enough information to find the height), drop an altitude to create a right triangle, use trigonometry (sine) to find the height, then apply the standard area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height}$.

The Tip: When a triangle isn't a right triangle but has an altitude already drawn to one side (shown by a right-angle marker), use the RIGHT triangle that altitude creates to find the height via sine — SOH-CAH-TOA (specifically 'SOH': $\sin = \text{opposite}/\text{hypotenuse}$) directly gives you the height using a known angle and the side length (hypotenuse) adjacent to that angle.

Why C is right: The altitude from Z creates a right triangle with the given angle $X=54^\circ$ and hypotenuse $XZ=19$. Using sine: $\text{height } (ZW) = XZ \times \sin(X) = 19 \times \sin(54^\circ)$. The base of the original triangle is $XY = 26$. Using the area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height} = (1/2) \times 26 \times 19\sin(54^\circ) = 13 \times 19\sin(54^\circ) = 247\sin(54^\circ)$.

Step-by-Step Solution:

- Step 1: Identify the altitude drawn from vertex Z down to side XY, creating a right angle at the foot of the altitude (let's call this point W).
- Step 2: In the right triangle formed (X, W, Z), the angle at X is given as 54° , and the hypotenuse of this small right triangle is $XZ = 19$.
- Step 3: Use the sine ratio to find the height ZW: $\sin(X) = \text{opposite}/\text{hypotenuse} = ZW/XZ$, so $ZW = XZ \times \sin(X) = 19 \times \sin(54^\circ)$.
- Step 4: Identify the base of the ORIGINAL triangle XYZ: this is side $XY = 26$.
- Step 5: Apply the triangle area formula: $\text{Area} = (1/2) \times \text{base} \times \text{height} = (1/2) \times 26 \times 19\sin(54^\circ)$.
- Step 6: Simplify: $(1/2) \times 26 = 13$, so $\text{Area} = 13 \times 19\sin(54^\circ) = 247\sin(54^\circ)$.
- Step 7: This matches choice C.

The Trap: Choice A (247) is the most commonly picked wrong answer — it correctly calculates $13 \times 19 = 247$ but completely forgets to multiply by $\sin(54^\circ)$, essentially treating XZ as if it were already the height (perpendicular to XY) rather than the hypotenuse of the smaller right triangle that needs the sine ratio applied.

POE — Eliminate the other three:

- ✗ A** — 247 forgets the $\sin(54^\circ)$ factor entirely — this would only be correct if XZ were already perpendicular to XY (which it's not; XZ is a slanted side, and its full length isn't the height).
- ✗ B** — 494 is double 247 (forgetting to apply the $1/2$ factor in the area formula, in addition to missing the $\sin(54^\circ)$ term) — a combination of two errors.
- ✗ D** — $494\sin(54^\circ)$ is double the correct answer — this comes from forgetting to apply the $(1/2)$ factor in the triangle area formula, using the full product of the sides instead of half of it.

Solving with Desmos:

Step 1: Open Desmos and calculate the height first: type $19 \cdot \sin(54 \cdot \pi / 180)$ (converting degrees to radians, since Desmos's sin function expects radians by default) — this gives the height ZW.

Step 2: Calculate the full area: type $(1/2) \cdot 26 \cdot 19 \cdot \sin(54 \cdot \pi / 180)$

Step 3: Press Enter — Desmos will evaluate this to a specific decimal number (approximately 199.85), which represents $247\sin(54^\circ)$ numerically, confirming choice C as the correct symbolic expression for the area.

Circles

12 questions · Easy 1 / Medium 3 / Hard 8

Circles — Q1 [Easy] [Grid-In] (ID: 186fdbca)

Question ID: 186fdbca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Easy

Question

A circle has a diameter of 12.6 centimeters. What is the radius of the circle, in centimeters?

Correct Answer: 6.3 (Student-Produced Response — no answer choices given)

The Rule: The radius of a circle is always exactly HALF of its diameter: $\text{radius} = \text{diameter} \div 2$.

The Tip: This is one of the most basic circle facts — memorize it so thoroughly you can apply it instantly. Whenever you see 'diameter,' your very next thought should be 'divide by 2 to get radius' (or multiply by 2 to go the other way).

Why this is the answer: Since $\text{radius} = \text{diameter} \div 2$, and the diameter is 12.6 centimeters: $\text{radius} = 12.6 \div 2 = 6.3$.

Step-by-Step Solution:

Step 1: Recall the relationship between diameter and radius: $\text{radius} = \text{diameter} \div 2$.

Step 2: Substitute the given diameter: $\text{radius} = 12.6 \div 2$.

Step 3: Calculate: $12.6 \div 2 = 6.3$.

Step 4: This is a grid-in question — enter 6.3 as your final answer.

The Trap: A common error on this problem is accidentally DOUBLING the diameter instead of halving it (perhaps confusing which direction the radius-diameter relationship goes), which would give an incorrect answer of 25.2 instead of the correct 6.3.

Solving with Desmos:

Step 1: Open Desmos and calculate the radius directly: type $12.6/2$

Step 2: Press Enter — Desmos will evaluate this to 6.3, confirming your grid-in answer.

Circles — Q2 [Medium] (ID: 408b39ea)

Question ID: 408b39ea

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

$$(x - 3)^2 + (y + 16)^2 = 289$$

The graph of the given equation is a circle in the xy -plane. The point (q, s) lies on the circle. Which of the following is NOT a possible value of q ?

Answer

- A. -17
- B. -3
- C. 3
- D. 17

Correct Answer: A

The Rule: For a circle equation in the form $(x-h)^2 + (y-k)^2 = r^2$, the center is at (h,k) and the radius is r (the square root of the right-hand side). Any point (q,s) on the circle must have its x -coordinate q within the range $[h-r, h+r]$, since the circle can't extend further left or right than one radius from its center.

The Tip: Once you identify the center and radius, find the circle's LEFTMOST and RIGHTMOST possible x -values by subtracting and adding the radius to the center's x -coordinate — any answer choice OUTSIDE this range cannot be a valid x -coordinate for a point on the circle.

Why A is right: From the equation, the center is $(3, -16)$ and the radius is $\sqrt{289} = 17$. This means any point on the circle must have an x -coordinate between $3-17=-14$ and $3+17=20$. Checking the choices: -17 falls OUTSIDE this range (since $-17 < -14$), making it NOT a possible value of q .

Step-by-Step Solution:

Step 1: Identify the center and radius from the equation $(x-3)^2 + (y+16)^2 = 289$: center = $(3, -16)$, radius = $\sqrt{289} = 17$.

Step 2: Determine the range of possible x -coordinates for any point on the circle: from (center x - radius) to (center x + radius), which is $(3-17)$ to $(3+17)$, or -14 to 20 .

Step 3: Check each answer choice against this range: -17 is LESS than -14 , meaning it falls outside the circle's possible x -range.

Step 4: Since -17 cannot be a valid x -coordinate for any point on this circle, it is NOT a possible value of q .

Step 5: This matches choice A.

The Trap: Choice D (17) is the most commonly picked wrong answer for students who confuse the RADIUS value (17) with a boundary condition, without correctly calculating the actual x -range (which extends all the way to 20 , well past 17) — 17 is actually well within the valid range.

POE — Eliminate the other three:

X B — -3 falls within the valid range (-14 to 20), so it IS a possible value of q — a point on the circle could have this x -coordinate.

X C — 3 is the x -coordinate of the CENTER itself, which is well within the valid range — definitely a possible value of q (in fact, points directly above/below the center have this x -coordinate).

X D — 17 falls within the valid range (-14 to 20), so it IS a possible value of q , even though it happens to match the radius value numerically — this is a coincidence, not a boundary violation.

Solving with Desmos:

Step 1: Open Desmos and type the circle equation exactly as given: $(x-3)^2+(y+16)^2=289$

Step 2: Press Enter — Desmos will graph the circle.

Step 3: Visually inspect the graph's leftmost and rightmost extent along the x-axis — you should see the circle spans from $x=-14$ to $x=20$.

Step 4: Check each answer choice against this visible range — you'll see $x=-17$ falls clearly to the LEFT of where the circle exists, confirming it's not a possible value of q and matching choice A.

Circles — Q3 [Medium] (ID: 267f82b5)

Question ID: 267f82b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

A circle in the xy -plane has the equation $(x - 19)^2 + (y - 17)^2 = 64k^2$, where k is a positive constant. Which of the following gives the center of the circle and its radius?

Answer

- A. The center is at $(19, 17)$, and the radius is 8.
- B. The center is at $(19, 17)$, and the radius is $8k$.
- C. The center is at $(19, 17)$, and the radius is 64.
- D. The center is at $(19, 17)$, and the radius is $64k$.

Correct Answer: B

The Rule: For a circle equation in the form $(x-h)^2 + (y-k)^2 = (\text{expression})^2$, the center is always at (h,k) , and the radius is the SQUARE ROOT of whatever is on the right-hand side — even if that right-hand side contains a variable or constant like k .

The Tip: Don't assume the right-hand side of a circle equation is already the radius itself — it's actually the radius SQUARED. Always take the square root of the entire right-hand expression (constants and all) to find the true radius, especially when that side contains extra variables.

Why B is right: From the equation $(x-19)^2 + (y-17)^2 = 64k^2$, the center is directly readable as $(19, 17)$. The radius is the square root of the right-hand side: $\sqrt{64k^2} = \sqrt{64} \times \sqrt{k^2} = 8 \times k = 8k$ (since k is stated to be positive, $\sqrt{k^2} = k$ without needing absolute value bars).

Step-by-Step Solution:

Step 1: Identify the center directly from the equation's structure $(x-19)^2 + (y-17)^2$: the center is $(19, 17)$.

Step 2: Identify the radius by taking the square root of the right-hand side: radius $= \sqrt{64k^2}$.

Step 3: Simplify this square root by taking the square root of each factor separately: $\sqrt{64} \times \sqrt{k^2} = 8 \times k$ (since k is given as positive, $\sqrt{k^2}$ simplifies directly to k , no absolute value needed).

Step 4: The radius is $8k$.

Step 5: This matches choice B: center $(19,17)$, radius $8k$.

The Trap: Choice C (radius 64) is the most commonly picked wrong answer — it uses the right-hand side value (64, ignoring the k^2 part) directly as if it were already the radius, without taking the necessary square root or accounting for the k^2 term.

POE — Eliminate the other three:

X A — This gives radius 8 (a constant), completely dropping the k variable — but the right side of the equation includes k^2 , which must be accounted for in the radius (giving $8k$, not just 8).

X C — This gives radius 64, using the right-hand side directly without taking the square root at all — but $64k^2$ is the radius SQUARED, not the radius itself.

✗ D — This gives radius $64k$, taking the square root of only the k^2 part but forgetting to also take the square root of the 64 (which becomes 8, not staying as 64).

Solving with Desmos:

Step 1: Since this problem involves the variable k , pick a simple test value to verify — let's use $k = 2$. Step 2: Open Desmos and type the equation with this value: $(x-19)^2 + (y-17)^2 = 64 \cdot 2^2$

Step 3: Press Enter — Desmos graphs this circle. Click on the circle or use the graph settings to inspect its radius — it should show a radius of 16 (since $8k = 8 \times 2 = 16$, matching choice B's formula).

Step 4: Compare this to what choice A would predict (radius 8, ignoring k) — you'll see the actual graphed radius (16) doesn't match constant 8, confirming that k must be included in the radius formula, verifying choice B.

Circles — Q4 [Medium] (ID: 11301fb6)

Question ID: 11301fb6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Medium

Question

The difference between the measure of angle A and the measure of angle B is $-\frac{5}{6}\pi$ radians. Which expression shows the difference between the measure of angle A and the measure of angle B , in degrees?

Answer

- A. $-\frac{5}{6}\pi \cdot \frac{360^\circ}{\pi}$
B. $-\frac{5}{6}\pi \cdot \frac{180^\circ}{\pi}$
C. $-\frac{5}{6}\pi \cdot \frac{\pi}{180^\circ}$
D. $-\frac{5}{6} \cdot \frac{180^\circ}{\pi}$

Correct Answer: C

The Rule: To convert an angle measure from RADIANS to DEGREES, multiply by the conversion factor $(180^\circ/\pi)$. This works because a full circle is both 2π radians and 360° , so π radians = 180° , giving the conversion ratio $180^\circ/\pi$.

The Tip: Memorize the conversion factor as a single fraction you can multiply by: radians $\times (180^\circ/\pi)$ = degrees. Keep the units visible in your head as you multiply — this helps confirm you're multiplying (not dividing) and using the correctly-oriented fraction.

Why C is right: To convert the given radian measure $(-5/6)\pi$ to degrees, multiply by the conversion factor $(180^\circ/\pi)$: $(-5/6)\pi \times (180^\circ/\pi)$. This matches the structure shown in choice C, using the correctly-oriented conversion fraction.

Step-by-Step Solution:

Step 1: Recall the radians-to-degrees conversion factor: multiply the radian measure by $(180^\circ/\pi)$.

Step 2: Apply this to the given difference, $(-5/6)\pi$ radians: multiply by $(180^\circ/\pi)$.

Step 3: Set up the expression: $(-5/6)\pi \times (180^\circ/\pi)$.

Step 4: This exact structure — the original radian expression multiplied by $(180^\circ/\pi)$ — matches choice C.

(For reference, if you wanted to actually simplify: the π terms cancel, giving $(-5/6) \times 180^\circ = -150^\circ$.)

The Trap: Choice B is the most commonly picked wrong answer — it uses the UPSIDE-DOWN conversion fraction $(180^\circ/\pi)$ written correctly in form, but let's check: B actually shows $180^\circ/\pi$ as well, so the key error for OTHER choices is using the wrong fraction orientation or the wrong numeric conversion constant (360 instead of 180).

POE — Eliminate the other three:

X A — This uses $360^\circ/\pi$ instead of $180^\circ/\pi$ — but the correct conversion factor for radians to degrees uses 180° (since π radians = 180° , not 360°).

X B — This uses the fraction the wrong way, dividing by degrees-per-something inconsistently — always double check that multiplying radians by $(180/\pi)$ gives degrees, matching the units correctly.

X D — This drops the π entirely from the original expression, changing it from $-(5/6)\pi$ to just $-(5/6)$ before applying a conversion — this loses essential information from the original radian measure.

Solving with Desmos:

Step 1: Open Desmos and calculate the numeric degree value directly: type $-(5/6)*\pi*(180/\pi)$

Step 2: Press Enter — Desmos will simplify this (the π terms cancel) and evaluate to -150 , confirming the angle is -150° in degree form.

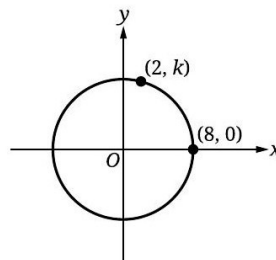
Step 3: Compare this calculation's STRUCTURE (original radian expression $\times 180/\pi$) to each answer choice — choice C matches this exact structure, confirming it's the correct expression.

Circles — Q5 [Hard] (ID: d12bca0a)

Question ID: d12bca0a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k ?

Answer

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

Correct Answer: D

The Rule: For any circle centered at the origin $(0,0)$, every point (x,y) on the circle satisfies $x^2 + y^2 = r^2$, where r is the radius. If you know one point on the circle, you can find the radius, then use that radius to find missing coordinates of other points.

The Tip: When a circle is centered at the origin and you're given one 'nice' point (like where it crosses an axis), use that point FIRST to find the radius — points on the x-axis or y-axis are the easiest way to directly read off the radius length.

Why D is right: Since the point $(8,0)$ lies on the circle centered at the origin, this tells us the radius directly: $r = 8$ (the distance from the origin to $(8,0)$ along the x-axis). Now use the circle equation $x^2 + y^2 = r^2 = 64$ with the point $(2,k)$: $2^2 + k^2 = 64$, so $k^2 = 64 - 4 = 60$, giving $k = \sqrt{60}$.

Step-by-Step Solution:

- Step 1: Use the given point (8,0) to find the radius: since this point is on the x-axis and the circle is centered at the origin, the distance from the origin to (8,0) IS the radius: $r = 8$.
- Step 2: Write the circle's equation using this radius: $x^2 + y^2 = 8^2 = 64$.
- Step 3: Substitute the other given point (2,k) into this equation: $2^2 + k^2 = 64$.
- Step 4: Simplify: $4 + k^2 = 64$.
- Step 5: Solve for k^2 : $k^2 = 64 - 4 = 60$.
- Step 6: Take the square root: $k = \sqrt{60}$ (taking the positive root, since the point appears above the x-axis in the figure).
- Step 7: This matches choice D.

The Trap: Choice C (8) is the most commonly picked wrong answer — it mistakenly uses the radius value (8) directly as k , without correctly applying the circle equation to account for the point's x-coordinate (2) as well.

POE — Eliminate the other three:

- X A** — 6 doesn't correspond to a correct calculation using the circle equation with the given points — likely a distractor value.
- X B** — 7 similarly doesn't align with the correct calculation $\sqrt{60} \approx 7.75$ — close but not exact, possibly from an estimation or rounding error rather than exact calculation.
- X C** — 8 is simply the radius value itself, mistakenly applied directly as k without accounting for the point (2,k)'s x-coordinate through the full circle equation.

Solving with Desmos:

- Step 1: Open Desmos and confirm the radius using the given point: type $\text{sqrt}(8^2+0^2)$ — this confirms radius = 8.
- Step 2: Solve for k using the circle equation with the point (2,k): type $\text{sqrt}(8^2-2^2)$
- Step 3: Press Enter — Desmos will evaluate this to $\sqrt{60}$ (approximately 7.746), confirming $k = \sqrt{60}$ and matching choice D.

Circles — Q6 [Hard] (ID: 8493fd10)

Question ID: 8493fd10

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

Answer

- A. 73
- B. 71
- C. 37
- D. 35

Correct Answer: B

The Rule: To find the center and radius of a circle from an equation NOT already in standard form $(x-h)^2+(y-k)^2=r^2$, use **COMPLETING THE SQUARE** on both the x -terms and y -terms separately to rewrite the equation into that standard form.

The Tip: When completing the square for a term like $x^2 + bx$, take half of the coefficient b , square it, and add (and immediately subtract) that value to keep the equation balanced — this systematic process works every time, even when the coefficient involves a square root like $\sqrt{t}$ here.

Why B is right: Complete the square for the x-terms ($x^2 + x\sqrt{t}$): take half of $\sqrt{t}$ (giving $\sqrt{t}/2$), square it (giving $t/4$), and rewrite as $(x + \sqrt{t}/2)^2 - t/4$. Complete the square for the y-terms ($y^2 - y$): take half of -1 (giving $-1/2$), square it (giving $1/4$), and rewrite as $(y - 1/2)^2 - 1/4$. The full equation becomes $(x + \sqrt{t}/2)^2 - t/4 + (y - 1/2)^2 - 1/4 - 65 = 0$, which rearranges to $(x + \sqrt{t}/2)^2 + (y - 1/2)^2 = 65.25 + t/4$. Since the radius is given as $\sqrt{83}$, meaning $\text{radius}^2 = 83$: set $65.25 + t/4 = 83$, and solve for t .

Step-by-Step Solution:

Step 1: Group the x-terms and y-terms separately: $(x^2 + x\sqrt{t}) + (y^2 - y) - 65 = 0$.

Step 2: Complete the square for the x-terms: take half of the coefficient of x (which is $\sqrt{t}$), giving $\sqrt{t}/2$; square this to get $t/4$. Rewrite: $x^2 + x\sqrt{t} = (x + \sqrt{t}/2)^2 - t/4$.

Step 3: Complete the square for the y-terms: take half of the coefficient of y (which is -1), giving $-1/2$; square this to get $1/4$. Rewrite: $y^2 - y = (y - 1/2)^2 - 1/4$.

Step 4: Substitute these completed-square forms back into the original equation: $(x + \sqrt{t}/2)^2 - t/4 + (y - 1/2)^2 - 1/4 - 65 = 0$.

Step 5: Move all the constant terms to the right side: $(x + \sqrt{t}/2)^2 + (y - 1/2)^2 = 65 + 1/4 + t/4 = 65.25 + t/4$.

Step 6: Since the radius is given as $\sqrt{83}$, the right side of the standard-form equation must equal 83 (radius squared): $65.25 + t/4 = 83$.

Step 7: Solve for t : $t/4 = 83 - 65.25 = 17.75$.

Step 8: Multiply both sides by 4: $t = 17.75 \times 4 = 71$.

Step 9: This matches choice B.

The Trap: Choice A (73) is the most commonly picked wrong answer — it typically results from a small arithmetic slip while combining the constant terms (65, $1/4$, and the target 83), such as forgetting to include the $1/4$ from completing the square on the y-terms.

POE — Eliminate the other three:

X A — 73 likely comes from a minor arithmetic error while combining the constant terms during the completing-the-square process — perhaps omitting the $1/4$ term from the y-completion.

X C — 37 doesn't correspond to a clean derivation from this equation — possibly a factor-of-2 error somewhere in the process.

X D — 35 similarly doesn't align with a correct completing-the-square derivation — likely from a more significant setup error.

Solving with Desmos:

Step 1: Since this problem involves solving for an unknown constant t , first set up the relationship you derived: $65.25 + t/4 = 83$. Open Desmos and type this as an equation: $65.25 + x/4 = 83$

Step 2: Press Enter — Desmos will solve this one-variable equation and plot the solution.

Step 3: Look for the solution — it should show $x = 71$, confirming $t = 71$ and matching choice B.

Question ID: a5ff1a96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

Answer

- A. $(x - 4)^2 + (y - 5)^2 = 18$
- B. $(x - 4)^2 + (y - 5)^2 = 36$
- C. $(x - 8)^2 + (y - 10)^2 = 9$
- D. $(x - 8)^2 + (y - 10)^2 = 18$

Correct Answer: B

The Rule: When two circles share the SAME CENTER (concentric circles) but have different radii, only the radius (and therefore the right-hand side of the equation, radius²) changes — the center coordinates (h,k) stay exactly the same in both equations.

The Tip: For 'same center, different radius' problems, copy the $(x-h)^2$ and $(y-k)^2$ parts UNCHANGED from the original equation, and only recalculate the right-hand side using the new radius — don't accidentally change the center coordinates.

Why B is right: Circle M has center (4,5) and radius² = 9, so radius = 3. Circle P has the same center (4,5) but a radius twice as large: $2 \times 3 = 6$. The new right-hand side is radius² = $6^2 = 36$, giving the equation $(x-4)^2 + (y-5)^2 = 36$.

Step-by-Step Solution:

- Step 1: Identify circle M 's center and radius from its equation $(x-4)^2 + (y-5)^2 = 9$: center = (4,5), radius = $\sqrt{9} = 3$.
- Step 2: Since circle P has the SAME center as circle M , its equation will also have $(x-4)^2$ and $(y-5)^2$ unchanged.
- Step 3: Calculate circle P 's radius: twice circle M 's radius = $2 \times 3 = 6$.
- Step 4: Calculate the new right-hand side (radius squared) for circle P 's equation: $6^2 = 36$.
- Step 5: Write circle P 's full equation: $(x-4)^2 + (y-5)^2 = 36$.
- Step 6: This matches choice B.

The Trap: Choice A (right side = 18) is the most commonly picked wrong answer — it results from doubling the ORIGINAL right-hand side ($9 \times 2 = 18$) directly, instead of correctly doubling the RADIUS first and then squaring that doubled radius (which gives 36, not 18) — area/radius-squared doesn't simply double when radius doubles, it quadruples.

POE — Eliminate the other three:

- X A** — 18 comes from doubling the original right-hand side ($9 \times 2 = 18$) directly — but doubling the RADIUS means the radius SQUARED (the right side) actually quadruples, not just doubles (since $(2r)^2 = 4r^2$, not $2r^2$).
- X C** — This keeps the same right side (9) as circle M but incorrectly changes the center coordinates to (8,10) — but a circle with 'the same center' should keep the ORIGINAL center coordinates (4,5), not double them.
- X D** — This combines both errors — incorrectly doubled center coordinates (8,10) AND an incorrectly doubled (rather than quadrupled) right-hand side (18).

Solving with Desmos:

- Step 1: Open Desmos and type circle M 's equation: $(x-4)^2 + (y-5)^2 = 9$
- Step 2: Press Enter — Desmos graphs this circle; you can confirm its radius is 3 by inspecting the graph.
- Step 3: Type circle P 's equation using your calculated values: $(x-4)^2 + (y-5)^2 = 36$
- Step 4: Press Enter — Desmos graphs this larger circle. Visually confirm both circles share the SAME center point, and that circle P 's radius (6) is exactly twice circle M 's radius (3), confirming choice B.

Circles — Q8 [Hard] [Grid-In] (ID: 149021da)

Question ID: 149021da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center P , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Correct Answer: 16 (Student-Produced Response — no answer choices given)

The Rule: The formula for arc length is: arc length = (central angle / 360°) $\times$ ($2\pi r$), where r is the radius. If you know the arc length and the central angle, you can solve this formula backwards to find the radius.

The Tip: Set up the arc length formula with the given angle and arc length, then solve for r algebraically — treat it just like any other equation with one unknown, isolating r on one side.

Why this is the answer: Using the arc length formula with angle 45° and arc length 4π : $4\pi = (45/360) \times 2\pi r$. Simplify $45/360$ to $1/8$: $4\pi = (1/8)(2\pi r) = \pi r/4$. Multiply both sides by 4: $16\pi = \pi r$. Divide both sides by π : $r = 16$.

Step-by-Step Solution:

Step 1: Recall the arc length formula: arc length = (central angle / 360°) $\times$ $2\pi r$.

Step 2: Substitute the given values: $4\pi = (45/360) \times 2\pi r$.

Step 3: Simplify the fraction $45/360$ by dividing both by 45: $45/360 = 1/8$.

Step 4: The equation becomes: $4\pi = (1/8) \times 2\pi r = (2\pi r)/8 = \pi r/4$.

Step 5: Multiply both sides by 4 to clear the fraction: $16\pi = \pi r$.

Step 6: Divide both sides by π : $r = 16$.

Step 7: This is a grid-in question — enter 16 as your final answer.

The Trap: A common error on this problem is forgetting to convert the angle fraction correctly ($45/360$ rather than accidentally using $45/180$ or another incorrect fraction), which would throw off the entire calculation.

Solving with Desmos:

Step 1: Open Desmos and set up the equation directly: type $4\pi = (45/360) \times 2\pi x$

Step 2: Press Enter — since this is a one-variable equation, Desmos will solve for x and plot the solution.

Step 3: Look for the solution — it should show $x = 16$, confirming the radius and matching your grid-in answer.

Circles — Q9 [Hard] (ID: 10e059a6)

Question ID: 10e059a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is 36 times the measure of angle A . What is the value of $\cos B$?

Answer

- A. 0
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{3}}{2}$

Correct Answer: D

The Rule: When one angle's measure is a MULTIPLE of another (like 'angle B is 36 times angle A'), first find the numeric value of the given angle, then multiply by that stated factor to find the second angle. Once you have the second angle's exact measure, apply the relevant trigonometric function directly.

The Tip: Keep angle measures in radians throughout the calculation if the problem starts in radians — converting to degrees and back introduces unnecessary opportunities for error. Trust the radian arithmetic, especially when working with fractions of π .

Why D is right: Angle $A = \pi/216$ radians. Angle $B = 36 \times \text{angle } A = 36 \times (\pi/216) = 36\pi/216$. Simplify this fraction: $36/216 = 1/6$, so angle $B = \pi/6$ radians. The value of $\cos(\pi/6)$ is a standard reference angle value: $\sqrt{3}/2$.

Step-by-Step Solution:

- Step 1: Identify the measure of angle A : $\pi/216$ radians (given).
- Step 2: Calculate angle B using the given relationship: angle $B = 36 \times \text{angle } A = 36 \times (\pi/216)$.
- Step 3: Simplify this multiplication: $36 \times \pi/216 = 36\pi/216$.
- Step 4: Reduce the fraction $36/216$ by dividing both by 36: $36/216 = 1/6$.
- Step 5: So angle $B = \pi/6$ radians.
- Step 6: Recall the standard value: $\cos(\pi/6) = \sqrt{3}/2$ (this is one of the standard reference angles, equivalent to 30°).
- Step 7: This matches choice D.

The Trap: Choice C ($\sqrt{2}/2$) is the most commonly picked wrong answer — it's the cosine value for a DIFFERENT standard angle ($\pi/4$, or 45°), confusing it with the correct angle $\pi/6$ (30°) found in this problem.

POE — Eliminate the other three:

- X A** — 0 would be the value of $\cos(\pi/2)$ (90°) — a completely different angle than the $\pi/6$ found in this problem.
- X B** — $1/2$ is actually the value of $\sin(\pi/6)$, not $\cos(\pi/6)$ — a common mix-up between sine and cosine values for the same standard angle.
- X C** — $\sqrt{2}/2$ is the value of $\cos(\pi/4)$ (45°) — this belongs to a different standard angle than the $\pi/6$ calculated here.

Solving with Desmos:

- Step 1: Open Desmos and calculate angle B first: type $36*(\pi/216)$ — this confirms angle $B = \pi/6$ (approximately 0.524 radians).
- Step 2: Calculate the cosine of this angle: type $\cos(36*(\pi/216))$
- Step 3: Press Enter — Desmos will evaluate this to approximately 0.866, which matches $\sqrt{3}/2$ (you can verify by typing $\text{sqrt}(3)/2$ separately to compare the decimal values), confirming choice D.

Circles — Q10 [Hard] [Grid-In] (ID: 9f9112ab)

Question ID: 9f9112ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle L is 60 degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n ?

Correct Answer: 3 (Student-Produced Response — no answer choices given)

The Rule: To convert an angle from DEGREES to RADIANS, multiply by the conversion factor $(\pi/180^\circ)$. A common shortcut for recognizable angles: 60° always equals $\pi/3$ radians — memorizing a few common conversions ($30^\circ=\pi/6$, $45^\circ=\pi/4$, $60^\circ=\pi/3$, $90^\circ=\pi/2$) can save time.

The Tip: When you see a degree measure that's a 'nice' fraction of 180° (like 60° , which is $180^\circ/3$), you can often recognize the radian equivalent directly from memory rather than doing the full conversion calculation each time.

Why this is the answer: Converting 60° to radians: $60^\circ \times (\pi/180^\circ) = 60\pi/180 = \pi/3$ (simplifying the fraction $60/180$ to $1/3$). Since the problem states angle L 's measure is π/n radians, and we've found it equals $\pi/3$, comparing directly: $\pi/n = \pi/3$, so $n = 3$.

Step-by-Step Solution:

Step 1: Convert the given angle measure from degrees to radians: $60^\circ \times (\pi/180^\circ)$.

Step 2: Simplify the fraction $60/180$ by dividing both by 60: $60/180 = 1/3$.

Step 3: So $60^\circ = \pi/3$ radians.

Step 4: Compare this to the given form π/n : since angle $L = \pi/3 = \pi/n$, this means $n = 3$.

Step 5: This is a grid-in question — enter 3 as your final answer.

The Trap: A common error on this problem is miscalculating the fraction $60/180$, perhaps simplifying incorrectly or forgetting to reduce it fully to $1/3$, which would lead to an incorrect value of n .

Solving with Desmos:

Step 1: Open Desmos and convert 60 degrees to radians directly: type $60*\pi/180$

Step 2: Press Enter — Desmos will simplify this to $\pi/3$ (or show the decimal equivalent, approximately 1.047).

Step 3: Since the problem states the angle equals π/n , and you've found it equals $\pi/3$, this directly confirms $n = 3$ for your grid-in answer.

Circles — Q11 [Hard] [Grid-In] (ID: 27f1db42)

Question ID: 27f1db42

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle F is defined by the equation $(x+8)^2 + (y+8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x+8)^2 + (y+8)^2 = p$, where p is a constant. What is the value of p ?

Correct Answer: 49 (Student-Produced Response — no answer choices given)

The Rule: When two circles share the SAME CENTER (concentric circles) and one radius is described as a certain amount MORE than the other (an additive relationship, not a multiplicative one), first find the original radius, then ADD the stated amount to get the new radius — don't confuse this with problems where the new radius is a MULTIPLE of the original.

The Tip: Read the relationship between the two radii VERY carefully — 'is 2 units greater than' means ADD 2 to the radius (not to the radius squared, and not multiply by 2) — this additive relationship is different from 'twice as large' problems, which require multiplication instead.

Why this is the answer: Circle F has $\text{radius}^2 = 25$ (from the equation), so $\text{radius} = 5$. Circle G's radius is 2 units GREATER than circle F's radius: $5 + 2 = 7$. Since circle G's equation uses p as the right-hand side (equivalent to radius^2), $p = 7^2 = 49$.

Step-by-Step Solution:

- Step 1: Identify circle F's radius from its equation $(x+8)^2 + (y+8)^2 = 25$: $\text{radius}^2 = 25$, so $\text{radius} = \sqrt{25} = 5$.
- Step 2: Apply the given relationship: circle G's radius is 2 units GREATER than circle F's radius (an ADDITION, not a multiplication): circle G's radius $= 5 + 2 = 7$.
- Step 3: Since circle G's equation is written as $(x+8)^2 + (y+8)^2 = p$, the value p represents circle G's radius SQUARED.
- Step 4: Calculate p : $p = 7^2 = 49$.
- Step 5: This is a grid-in question — enter 49 as your final answer.

The Trap: A common error on this problem is confusing this ADDITIVE relationship ('2 units greater') with a MULTIPLICATIVE one (like 'twice as large'), which would incorrectly lead to a radius of 10 (5×2) instead of the correct 7 ($5 + 2$), and consequently an incorrect p value of 100 instead of 49.

Solving with Desmos:

- Step 1: Open Desmos and calculate circle F's radius first: type $\text{sqrt}(25)$ — this confirms $\text{radius} = 5$.
- Step 2: Calculate circle G's radius using the additive relationship: type $5+2$ — this confirms $\text{radius} = 7$.
- Step 3: Calculate p (radius squared): type 7^2
- Step 4: Press Enter — Desmos will evaluate this to 49, confirming your grid-in answer.

Circles — Q12 [Hard] [Grid-In] (ID: 44d67c6c)

Question ID: 44d67c6c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n ?

Correct Answer: -2 (Student-Produced Response — no answer choices given)

The Rule: To find the center and radius of a circle from an equation with x^2 and y^2 terms plus linear terms (not in standard form), use COMPLETING THE SQUARE on both the x -terms and y -terms. Once you have the standard form, you can relate the radius to any given point on a related (same-center) circle using the distance formula.

The Tip: When a problem involves TWO related circles (same center, different size) and gives you an actual POINT on the second circle, use the distance formula from the shared center to that point to find the second circle's radius directly — this is often more reliable than trying to track scale factors through multiple algebraic steps.

Why this is the answer: Complete the square on x^2-10x (giving $(x-5)^2-25$) and y^2-8y (giving $(y-4)^2-16$). The equation becomes $(x-5)^2 + (y-4)^2 = 25+16+14n = 41+14n$, so circle A has center (5,4) and $\text{radius}^2 = 41+14n$. Circle B has the same center (5,4) with a diameter twice circle A's — meaning circle B's radius is twice circle A's radius. Using the given point (11,8) on circle B, calculate circle B's actual radius via the distance formula: $\sqrt{[(11-5)^2+(8-4)^2]} = \sqrt{(36+16)} = \sqrt{52}$, so circle B's $\text{radius}^2 = 52$. Since circle B's radius = $2 \times$ circle A's radius, circle B's $\text{radius}^2 = 4 \times$ circle A's radius^2 : $52 = 4 \times (41+14n)$. Solving: $13 = 41+14n$, so $14n = -28$, giving $n = -2$.

Step-by-Step Solution:

Step 1: Complete the square for the x-terms in $x^2 - 10x$: take half of -10 (giving -5), square it (giving 25), rewrite as $(x-5)^2 - 25$.

Step 2: Complete the square for the y-terms in $y^2 - 8y$: take half of -8 (giving -4), square it (giving 16), rewrite as $(y-4)^2 - 16$.

Step 3: Substitute back into the original equation: $(x-5)^2 - 25 + (y-4)^2 - 16 - 14n = 0$.

Step 4: Rearrange to standard form: $(x-5)^2 + (y-4)^2 = 25 + 16 + 14n = 41 + 14n$. This means circle A has center (5,4) and $\text{radius}^2 = 41+14n$.

Step 5: Circle B shares the same center (5,4) but has TWICE the diameter of circle A — meaning circle B's radius is also twice circle A's radius.

Step 6: Use the given point (11,8) on circle B to find circle B's actual radius via the distance formula from the center (5,4): distance = $\sqrt{[(11-5)^2 + (8-4)^2]} = \sqrt{(36+16)} = \sqrt{52}$. So circle B's $\text{radius}^2 = 52$.

Step 7: Since circle B's radius = $2 \times$ circle A's radius, squaring both sides: circle B's $\text{radius}^2 = 4 \times$ circle A's radius^2 . Substitute: $52 = 4 \times (41+14n)$.

Step 8: Divide both sides by 4: $13 = 41 + 14n$.

Step 9: Solve for n: $14n = 13 - 41 = -28$, so $n = -28/14 = -2$.

Step 10: This is a grid-in question — enter -2 as your final answer.

The Trap: A common error on this problem is forgetting that DOUBLING the diameter means the radius SQUARED (used in the equation) actually QUADRUPLES, not just doubles — using a factor of 2 instead of 4 when relating circle A's and circle B's radius-squared values would lead to an incorrect value of n.

Solving with Desmos:

Step 1: Open Desmos and complete the square verification: type the original equation $x^2+y^2-10x-8y-14*n=0$ with n as a slider — Desmos will offer to create a slider.

Step 2: Calculate circle B's radius using the given point and center: type $\text{sqrt}((11-5)^2+(8-4)^2)$ — this confirms circle B's radius is $\sqrt{52}$.

Step 3: Set up the relationship: circle B's radius^2 (52) should equal 4 times circle A's radius^2 ($41+14n$). Type: $52=4*(41+14x)$ and solve for x using Desmos (treating it as an equation to solve).

Step 4: Desmos will show the solution $x = -2$, confirming $n = -2$ for your grid-in answer.

June Resit 2026 - Math (Part 1)

Εκφωνήσεις & αναλυτικές λύσεις

SAT Math - Σημειώσεις

Άσκηση 1 (Exponential - ισοδύναμες μορφές)

The function $m(x) = c(1.8^x + 1.8^d)$, $c, d \in \mathbb{Z}$, $0 < c < d$. The functions $s(x), t(x)$ are given by

$$\text{I. } s(x) = c \cdot (1.8^x + u) \quad \text{II. } t(x) = c \cdot 1.8^x + v$$

Which of the two has the same y -intercept as $m(x)$?

A) I B) II C) I & II D) Neither

Λύση

Το y -intercept κάθε συνάρτησης είναι η τιμή της στο $x = 0$, και $1.8^0 = 1$:

$$m(0) = c(1 + 1.8^d) = c + c \cdot 1.8^d$$

$$s(0) = c(1 + u) = c + cu$$

$$t(0) = c \cdot 1 + v = c + v$$

Για να ισχύει $s(0) = m(0)$ πρέπει $u = 1.8^d$, και για να ισχύει $t(0) = m(0)$ πρέπει $v = c \cdot 1.8^d$. Οι u, v είναι τυχαίες σταθερές, άρα καμία από τις δύο ισότητες δεν ισχύει αναγκαστικά.

Answer / Απάντηση: D) Neither

Σημείωση: Αν διαβάσουμε την εκφώνηση ότι οι s, t είναι ταυτοτικά ίσες με την m , τότε $u = 1.8^d$, $v = c \cdot 1.8^d$ και θα είχαν το ίδιο y -intercept (C). Η λύση του αρχείου εννοεί u, v ελεύθερες σταθερές, γι' αυτό η απάντηση είναι D.

Άσκηση 2 (Vieta)

$120x^2 - (15p + 21q)x + pq = 0$, $p, q \in \mathbb{R}_{>0}$. The sum of solutions of the quadratic is in the form $k \cdot (5p + 7q)$. Find k .

Λύση

Για $ax^2 + bx + c = 0$ το άθροισμα των ριζών είναι $S = -\frac{b}{a}$ (τύποι Vieta). Εδώ $a = 120$, $b = -(15p + 21q)$:

$$S = -\frac{-(15p + 21q)}{120} = \frac{15p + 21q}{120} = \frac{3(5p + 7q)}{120} = \frac{1}{40}(5p + 7q).$$

Συγκρίνοντας με τη μορφή $k(5p + 7q)$ παίρνουμε

Answer / Απάντηση: $k = \frac{1}{40}$

Άσκηση 3 (Εκθετική - y -τομή)

$f(x) = -a^x + b$. Let $g(x) = f(x) - 9$, which has y -intercept at $(0, -\frac{31}{4})$. If $a \cdot b = \frac{225}{4}$, find a and b .

Λύση

Αφού $a^0 = 1$:

$$g(0) = f(0) - 9 = (-1 + b) - 9 = b - 10.$$

Η y -τομή είναι $-\frac{31}{4}$, άρα

$$b - 10 = -\frac{31}{4} \Rightarrow b = 10 - \frac{31}{4} = \frac{9}{4}.$$

Από το γινόμενο:

$$a = \frac{225/4}{b} = \frac{225/4}{9/4} = 25.$$

Έλεγχος: $ab = 25 \cdot \frac{9}{4} = \frac{225}{4} \checkmark$ και $g(0) = \frac{9}{4} - 10 = -\frac{31}{4} \checkmark$

Answer / Απάντηση: $a = 25, b = \frac{9}{4}$

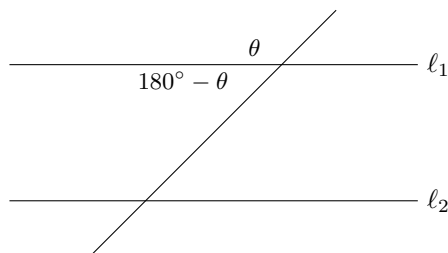
Σημείωση: στη λύση του αρχείου είχε παραλειφθεί το -9 ($g(0) = 1 + b$), που έδινε λάθος τιμές. Εδώ η g υπολογίζεται σωστά.

Άσκηση 4 (Παράλληλες & τέμνουσα)

A transversal line intersects two parallel lines, creating 4 acute and 4 obtuse angles. The measure of one of these angles is $(9x - 150)$ degrees. The sum of four of these angles is k degrees. Which of the following is **NOT** a possible value for k ?

(A) $18x - 120$ (B) $-18x + 1140$ (C) 360 (D) $-36x + 1320$

Λύση



Όταν μια τέμνουσα κόβει δύο παράλληλες, και οι οκτώ γωνίες παίρνουν μόνο δύο τιμές: θ (4 γωνίες) και $180^\circ - \theta$ (4 γωνίες), με $\theta = 9x - 150$. Άρα $180^\circ - \theta = 330 - 9x$.

Αθροίζοντας τέσσερις γωνίες, έστω j από αυτές ίσες με θ και $4 - j$ ίσες με $180^\circ - \theta$:

j	Άθροισμα k
4	$4\theta = 36x - 600$
3	$3\theta + (180 - \theta) = 2\theta + 180 = 18x - 120$
2	$2\theta + 2(180 - \theta) = 360$
1	$\theta + 3(180 - \theta) = 540 - 2\theta = 840 - 18x$
0	$4(180 - \theta) = 1320 - 36x$

Οι επιλογές: (A) $18x - 120 \checkmark$ (εμφανίζεται), (C) $360 \checkmark$, (D) $1320 - 36x \checkmark$. Η (B) $-18x + 1140$ δεν εμφανίζεται (η σωστή τιμή είναι $-18x + 840$).

Answer / Απάντηση: B

Άσκηση 5 (Τετραγωνική - τομές με x -άξονα)

The quadratic $g(x) = ax^2 + bx + c$ crosses the x -axis at $(x_1, 0), (x_2, 0)$ where $x_1 + x_2 = -6$, $x_1x_2 = -27$. If $a > 1$ and $a \in \mathbb{Z}$, what could be the value of $b - a$?

(A) 10 (B) 5 (C) 8 (D) 12

Λύση

Βήμα 1 - οι ρίζες. Οι x_1, x_2 είναι ρίζες της $t^2 - (x_1 + x_2)t + x_1x_2 = t^2 + 6t - 27 = (t + 9)(t - 3)$, άρα $x_1, x_2 = -9, 3$ (με Desmos: τομή των $y = -6 - x$ και $y = -27/x$).

Βήμα 2 - η μορφή της g .

$$g(x) = a(x + 9)(x - 3) = a(x^2 + 6x - 27) = ax^2 + 6ax - 27a.$$

Άρα $b = 6a$, και

$$b - a = 5a.$$

Βήμα 3 - περιορισμός. Αφού $a \in \mathbb{Z}$ και $a > 1$, ισχύει $a \geq 2$, άρα $b - a \in \{10, 15, 20, \dots\}$: πολλαπλάσιο του 5 και τουλάχιστον 10.

- (A) $10 = 5 \cdot 2$ ✓ (B) $5 \Rightarrow a = 1 \times$ (θέλουμε $a > 1$)
- (C) 8, (D) 12: δεν είναι πολλαπλάσια του 5 ×

Answer / Απάντηση: A) 10

Άσκηση 6 (Συνάρτηση τετραγωνικής ρίζας)

$g(x) = k\sqrt{x + m}$, the point $(-49, 0)$ lies on the curve and $g(32) < 0$. Which of the following must be true about g ?

(A) $k < m$ (B) $k > m$ (C) $g(0) = 49$ (D) $g(0) = -49$

Λύση

Από το σημείο $(-49, 0)$: $g(-49) = k\sqrt{m - 49} = 0$, άρα $k = 0$ ή $m = 49$.

Αν $k = 0$ τότε $g \equiv 0$ και $g(32) = 0$, ενώ δίνεται $g(32) < 0$. Άρα $m = 49$ (και $k \neq 0$).

Τώρα $g(32) = k\sqrt{32 + 49} = k\sqrt{81} = 9k < 0 \Rightarrow k < 0$.

Έτσι $k < 0 < 49 = m$, δηλαδή $k < m$. Για τις υπόλοιπες: $g(0) = k\sqrt{49} = 7k$, που είναι αρνητικός αλλά όχι απαραίτητα -49 (ούτε 49), άρα οι (C),(D) δεν ισχύουν αναγκαστικά.

Answer / Απάντηση: A) $k < m$

Άσκηση 7 (Παραγοντοποίηση)

$x(9x - 2k) + m(9x - 2k) - (x + m) = 0$, with k, m positive constants. The equation has a solution at $x = 50$. Find k .

Λύση

Βγάζουμε κοινό παράγοντα το $(9x - 2k)$ από τους δύο πρώτους όρους:

$$(x + m)(9x - 2k) - (x + m) = 0 \Rightarrow (x + m)(9x - 2k - 1) = 0.$$

Άρα $x = -m$ ή $x = \frac{2k + 1}{9}$. Αφού $m > 0$, το $-m$ είναι αρνητικό και δεν μπορεί να είναι 50. Επομένως

$$\frac{2k + 1}{9} = 50 \Rightarrow 2k + 1 = 450 \Rightarrow k = \frac{449}{2} = 224.5.$$

Έλεγχος (ανεξαρτησία από το m): $9 \cdot 50 - 2(224.5) - 1 = 450 - 449 - 1 = 0$ ✓

Answer / Απάντηση: $k = 224.5$

Άσκηση 8 (Κυβική - πρόσημο)

$h(x) = -5(x-u)(x-v)(x-w)$. For $x \in (u, w)$ (that is $u < x < w$) $h(x)$ is negative. The graph of $y = h(x)$ contains the point (m, n) , $m, n \in \mathbb{R}$. If $n = -12$, which of the following is/are true?

- (I) $m < w$ (II) $u < m < v$ (III) $m > u$
 (A) Only I (B) Only II (C) Only I, II (D) Only I, III

Λύση

Για $x \in (u, w)$ ισχύει $x - u > 0$ και $x - w < 0$, άρα $(x - u)(x - w) < 0$ και $-5(x - u)(x - w) > 0$. Για να είναι $h(x) < 0$ πρέπει ο τρίτος παράγοντας $x - v$ να είναι αρνητικός, δηλαδή $x < v$, για κάθε x του (u, w) . Άρα $v \geq w$, και τα πρόσημα της h είναι:

$x < u$	$u < x < w$	$w < x < v$	$x > v$
+	-	+	-

Αφού $h(m) = -12 < 0$, το m ανήκει σε περιοχή αρνητικού προσήμου, δηλαδή $m \in (u, w) \cup (v, \infty)$. Και στις δύο περιπτώσεις $m > u$, άρα η **(III) ισχύει πάντα**. Οι (I) και (II) ισχύουν μόνο όταν το m είναι στο (u, w) , ενώ αν $m > v$ δεν ισχύουν.

Answer / Απάντηση: D) Only I, III (κλειδί του αρχείου)

Προσοχή: με την εκφώνηση όπως έχει μεταφερθεί, η μόνη πάντα αληθής πρόταση είναι η (III) και καμία επιλογή δεν είναι ακριβώς "Only (III)". Η D είναι η πλησιέστερη (και το κλειδί της εξέτασης), αλλά αξίζει έλεγχος της πρωτότυπης εκφώνησης.

Άσκηση 9 (Εκθετική)

$f(x) = c \cdot d^x$, $c, d > 0$. If $f(r) = k$ and $f(r+1) = k - 0.64k$, find d .

Λύση

Η νέα τιμή είναι $f(r+1) = k - 0.64k = 0.36k$. Ο λόγος διαδοχικών τιμών της εκθετικής είναι σταθερός και ίσος με τη βάση:

$$\frac{f(r+1)}{f(r)} = \frac{cd^{r+1}}{cd^r} = d.$$

Άρα

$$d = \frac{0.36k}{k} = 0.36.$$

(Μείωση 64% ανά βήμα σημαίνει πολλαπλασιασμός με $1 - 0.64 = 0.36$.)

Answer / Απάντηση: $d = 0.36$

Άσκηση 10 (Ρίζες τετραγωνικών)

$f(x) = x^2 + 3x - 10$ has solutions $r < s$; $g(x) = x^2 + 4x - 21$ has solutions $u < v$. $h(x) = x^2 + 7x + c$ has solutions $s + u$ and $v + r$ and can be rewritten as $(x + a)(x + b)$. Find c .

Λύση

Ρίζες της f : $x^2 + 3x - 10 = (x + 5)(x - 2) = 0 \Rightarrow r = -5, s = 2$.

Ρίζες της g : $x^2 + 4x - 21 = (x + 7)(x - 3) = 0 \Rightarrow u = -7, v = 3$.

Ρίζες της h :

$$s + u = 2 - 7 = -5, \quad v + r = 3 - 5 = -2.$$

Άρα $h(x) = (x + 5)(x + 2) = x^2 + 7x + 10$. **Έλεγχος:** άθροισμα ριζών $-5 + (-2) = -7 = -(7) \checkmark$

Answer / Απάντηση: $c = 10$